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How Rich are Mixture Attitudes? Theory and Experimental Evidence*

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Abstract

Preferences are often assumed to display global mixture attitudes, i.e., convex preferences are globally mixture-loving. We study preferences that can be both mixture-loving and averse. We provide a hierarchy capturing departures from one-sidedness in a general environment encompassing consumption, risk, social allocation, and others. The farthest departure is *strong non-betweenness* (SNB), which allows indifference curves to be neither convex nor concave. We show that SNB is generic within this class. Our results distinguish familiar models and show how to identify this behavior experimentally. A comprehensive review reveals that past experimental designs can seldom test for it. In a high-stakes experiment involving lotteries over holidays, 42% of subjects display SNB. Further theoretical and experimental results show that SNB arises from the interaction between preferences for certainty and randomization, which drive opposite mixture attitudes at different mixture weights. We see SNB as the behavioral counterpart of facing rich multi-dimensional mixture tradeoffs, and our results suggest it may be more prevalent than previously thought.

JEL Codes: C91, D81, D11

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1 Introduction

Economic environments often feature ways to combine objects of choice. For example, weighted averages of consumption bundles, social allocations, production inputs, as well as probabilistic mixtures of lotteries, portfolios, and strategies. These *mixture operations* are fundamental as they describe the geometry of the space but, more importantly, because preferences are often defined via *mixture attitudes*.

Attitudes towards mixtures have been widely studied across environments. Convex preferences, which weakly favor mixtures, are associated with diminishing marginal rates of substitution in consumer theory, diversification, consumption smoothing over time, inequity aversion, and hedging under ambiguity. Conversely, concave preferences, which weakly disfavor mixtures, capture risk seeking, specialization, and optimism under ambiguity. Expected utility and betweenness preferences satisfy both conditions.[1]

Most theories assume globally convex or concave preferences, which are either mixture loving or mixture averse. This is a strong assumption. For instance, the same individual may buy both insurance and lottery tickets (Friedman and Savage, 1948), substitutes may become complements as the budget increases, and an agent may be risk averse for gains but risk loving for losses (Kahneman and Tversky, 1979; Tversky and Kahneman, 1992). In this paper, we depart from this assumption and study preferences that are neither convex nor concave. We say that such preferences display *mixed mixture attitudes* (MMA henceforth). To motivate them, consider the following thought experiment.

Suppose Alice has two trip destinations: Chamonix and Malaga. She is indifferent between Chamonix and Malaga. Suppose there are also two "surprise trips" which are 1) a 50-50 chance of Chamonix-Malaga and 2) a 90-10 chance of Chamonix-Malaga, with the trip destination revealed before departure. Many theories allow Alice to prefer the 50-50 mixture to Chamonix. This could be motivated by a preference for randomization and Alice liking to be surprised. Indeed, such surprise trips are a real-life commodity and are frequently purchased. Suppose, however, that Alice also prefers going to Chamonix for sure over the 90-10 lottery. She prefers to randomize if it is random enough, else she prefers to know for sure. This can naturally be motivated by a preference for certainty, which is strongest when probabilities are close to 100-0. However, the joint preference pattern above cannot be accommodated by any globally convex or concave preferences.

We see two important gaps in our understanding of the above preferences. First, while some existing models can accommodate MMA, no general framework organizes these preferences. Second, we know little about the prevalence of such preferences and the mechanisms

[1]For consumer theory, see Debreu (1959) and Mas-Colell, Whinston and Green (1995); for diversification, Markowitz (1952); for consumption smoothing, Samuelson (1937); for inequity aversion, Fehr and Schmidt (1999) and Bolton and Ockenfels (2000); for ambiguity hedging, Schmeidler (1989) and Gilboa and Schmeidler (1989); for ambiguity attitudes, Ghirardato, Maccheroni and Marinacci (2004); for expected utility, von Neumann and Morgenstern (1944); and for betweenness preferences, Chew (1983; 1989), Dekel (1986), and Gul (1991).

that generate them. Our extensive literature audit across domains such as risk, uncertainty, consumption, time, and social preferences shows that only a very small fraction (4%) of existing experiments can, even in principle, directly identify MMA.

We address these gaps on both theoretical and empirical grounds. On the theoretical end, we develop a hierarchy that captures how far preferences depart from global one-sidedness, establish the economic implications of these departures, and classify existing models according to the attitudes they accommodate. We further show that the strongest departure, strong non-betweenness, is generic among MMA preferences. On the empirical end, we document the widespread prevalence of these attitudes in a high-stakes experiment involving surprise trips. We show that these choices are intentional rather than driven by noisy or stochastic choice, and arise from the interaction between preferences for randomization and certainty.

We begin with our theoretical framework. The choice space is $\mathcal{X} \subseteq [0,1]^D$, convex and compact. The mixture operation is simply convex addition. This general and mathematically transparent framework nests many economic environments, including consumption bundles, social allocations, lotteries, and Anscombe-Aumann acts. For example, mixtures correspond to weighted averages of consumption bundles, probabilistic mixtures of lotteries, and state-wise mixtures of acts. We impose four basic rationality axioms: completeness, transitivity, mixture continuity, and strict monotonicity. While MMA preferences may arise from noisy or irrational choices, we are interested in their shape and provenance from intentional choices under a rational benchmark.

Within this framework, we distinguish three increasingly strong departures from global one-sidedness. MMA requires both mixture-loving and mixture-averse choices, potentially for *different pairs* of objects. *Weak non-betweenness* (WNB) requires both attitudes for mixtures of the *same pair* of objects, whereas *strong non-betweenness* (SNB) further requires that the *same indifferent pair*. To illustrate, suppose Alice prefers a 50-50 lottery between Chamonix and Malaga to either destination for sure, but prefers either Lisbon or Rome for sure to a 50-50 lottery between them, while being indifferent between the destinations in each pair. These choices display MMA, but need not display WNB. Now suppose Alice strictly prefers Chamonix to Malaga, but prefers a 90-10 Chamonix-Malaga lottery to Chamonix and Malaga to a 10-90 lottery. She then displays WNB. Finally, Alice's original preferences display SNB: she is indifferent between Chamonix and Malaga, prefers the 50-50 lottery to either, but prefers either to the 90-10 lottery. We show that these three notions form a strict hierarchy and classify existing models according to the attitudes they accommodate. Importantly, each level of this hierarchy imposes different restrictions on how mixture attitudes can change.

In particular, we show that WNB without SNB restricts the order in which mixture attitudes can change along the same pair of objects. Moving from the worse to the better endpoint, if a mixture is worse than the worse endpoint and another is better than the better endpoint, the former must come first. Conversely, if a peak above the better endpoint precedes a dip be-

low the worse endpoint, then the preference necessarily displays SNB. This *peak-dip condition* characterizes SNB even when the original endpoints are not indifferent, making it easier to identify experimentally. More generally, the hierarchy shows that SNB permits substantially richer mixture attitudes than MMA or WNB alone. However, since SNB requires opposing mixture attitudes even between indifferent objects, one might expect such preferences to be particularly demanding.

We show that this is not the case. Among MMA preferences, those displaying SNB form an open and dense set, even when we restrict attention to smooth preferences. The intuition is simple. Without SNB, moving from a mixture-loving to a mixture-averse indifference curve requires passing through a chord on which mixing is entirely neutral. By arbitrarily slightly distorting this neutral chord, we obtain SNB while preserving MMA. Thus, once we allow even the weakest departure from global one-sidedness, SNB is generically the norm. Importantly, this density result is special to MMA, as SNB is not dense among all twice continuously differentiable preferences. This raises the question of what happens when neutrality is not accidental, but imposed structurally in a partial manner.

We investigate this question through certainty independence, a common axiom in theories under uncertainty, which implies mixture neutrality with respect to constant acts. We show, in our framework, that even a weak form of certainty independence is sufficient for MMA to imply SNB. In particular, in standard Anscombe–Aumann environments satisfying certainty independence, any departure from global one-sidedness necessarily generates SNB. Hence, while our genericity result shows that SNB is the norm among MMA preferences, certainty independence provides an important class of environments in which it is unavoidable. Together, these results suggest that SNB is considerably less exceptional than its definition might suggest. This leaves open the question of what economic considerations can give rise to such rich preferences.

Finally, we investigate how SNB can arise from the aggregation of opposing motives. Many economic choices involve multiple considerations that favor different attitudes towards mixing. An investor may value diversification while disliking the cost of holding multiple positions. Similarly, Alice may enjoy randomization while also valuing certainty. We study the aggregation of a mixture-loving and a mixture-averse preference under continuous and strictly increasing aggregation rules. We show that there exist pairs of such preferences whose aggregation necessarily generates SNB, independently of the aggregation rule. The intuition is that the relative strength of the two motives changes along the mixture. In Alice's case, the preference for randomization may dominate around the 50-50 lottery, while the preference for certainty dominates near the endpoints. This provides a natural explanation for the preference pattern in our example and in our experiment.

Our theoretical results and the thought experiment suggest that MMA preferences may arise naturally, but tell us little about their empirical prevalence. Existing experimental evi-

dence is similarly limited. This may reflect the limitations of experimental designs rather than an absence of such preferences. To investigate this gap, we conducted a systematic literature audit of 617 experimental studies across seven domains, including risk, consumption, time, portfolio choice, charitable giving, effort allocation, and social preferences. We find that only 23 studies (4%) can, in principle, identify MMA, and only 10 (1.6%) can identify SNB. The main limitation is that most designs elicit only weak preferences or choices over a single mixture weight, whereas identifying richer attitudes requires multiple strict comparisons from the same individual. Thus, despite extensive experimental work on mixture attitudes, we know remarkably little about how prevalent MMA preferences are.

We therefore turn to an experiment that allows us to identify these attitudes and investigate their origins. We offer a reanalysis of the high-stakes experiment of Toussaert (2024), in which subjects choose between sure holiday destinations and lotteries over trips whose destinations are revealed shortly before departure. We focus on surprise trips for several reasons. First, trips are realistic and familiar objects of choice while surprise trips, and other surprise experiences, have been gaining popularity.[2] Second, trips are multidimensional: two destinations can be equally attractive while offering very different experiences. Finally, delayed revelation creates competing motives. Subjects may enjoy imagining the possibilities and being surprised, while also preferring certainty to avoid anxiety and facilitate preparation. This provides a natural environment for studying the aggregation of opposing motives suggested by our theory.

Our empirical analysis addresses five questions: how rich mixture attitudes are, which mixtures subjects prefer, whether these choices are intentional or noisy, what motivates them, and how multidimensionality affects them. The experiment provides several complementary measures to answer these questions.

The experiment involved 83 respondents who faced a real possibility of winning a four-day European trip worth £420. Subjects first ranked ten destinations and reported their monetary valuations. They then completed 45 decision problems, primarily choosing between a sure destination and lotteries over trips. Subjects also chose between sure trips so that we could check consistency of rankings with binary choice and rule out noisy rankings. The binary lotteries varied the probability of the better-ranked destination between 10%, 50%, and 90%, allowing us to identify mixture attitudes along the same chord. Additional tasks elicited subjects' willingness to construct their own lotteries, delay learning their destination, and pay to preserve the surprise. Subjects also reported their motives for and against uncertainty and made analogous choices over monetary lotteries.

We summarize our five main findings below.

1. *How rich are mixture attitudes?* We use rankings, valuations, and lottery choices to clas-

[2]The global surprise-trip planning market was estimated at USD 1.4 billion in 2024 and is projected to reach USD 5.1 billion by 2033 (Growth Market Reports, 2025). Additionally, 52% of respondents to a Booking.com survey expressed interest in taking a surprise trip.

sify subjects according to our hierarchy. Of the 79 classifiable respondents, 36 display MMA, of whom 35 display SNB. Thus, around 42% of our full sample displays SNB.

2. *Which mixtures do people prefer?* Mixture loving is most common at 50-50, particularly between similarly ranked destinations. In contrast, mixture aversion is most common when the probability of the better destination is small. The most common pattern is similar to the thought experiment: preferring a 50-50 lottery to the better destination, but preferring that destination to a 90-10 lottery. Thus, we observe the *peak then dip* behavior that characterizes SNB.

3. *Are these preferences intentional or noisy?* Our theoretical framework considers preferences satisfying standard rationality axioms, but similar choice patterns might also arise from stochastic errors. We show that the identified mixture attitudes predict behavior in other tasks, including lotteries over more destinations and subjects' willingness to construct their own lotteries. Moreover, standard stochastic-choice models fail to reproduce the observed patterns. This suggests that these attitudes reflect deliberate preferences rather than merely noisy choices.

4. *Where does the (dis)taste for randomization come from?* Subjects who enjoy anticipation, surprise, and imagining possibilities are more mixture loving, while concerns about preparation and disappointment are more mixture averse. Moreover, 74% of subjects who construct their own lotteries choose to delay learning their destination, and 54% are willing to give up at least £5 to preserve the surprise. These findings are consistent with competing preferences for randomization and certainty.

5. *How does multidimensionality affect mixing?* Money is one-dimensional: more is always better. Trips, however, can be equally attractive while offering different experiences. We investigate whether this distinction matters by comparing lotteries over trips with analogous lotteries over money. Mixture lovingness almost disappears: the share preferring a 50-50 lottery to the better sure outcome falls from 53% for trips to 4% for money. We distinguish between *vertical* attributes, for which the subject has a clear preference, and *horizontal* attributes, to which the subject is ambivalent. For example, a subject may prefer warmer to colder climate, but be open to both the beach and the mountains. We find that horizontal differences are associated with more mixture loving, whereas vertical attributes predict valuations of sure trips. Thus, the attitude towards randomization appears to depend not only on uncertainty about the value of the outcome, but also on uncertainty about the experience itself.

Related Literature. Our work firstly speaks to literatures in many economic environments that feature mixture operations, from consumption to risk, uncertainty, social allocations, and many others. In all those settings, mixture attitudes are typically held fixed and globally one-sided. This is partly due to tractability, since convexity keeps optimization and equilibrium

well behaved. It also partly reflects that these theories typically rely on one particular motive, such as diversification (Markowitz, 1952; Dekel, 1989), hedging (Gilboa and Schmeidler, 1989; Maccheroni, Marinacci and Rustichini, 2006), caution (Cerreia-Vioglio, Dillenberger and Ortoleva, 2015), or fairness (Fehr and Schmidt, 1999; Saito, 2013). Our contribution here is to study departures from these restrictions in a general theoretical environment which nests these domains akin to the approach taken in Herstein and Milnor (1953). We offer a hierarchy of mixed attitudes, based on how much they depart from global one-sidedness. This hierarchy applies to all these domains as it only relies on the shapes of indifference curves and is independent of particular psychological or domain-specific channels. On a more general level, we highlight that mixed attitudes can naturally arise out of deliberate and intentional aggregation of multiple motives that have opposite mixture attitudes. Our literature audit shows that their apparent experimental rarity is driven by design capabilities.

Second, a rich literature on risk and uncertainty has developed models that accommodate mixed mixture attitudes (Machina, 1982; Quiggin, 1982; Chew, Epstein and Segal, 1991; Tversky and Kahneman, 1992; Ghirardato, Maccheroni and Marinacci, 2004; Klibanoff, Marinacci and Mukerji, 2005). Our hierarchy organizes these models and clarifies the restrictions they impose. For example, quadratic utility accommodates MMA, but not WNB (Machina, 1982; Chew, Epstein and Segal, 1991). In contrast, we show that a weak form of certainty independence (Gilboa and Schmeidler, 1989) implies that MMA must take its strongest form, SNB. This applies, in particular, to α-maxmin (Ghirardato, Maccheroni and Marinacci, 2004) and Choquet expected utility (Schmeidler, 1989). Thus, models that accommodate mixed attitudes can nevertheless impose very different restrictions on preferences.

Previous experiments have also provided suggestive evidence of mixed attitudes in risky choice (Chew and Waller, 1986; Camerer, 1989; Camerer and Ho, 1994; Bernasconi, 1994; McGranaghan et al., 2026). However, most were not designed to distinguish their different forms and to test the strongest departures. Our literature audit shows that existing studies have largely focused on monetary lotteries, elicited only weak comparisons or a single mixture weight, or lacked sufficient power to identify richer attitudes. Additionally, there is indirect evidence of mixed attitudes via indirect estimation of the probability weighting function.[3] Our contribution is to document the prevalence of SNB in a realistic, high-stakes setting. Moreover, our theoretical characterization allows us to identify SNB through a simple extension of standard experimental methods.

Finally, our work connects two literatures that have largely developed separately. A preference for certainty has been documented since Allais (1953), through the certainty effect (Kahneman and Tversky, 1979), disproportionate preferences for certain payments (Andreoni and Sprenger, 2012), and lotteries valued below their worst outcome (Gneezy, List and Wu,

[3] Wu and Gonzalez (1996); Gonzalez and Wu (1999); Abdellaoui (2000); Bleichrodt and Pinto (2000) find evidence of probability weighting that is neither convex nor concave, which would imply MMA *given specific models*, but they do not document direct choices that reveal MMA.

2006). It has been modeled through disappointment (Gul, 1991), preferences over the timing of uncertainty resolution (Dillenberger, 2010), and caution (Cerreia-Vioglio, Dillenberger and Ortoleva, 2015). Conversely, a preference for randomization has been documented through repeated choices (Agranov and Ortoleva, 2017; Agranov, Healy and Nielsen, 2023), convex budgets (Feldman and Rehbeck, 2022), and high-stakes field decisions (Dwenger, Kübler and Weizsäcker, 2018). It has also been modeled as a deliberate preference (Machina, 1985; Cerreia-Vioglio et al., 2019). Common explanations include indecision (Agranov and Ortoleva, 2017; Dwenger, Kübler and Weizsäcker, 2018; Halevy, Walker-Jones and Zrill, 2025) and hedging (Raiffa, 1961; Saito, 2015). However, relatively little attention has been paid to how preferences for certainty and randomization interact.[4] We instead consider the aggregation of two opposing motives. A preference for certainty may dominate near the endpoints, while a preference for randomization may dominate near an even split. Our aggregation results show how this interaction can generate SNB, and our experiment documents precisely this pattern.

Our experiment also sheds light on the origins of these motives. In particular, the preference for randomization appears related to anticipatory utility from uncertainty over different experiences (Loewenstein, 1987; Caplin and Leahy, 2001). This is consistent with evidence that individuals enjoy imagining uncertain positive outcomes (Lee and Qiu, 2009; Kocher, Krawczyk and van Winden, 2014) and may prefer mystery goods to known goods of equal expected value (Buechel and Li, 2023). Unlike standard explanations based on indecision, hedging, or domain-specific risk attitudes (Dillenberger and Lu, 2026), this mechanism can explain why a lottery is preferred to even its best possible outcome. Together with the preference for certainty, it provides a natural explanation for the rich mixture attitudes we observe.

The paper is structured as follows. We introduce the theoretical framework and our theoretical results in **Section 2**. **Section 3** goes over our literature audit. **Section 4** details our experimental design and empirical findings. **Section 5** discusses further directions and other areas of interest for our findings.

2 Theory

2.1 General Framework

We consider a general framework that is flexible enough to handle many common economic environments in which mixing may take place. Each object of choice is described by $d \geq 2$

[4]Existing models connecting the two typically rely on a single motive. For example, cautious expected utility explains both the certainty effect and a preference for randomization through caution, which generates convex preferences (Cerreia-Vioglio, Dillenberger and Ortoleva, 2015; Cerreia-Vioglio et al., 2019). Similarly, McGranaghan et al. (2026) document the coexistence of common-ratio effects and an attraction to mixing. However, their model of upside potential is a form of quadratic utility, which by our results can only accommodate opposite mixture attitudes on different chords.

dimensions, each measured on a normalized scale on which higher values are better. The choice space is a compact, convex set $\mathcal{X} \subseteq [0,1]^D$ that contains the diagonal

$$b(t) := t\mathbf{1} = (t, \ldots, t), \qquad t \in [0,1],$$

which we use as a common yardstick to calibrate preferences. This framework nests many economic environments. A finite lottery can be represented by its upper-tail probabilities, a finite Anscombe–Aumann act by the vector of its statewise upper-tail probabilities,[5] and consumption bundles or social allocations by one coordinate per good or individual. Our results therefore apply straightforwardly across these environments and characterize mixture attitudes through their geometric properties alone.

Our analysis rests on two important notions defined on $\mathcal{X}$. The first one is *mixing*, which we define through the natural operation of convex combination: for $\alpha \in [0,1]$, we write

$$x\alpha y := \alpha x + (1-\alpha)y.$$

In the environments above, this corresponds to probabilistic mixing of lotteries, statewise mixing of acts, and proportional combination of bundles or allocations. The second is *dominance*: for $x, y \in \mathcal{X}$, write $x \geq y$ when $x_i \geq y_i$ in every dimension, and write $x > y$ when $x \geq y$ and $x \neq y$. We study preferences on $\mathcal{X}$ that satisfy the following basic rationality postulates.

Definition 1 (Basic Axioms)**.**

- *Complete:* $\forall x, y \in \mathcal{X}$, $x \succsim y$ *or* $y \succsim x$.
- *Transitive:* $\forall x, y, z \in \mathcal{X}$, $x \succsim y$ *and* $y \succsim z$ *implies* $x \succsim z$.
- *Mixture Continuous:* $\forall x, y, z \in \mathcal{X}$, *the sets* $\{\alpha \in [0,1] \mid x \succsim y\alpha z\}$ *and* $\{\alpha \in [0,1] \mid x \precsim y\alpha z\}$ *are closed.*
- *Strictly Monotone:* $\forall x, y \in \mathcal{X}$, *if* $x \geq y$, *then* $x \succsim y$; *if* $x > y$, *then* $x \succ y$.

Importantly, strict monotonicity implies that $x \succ x\alpha y \succ y$ whenever $x > y$ and $\alpha \in (0,1)$. Monotonicity is a natural assumption in many environments. In probabilistic settings, it is stochastic dominance, while in social choice, it is Pareto dominance. Hence, rich mixture attitudes can only arise from trade-offs between dimensions (requiring $d \geq 2$). As dominance leaves these trade-offs unresolved, incompleteness is present in our framework through the dominance relation. We however require the preference relation to be complete, since the

[5] The simplex does not fit our framework directly: since probabilities sum to one, its only diagonal point is $(1/n, \ldots, 1/n)$, and raising one coordinate requires lowering another, so its coordinates are not monotone. Instead, rank the outcomes $x_1 \succ \cdots \succ x_n$ and map a lottery $(p_1, \ldots, p_n)$ to its upper-tail probabilities $(p_1,\ p_1 + p_2,\ \ldots,\ \sum_{k=1}^{n-1} p_k)$. This map is bijective and linear, so mixing is preserved; preferences that respect first-order stochastic dominance are monotone in the new coordinates; and the diagonal $(t, \ldots, t)$, $t \in [0,1]$, is the lottery $(x_1, t;\ x_n, 1-t)$. For a finite-state Anscombe–Aumann act, we can apply the map state by state and the diagonal then consists of constant acts.

decision maker must ultimately make a choice. Mixed mixture attitudes could arise from incomplete preferences, stochastic choice, or other departures from these axioms. We choose to maintain the rationality axioms because our aim is to understand how minimal departures from one-sided mixture attitudes, within the rationality benchmark, can generate rich choice behavior.

2.2 A Hierarchy of Mixture Attitudes

Standard theory typically imposes a global restriction on mixture attitudes: convexity, concavity, or both. Classic models, such as expected utility, impose both convexity and concavity on *probabilistic* mixtures.[6] Betweenness preferences (Chew, 1983; Dekel, 1986) retain both while relaxing independence. Convexity alone is also ubiquitous. Standard consumer preferences such as Cobb–Douglas, Leontief, and CES preferences are convex (Cobb and Douglas, 1928; Dixit and Stiglitz, 1977), as are maxmin expected utility and variational preferences under uncertainty (Gilboa and Schmeidler, 1989; Maccheroni, Marinacci and Rustichini, 2006). Conversely, concavity captures a global preference against mixtures. Simple specialization-loving preferences, such as $u(x) = \max_i x_i$ have this property; under uncertainty, maxmax expected utility and smooth ambiguity with a convex second-order aggregator provide analogous examples. We provide their definitions here.

Definition 2 (Basic Mixture Attitudes)**.**

- $\succsim$ *is convex if* $\forall x \succsim y$ *and* $\alpha \in [0,1]$, $x\alpha y \succsim y$.
- $\succsim$ *is concave if* $\forall x \succsim y$ *and* $\alpha \in [0,1]$, $x\alpha y \precsim x$.

But many real-world attitudes are not globally one-sided. An agent may be risk-loving over losses and risk-averse over gains; two goods may be substitutes for some prices and complements in others. Models that relax both conditions have received less attention, and we actually know fairly little about the structure of such preferences. Consumer theory tends to lump all non-convex preferences into a single category, and decision theory similarly has not investigated this class of preferences in depth. In particular, are there weaker and stronger departures from these global mixture attitudes? What do these departures look like, and are they pathological or reasonable? Are these departures economically prevalent and relevant?

The most general departure from one-sidedness is to display both attitudes somewhere. We call this mixed mixture attitudes (MMA).

Definition 3 (Mixed Mixture Attitudes)**.** *A preference* $\succsim$ *displays mixed mixture attitudes (MMA) if there exist pairs* $x^+ \succsim y^+$ *and* $x^- \succsim y^-$ *and weights* $\alpha, \beta \in (0,1)$ *such that*

$$x^+\alpha y^+ \succ x^+, \quad \text{and} \quad x^-\beta y^- \prec y^-.$$

[6] Yaari (1987) is a well known exception and considers rank weighted prize mixtures.

Crucially, these pairs can be distinct and hence both mixture attitudes may exist but may be separated. A stronger notion requires both attitudes to appear for the same pair of objects. We call this weak non-Betweenness (WNB).

Definition 4 (Weak Non-Betweenness). $\succsim$ *displays Weak Non-Betweenness (WNB) if there exist* $x \succsim y$ *and* $\alpha, \beta \in (0,1)$ *such that*

$$x\alpha y \succ x \succsim y \succ x\beta y.$$

Finally, our strongest notion requires both attitudes to appear for a pair of indifferent objects. We call this strong non-Betweenness (SNB).

Definition 5 (Strong Non-Betweenness). $\succsim$ *displays Strong Non-Betweenness (SNB) if there exist* $x \sim y$ *and* $\alpha, \beta \in (0,1)$ *such that*

$$x\alpha y \succ x \sim y \succ x\beta y.$$

Denote by $\mathcal{P}^M, \mathcal{P}^W, \mathcal{P}^S$ the set of preferences that satisfy our Basic Axioms and display MMA, WNB, and SNB, respectively. It is immediate that

$$\mathcal{P}^S \subseteq \mathcal{P}^W \subseteq \mathcal{P}^M.$$

MMA lets the two signs occur on unrelated chords. WNB places them on the same chord. SNB additionally holds the endpoint value fixed. The hierarchy therefore measures how freely the sign of the mixture attitude can move through the choice space. It is not automatic that these definitions are behaviorally distinct, especially given our basic axioms. Under mixture continuity, one can replace $x \succsim y$ by $x \sim y$ in **Definition 2** and obtain an equivalent definition of convex and concave preferences. Hence, it may not be immediately obvious that WNB and SNB are distinct axioms. However, it turns out they are distinct in behaviorally significant manners.

Proposition 1. $\mathcal{P}^S \subsetneq \mathcal{P}^W \subsetneq \mathcal{P}^M$ *even if all Basic Axioms hold.*

We illustrate the difference between these three classes of preferences via **Figure 1** and **Figure 2**, the indifference curves (ICs henceforth) and value functions of chords for preferences that satisfy the three properties, respectively. **Figure 1** showcases three sets of ICs of preferences satisfying MMA without WNB, WNB without SNB, and SNB, respectively. The value functions give intuition for **Proposition 1**. In **Figure 2a**, the equivalence between the $\succsim$ and $\sim$ definitions of convexity and concavity is immediate. Note that if a pair $x \succ y$ displays convexity (mixture loving) then one can move y continuously until a point $\hat{y} \sim x$ without losing convexity. However, in **Figure 2b**, we see that moving y to a point $\hat{y} \sim x$ is always possible, but because mixture attitudes are not global, you may have now excluded

some mixture signs that existed in the original extended chord. Thus SNB is strictly more demanding than WNB.

In the following, we will discuss and highlight what each of these three classes of preferences in our hierarchy imposes on choice observables.

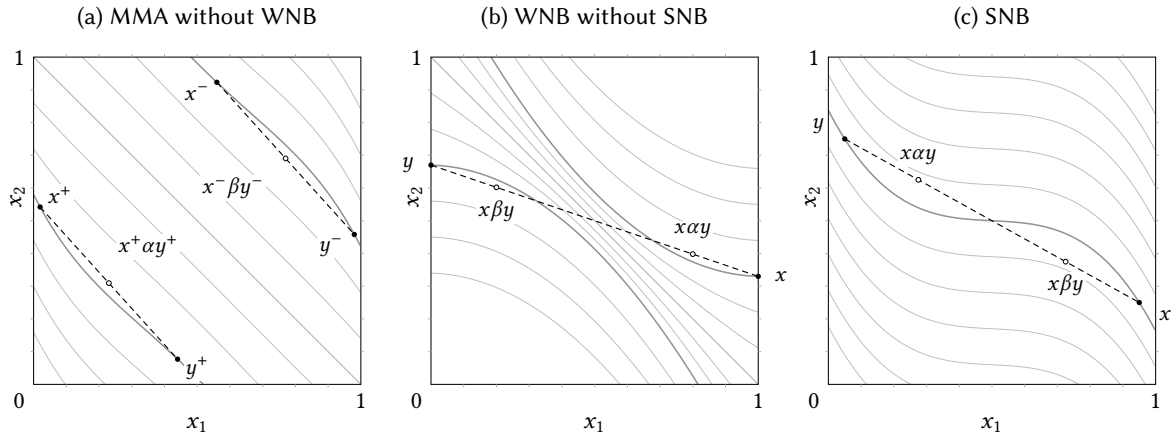


Figure 1: Three strictly monotone indifference maps on $[0,1]^2$. In panel (a), the lower dashed chord witnesses mixture lovingness for $x^+ \sim y^+$, while the upper dashed chord witnesses mixture aversion for $x^- \sim y^-$. In panel (b), the dashed chord satisfies $x\alpha y \succ x \succsim y \succ x\beta y$. In panel (c), the endpoints satisfy $x \sim y$ and the marked mixtures satisfy $x\alpha y \succ x \sim y \succ x\beta y$.

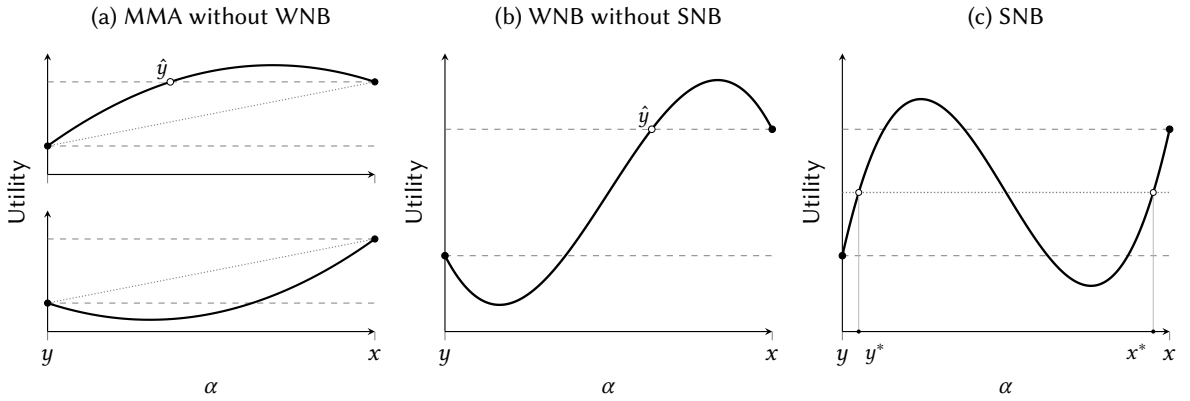


Figure 2: Utility of mixtures $x\alpha y$ with $x \succ y$, where α is the weight on x. Panel (a) shows two one-sided benchmark chords displaying mixture lovingness and mixture aversion. In panel (b), the dip precedes the peak, the only pattern WNB without SNB permits. In panel (c), the peak precedes the dip. The marked points $\hat{y}$ in the upper graph of panel (a) and in panel (b) satisfy $\hat{y} \sim x$.

How different are WNB and MMA? Figure 1a showcases ICs that allow for MMA but not WNB. The chords from x^+ to y^+ and x^- to y^- are strictly mixture loving and averse, respectively. However, the ICs are also drawn in such a way that no chord, no matter the origin and orientation, can ever cross two ICs of opposite curvature twice each. On the other hand, WNB without SNB allows for such crossings but fixes ICs to be only of one sign. In Figure 1b, we see precisely that for a single chord from x to y to display both mixture attitudes, it must cross two ICs, each twice.

What restrictions do these choice behaviors impose? First, while MMA can display ICs that are mixture loving initially from (0,0) and then become mixture averse at (1,1), WNB can never occur in such a transition region.[7] Second, it is possible for a non-WNB MMA preference to have mixture averse low-value ICs and mixture loving high-value ICs. Nonetheless, if this happens, the curvature of the ICs, when the two regions meet, must be small, else a WNB chord becomes possible.

We illustrate how WNB is a stronger departure than MMA from one-sidedness via the perspective of consumption bundles. Pick any two bundles x and y that coincide in their MRS, which in turn coincides with a price vector p. Then, with globally convex or concave preferences, we know that $x \succ y$ implies $x \cdot p \geq y \cdot p$. The preferred bundle must cost at least as much at the same prices. This is still true for MMA preferences without WNB, but not for WNB ones. We relegate the exact statement and its proof to **Section A.2**.

To summarize, MMA without WNB implies a fairly mild departure from globally one-sided preferences, as the different attitudes are neatly compartmentalized in different regions. In addition, when these different regions meet, the curvature is close to linear. However, WNB necessitates more pronounced curvature in the meeting regions and also requires a specific directionality of curvature when moving from worse to better outcomes.

How different are WNB and SNB? For WNB to occur, we must go from a lower valued mixture averse IC to a higher valued mixture loving IC. But can it occur in other ways without triggering SNB? The answer is no. The next proposition makes this precise.

Proposition 2 (Ordered mixture attitudes)**.** *If $\succsim$ satisfies the basic axioms and displays WNB but not SNB, then for every $x \succsim y$, exactly one of the following holds:*

1. *For every $\eta \in [0,1]$, $x\eta y \succsim y$, or for every $\eta \in [0,1]$, $x\eta y \precsim x$.*
2. *$x \succ y$ and there exist $0 < \alpha < \beta < 1$ such that:*
   - *(a) if $\eta \in (0,\alpha)$, then $x\eta y \prec y$;*
   - *(b) if $\eta \in [\alpha,\beta]$, then $x \succsim x\eta y \succsim y$;*
   - *(c) if $\eta \in (\beta,1)$, then $x\eta y \succ x$.*

The first case covers mixtures that are strictly one-sided (loving only or averse only), which can naturally coexist alongside WNB chords elsewhere in the choice space. The second case provides the central behavioral implication: whenever both mixture attitudes occur on the same chord, mixture lovingness (case c) must occur at a higher mixture weight on the preferred option than mixture aversion (case a), meaning $\alpha < \beta$.

[7] WNB preferences can of course display such a shift in one region while having the WNB certifying behavior elsewhere.

To see the intuition, note that if ICs have the shape of **Figure 1b**, then the value function along the chord must take the form shown in **Figure 2b**. In the absence of SNB, a horizontal utility level can intersect the mixture path in at most two connected components. Consequently, the value function must exhibit mixture aversion at low mixture weights before transitioning to mixture lovingness at high mixture weights. The opposite pattern—where a peak precedes a dip—inevitably forces a horizontal utility level to cross the value function three times as in **Figure 2c**, yielding an indifference pair that displays both mixture attitudes and thereby triggers SNB. More generally, if we have more than one peak and one dip, it must be SNB. So highly non-linear functions all belong to the SNB class. We relegate the proof to **Section A.3**.

Operationalizing SNB. SNB requires indifferent endpoints, which are hard to obtain experimentally, so the property is hard to test. We provide an alternative definition to overcome this difficulty.

Definition 6. *$\succsim$ displays SNB* if there exist $x \succsim y$ and $\alpha, \beta \in (0,1)$ with $\alpha < \beta$ such that*

$$x\alpha y \succ x \succ x\beta y \quad \text{or} \quad x\alpha y \succ y \succ x\beta y$$

SNB* differs from SNB and WNB in important ways. First, it is a property evaluated on an arbitrary chord without requiring prior knowledge of indifference. Second, it compares mixtures relative to a single endpoint (x or y). Finally, it formalizes the exact "peak-then-dip" mixture pattern that WNB without SNB explicitly rules out. Crucially, under our basic axioms, SNB* and SNB are logically equivalent.

Proposition 3. *If $\succsim$ satisfies the basic axioms then $\succsim$ displays SNB* if and only if $\succsim$ displays SNB.*

We highlight this result because it yields a significantly simpler and less demanding experimental identification of SNB behavior. In **Figure 2c**, both endpoints directly satisfy the definition of SNB*. By continuity, the value function crosses the middle utility level three times, generating the pair $x^* \sim y^*$ that satisfies the original definition of SNB. Conversely, taking any subchord $x^* \sim y^*$ satisfying SNB, extending the chord outwards naturally recovers SNB* at the extended endpoints. This equivalence removes the need for the experimenter to calibrate exact indifference pairs in the lab to test for SNB.

What do these types of mixture attitudes reflect about the decision-maker's preference? Mixture attitudes reflect the discrepancy between an agent's preference and the geometry of the mixture operation. The three properties capture the degree and flexibility of this discrepancy. In its strongest form, SNB imposes essentially no restriction about the shape or form of the discrepancy. However, MMA and WNB without SNB impose far more stringent restrictions.

Genericity of SNB. We ask next if there are purely geometric reasons to expect SNB, given that it violates many natural properties we have come to expect from one-sided preferences. We first endow the preference space with a natural topology. Note that if $\succsim$ satisfies the basic axioms, then it can be represented by a function $U_{\succsim}$ such that $U_{\succsim}(x) = t$ if and only if $x \sim b(t)$. $U_{\succsim}$ is continuous and strictly increasing in the coordinate-wise order. We call $U_{\succsim}$ the calibrated utility as it assigns to each outcome its equivalent value on the diagonal given $\succsim$. Write $\mathcal{U}^0$ for the set of calibrated utilities from preferences satisfying our basic axioms. For $k \geq 1$, let $\mathcal{U}^k$ contain the preferences whose calibrated utilities are at least k times continuously differentiable, endowed with the C^k topology.

We show that, in a geometric sense, SNB is not knife-edge or pathological amongst the class of MMA preferences.

Theorem 1.

1. *$\mathcal{P}^S \cap \mathcal{U}^k$ is open and dense in $\mathcal{P}^M \cap \mathcal{U}^k$ for all $k \geq 0$.*
2. *$\mathcal{P}^S \cap \mathcal{U}^k$ is not dense in $\mathcal{U}^k$ for all $k \geq 2$.*

Therefore, we show that SNB preferences are dense in MMA preferences. This holds even under arbitrarily strong C^k topologies. By contrast, SNB is not dense among all preferences satisfying our Basic Axioms for $k \geq 2$. While SNB can be generated by arbitrarily small distortions of indifference curves in weaker topologies, such distortions need not be small in C^k for $k \geq 2$. The density of SNB is therefore, in this sense, special to MMA preferences.

The proof has a simple intuition. Suppose that a preference $\succsim$ displays MMA but not SNB. There must then exist two distinct indifferent pairs, (x, y) and (x', y'), such that $\succsim$ is strictly mixture loving along one chord and strictly mixture averse along the other. By continuity, we can construct a path from (x, y) to (x', y') while preserving indifference between the endpoints at every point along the path. If SNB never occurs, the attitude toward mixing cannot change directly from loving to averse along this path. It must therefore pass through an intermediate indifferent pair (x^*, y^*) at which the preference is neutral toward mixing. The proof then shows that such exact neutrality is non-generic: it can be eliminated by an arbitrarily small perturbation. Hence, distorting the chord of a mixture neutral indifferent pair, rather than an arbitrary chord, is what allows for the stronger density result.

2.3 Models of Rich Mixture Attitudes

In this section, we discuss how various models of choice can accommodate mixed mixture attitudes. We focus here on models of choice under risk and uncertainty because the literature is richest here for models of mixture attitudes.[8]

[8]We focus on the attitudes towards *probabilistic* mixtures that these models exhibit. Yaari (1987) is a well-known exception and considers rank-weighted prize mixtures. The models we study can have, in general, attitudes towards *prize* mixtures that satisfy SNB.

Certainty independence and models under uncertainty. **Theorem 1** shows that MMA preferences without SNB can be approximated by preferences displaying SNB. Its proof shows that the set of preferences that are mixture neutral *somewhere* is meagre. One might therefore wonder if the result changes when considering models that allow for mixture neutrality partially in a structured manner. Certainty independence, a common axiom in decision theory under uncertainty, provides a natural test: it requires comparisons to be preserved when objects are mixed with a common certainty.

Call $n \in \mathcal{X}$ a *neutral benchmark* if, for every $x, y \in \mathcal{X}$ and $\gamma \in (0,1)$,

$$x \succsim y \quad \Longleftrightarrow \quad x\gamma n \succsim y\gamma n.$$

Certainty independence makes every constant act a neutral benchmark. We require only one such benchmark in the relative interior of the choice space.

Proposition 4. *Suppose $\succsim$ satisfies the Basic Axioms and admits a neutral benchmark $n \in \mathrm{ri}(\mathcal{X})$. Then*

$$\succsim \in \mathcal{P}^M \quad \Longleftrightarrow \quad \succsim \in \mathcal{P}^S.$$

The intuition is that we can contract a pair, while preserving its mixture attitude, toward a common neutral benchmark. Hence, a chord displaying mixture loving and another displaying mixture aversion can both be brought into an arbitrarily small neighborhood of the same interior point. Around that point, our assumption makes the local preference geometry homogeneous: the shape of an indifference surface is preserved as one moves toward or away from the benchmark. Opposite curvature on two chords therefore cannot be separated by an isolated flat region. They must generate a chord along which the attitude toward mixing itself changes sign, which is precisely strong non-Betweenness.

This result immediately applies to α-maxmin, Choquet expected utility, and, more generally, invariant biseparable preferences (Hurwicz, 1951; Schmeidler, 1989; Ghirardato, Maccheroni and Marinacci, 2004). In their Anscombe–Aumann representations, interior constant utility profiles provide the required benchmarks. These models can therefore be globally one-sided, but whenever they display MMA, they must display SNB. Hence, certainty independence imposes enough structure to rule out MMA and WNB without SNB. There is therefore at least one natural class of models where SNB is unavoidable.

But there are also models with greater flexibility. Smooth ambiguity (Klibanoff, Marinacci and Mukerji, 2005) and dual-self variational preferences (Chandrasekher et al., 2022) do not generally satisfy certainty independence. Mixture attitudes therefore need not be preserved when acts are mixed with a common certainty. This allows mixture-loving and mixture-averse attitudes to remain in different parts of the choice space without necessarily generating SNB. Consequently, these models are flexible enough to accommodate all three levels of our hierarchy, including MMA without WNB and WNB without SNB, on our bounded domain.

Models under risk. Are there models that can instead *only* display the weaker departures from one-sidedness? We give two models that respectively allow for MMA without WNB and WNB without SNB.

Example (Quadratic Utility) Let $\boldsymbol{p} = (p_1, ..., p_n)$ be a lottery over finite outcomes. Quadratic Utility has the value function

$$V(\boldsymbol{p}) = \sum_{i,j} \phi_{ij} p_i p_j,$$

where $\phi_{ij} = \phi_{ji}$. Write $\Phi = (\phi_{ij})$, so that $V(\boldsymbol{p}) = \boldsymbol{p}'\Phi\boldsymbol{p}$. Then, quadratic utility satisfies

$$V(\boldsymbol{p}\alpha\boldsymbol{q}) = \alpha V(\boldsymbol{p}) + (1-\alpha)V(\boldsymbol{q}) + K(\boldsymbol{p},\boldsymbol{q})\,\alpha(1-\alpha), \qquad K(\boldsymbol{p},\boldsymbol{q}) = -(\boldsymbol{p}-\boldsymbol{q})'\Phi(\boldsymbol{p}-\boldsymbol{q}).$$

Hence it displays MMA whenever K takes both signs across indifferent pairs, but it can never display WNB as the sign of K is fixed for any mixture.

Quadratic utility forces the mixture attitudes to depend only on the pairs being mixed. Therefore, the discrepancy between preference and mixture operation is not at the level of the mixture weight but purely at the level of objects. For quadratic utility, ϕ_{ij} describes pairwise interactions between outcomes. It is fixed for all pairs, does not depend on how much probability is transferred, and the overall mixture attitude is the sum of these gains and losses. However, such a model cannot enable a subject to be mixture loving for 50-50 weights but mixture averse for extreme weights. Similarly, we can show that a generalization of quadratic utility via the axiom of mixture symmetry (Chew, Epstein and Segal, 1991) yields preferences that can generate MMA and WNB but not SNB.

There are also models in risk that force the equivalence between SNB and MMA. Rank-dependent expected utility (Quiggin, 1982) is such a class of models. Its representation applies the same weighting function w to every upper-tail probability:

$$V(p) = u_1 + \sum_{i=2}^{n} (u_i - u_{i-1}) w(P_i), \qquad P_i = \Pr(x \geq x_i).$$

With at least four ordered prizes, MMA is equivalent to SNB. For MMA to occur, the weighting function w must be neither globally convex nor concave. With three prizes, mixture-loving and mixture-averse indifference curves can remain separated. However, with four prizes, the two intermediate prizes impose different utility weights on the same weighting function, ruling out the possibility of separation. On the other hand, cumulative prospect theory is more flexible because it evaluates gains and losses separately with possibly different weighting functions. Importantly, this flexibility comes from probability weighting across gains and losses, not from loss aversion alone.

These differences also clarify the overlap between models. Masatlioglu and Raymond

(2016) showed that quadratic utility and rank-dependent expected utility have a non-empty intersection *if* preferences are one-sided. We show here that the intersection is empty whenever $n \geq 4$ and preferences are not one-sided.

We end with Table 1, which summarizes the mixture attitudes of a comprehensive list of models in the domain of decision under risk and uncertainty that can generate MMA preferences for probabilistic mixtures.

Table 1: Exclusive mixture-attitude patterns across models

Model	Key feature	MMA only	WNB only	SNB
Panel A. Risk				
Quadratic utility (Machina, 1982; Chew et al., 1991)	Quadratic in probabilities	✓	✗	✗
Mixture symmetry (Chew et al., 1991)	Symmetry on indifferent chords	✓	✓	✗
RDEU, $n = 3$ (Quiggin, 1982)	Common probability weighting	✓	✗	✓
RDEU, $n \geq 4$ (Quiggin, 1982)	Common probability weighting	✗	✗	✓
RDEU with Prelec w, $n \geq 3$ (Prelec, 1998)	Parametric probability weighting	✗	✗	✓
Bell-type disappointment, $n \geq 3$ (Bell, 1985)	Linear, mean-referenced disappointment	✗	✗	✓
CPT, unrestricted weights (Tversky and Kahneman, 1992)	Separate gain/loss weighting	✓	✓	✓
Panel B. Uncertainty				
α-maxmin EU (Hurwicz, 1951)	Weighted best/worst expected utilities	✗	✗	✓
Choquet EU (Schmeidler, 1989)	Nonadditive beliefs	✗	✗	✓
Invariant biseparable (Ghirardato et al., 2004)	Certainty independence	✗	✗	✓
Smooth ambiguity (Klibanoff et al., 2005)	Unrestricted increasing aggregator	✓	✓	✓
Variational minus UA (Chandrasekher et al., 2022)	Optimism over penalized prior sets	✓	✓	✓

Note. A non-exhaustive selection of models satisfying the Basic Axioms on the stated domains. A ✓ indicates that some specification belongs to the corresponding set; a ✗ indicates that none does. Here n is the number of ordered prizes; states are unrestricted in number unless specified. UA denotes uncertainty aversion. We give further details in the Online Appendix.

2.4 Motive Aggregation

What can give rise to such attitudes? In this section, we show that aggregations of multiple preference motives naturally lead to mixed mixture attitudes.

Let $\mathcal{L}$ and $\mathcal{A}$ denote, respectively, the sets of preferences satisfying the Basic Axioms that are convex but not concave, and concave but not convex. Thus, the first class is one-sided mixture loving and the second is one-sided mixture averse, with a strict attitude somewhere in each case. We consider a decision maker who is motivated by two sets of motives. Her mixture loving and averse motives are summarized by some $\succsim_\ell \in \mathcal{L}$ and $\succsim_a \in \mathcal{A}$. For example, an investor may value diversification across stocks while disliking the cost of tracking or

trading a larger number of positions. More generally, the benefits and costs associated with mixing need not respond to changes in the mixture in the same way. We study what happens when two such opposing considerations are aggregated into a single preference.

Let $\mathcal{F}$ be the set of continuous aggregation rules $F : [0,1]^2 \to [0,1]$ that are strictly increasing in each argument and satisfy

$$F(t,t) = t \qquad \forall t \in [0,1].$$

Given $(\succsim_\ell, \succsim_a, F) \in \mathcal{L} \times \mathcal{A} \times \mathcal{F}$, write $U_{\succsim_i}$ for the calibrated utility of input i and define the aggregate preference $\succsim_F$ by

$$x \succsim_F y \quad \Longleftrightarrow \quad F\big(U_{\succsim_\ell}(x), U_{\succsim_a}(x)\big) \geq F\big(U_{\succsim_\ell}(y), U_{\succsim_a}(y)\big).$$

We show that there exist pairs of motives that must be aggregated into SNB preferences by *any* continuous and increasing aggregation rule.

Proposition 5. *There exist* $\succsim_\ell \in \mathcal{L}$ *and* $\succsim_a \in \mathcal{A}$ *such that* $\forall F \in \mathcal{F}$, $\succsim_F$ *displays SNB.*

We give a simple intuition. If $\succsim_a$ is strictly mixture averse in a part of the chord where $\succsim_\ell$ is mixture neutral and vice versa, then monotonicity of F suffices to produce SNB. Naturally, a version of this argument, where the strengths of the mixture attitudes of the two preferences are mismatched in different areas of a chord while the aggregation rule does not disproportionately favor one motive, will also imply SNB.

3 Literature Audit

A natural question is whether the experimental literature has already documented MMA, WNB or SNB. We conducted a literature audit across domains with a natural mixture operation. We identified papers through Web of Science searches, supplemented by backward and forward citation searches and searches of IDEAS/RePEc, SSRN, and Google Scholar.[9] We screened for experiments that might permit comparisons between alternatives and their mixtures, and reviewed papers only when the design could be inspected in an article, an earlier manuscript or an experimental appendix. Related elicitation designs and methodological papers were retained even when they could not identify the attitudes we study.

The reviewed studies span risk and randomization, consumption, time preferences, portfolios, charitable giving, effort and interpersonal allocation. They address varied questions, and few set out to identify mixed mixture attitudes. Despite this variety, their mixture operations fall into two broad groups. Probabilistic mixtures average probability distributions,

[9]The database searches were conducted on 26–27 August 2026, with supplementary searches and paper reviews through 15 October 2026. The search strings, screening criteria and paper-level classifications are reported in the online appendix.

including state-by-state for Anscombe–Aumann acts. Outcome mixtures average quantities, such as consumption at each date, payoffs in each state or allocations across people.[10] For example, consider giving \$100 to oneself or to another subject. A probabilistic mixture gives the full \$100 to oneself with probability p and to the other subject with probability $1 - p$. An outcome mixture instead gives $100p$ dollars to oneself and $100(1 - p)$ dollars to the other subject.

Table 2 summarises which mixture attitudes these designs can identify. A design is classified as capable if some possible pattern of responses identifies the attitude under the maintained assumptions, whether or not the paper reports such a pattern.[11] The convex/concave column records local weak comparisons, rather than establishing global convexity or concavity. These comparisons are common across domains, while designs capable of identifying MMA, WNB or SNB are much rarer.

Across these domains, preferences are typically elicited through binary or finite-menu choices, rankings, choices of randomization weights or allocations from convex budgets, and valuations or monetary trade-offs. The latter include certainty equivalents, willingness to pay or accept, and switching points in price lists. The limitations of these methods for our purposes follow from three identification requirements.

First, non-concavity requires an interior mixture to be strictly better than both endpoints, while non-convexity requires it to be strictly worse than both. Choice alone generally reveals only weak preference. For example, choosing an interior point from a mixture segment shows that it is weakly preferred to either endpoint, but does not establish non-concavity. Choosing an endpoint shows that it is weakly preferred to the available mixtures, but does not establish non-convexity: the mixtures need not be worse than the unchosen endpoint, and neither comparison need be strict.

Second, MMA requires both violations for the same subject under the same preference. Non-concavity in one group and non-convexity in another do not establish MMA. Individual identification therefore requires within-subject comparisons. Even then, comparisons across treatments are informative only if they concern the same maintained preference. Few designs can recover both strict violations jointly.

Third, WNB and SNB require opposite strict mixture attitudes along the same chord. The design must recover a mixture strictly above both endpoints and another strictly below them, at different weights of the same pair of objects. Direct identification of SNB additionally requires indifferent endpoints, although the SNB* equivalence permits indirect identification. Many designs examine only one mixture weight for a given pair. A single choice from a convex budget also leaves comparisons among unchosen allocations unresolved. Further comparisons along the same chord are therefore needed to recover these reversals.

[10] For effort allocations, where less work is preferred, we reverse the direction of monotonicity.

[11] We allow indirect identification through mixture continuity, transitivity and dominance where applicable, including identification of SNB through its equivalence with SNB*. The classifications were produced with ChatGPT 6 Astra and await verification by a human research assistant.

Design capabilities							Illustrative examples	
Domain	N	**Convex/Concave**	**Non-Concave**	**Non-Convex**	**MMA**	**WNB / SNB**	**Object**	**Operation type**
Randomization and lotteries	54	**48** (89%)	**21** (39%)	**10** (19%)	**7** (13%)	**3** (6%)	Lotteries / AA acts $p \in \Delta(O),\ f \in \Delta(O)^S$	Probability mixing Statewise probability mixing
Consumption budgets	23	**15** (65%)	**12** (52%)	**4** (17%)	**4** (17%)	**4** (17%)	Commodity quantities $x, y \in \mathbb{R}^K_+$	Goods-quantity mixing
Time preferences	85	**74** (87%)	**8** (9%)	**3** (4%)	**3** (4%)	**0** (0%)	Dated payments / work $c, e \in \mathbb{R}^T$	Dated-amount mixing Dated-workload mixing
Portfolio and asset choice	98	**88** (90%)	**15** (15%)	**1** (1%)	**1** (1%)	**0** (0%)	State-contingent payoffs $f \in \mathbb{R}^S$	Statewise payoff mixing
Charitable giving	111	**99** (89%)	**8** (7%)	**2** (2%)	**2** (2%)	**2** (2%)	Donor–charity allocations $z \in \mathbb{R}^{K+1}_+$	Donation-allocation mixing
Effort allocation	57	**27** (47%)	**1** (2%)	**1** (2%)	**1** (2%)	**0** (0%)	Task counts / work time $e, f \in \mathbb{R}^K_+$	Task-count mixing Time-allocation mixing
Social/interpersonal allocation	189	**172** (91%)	**25** (13%)	**9** (5%)	**5** (3%)	**1** (1%)	Interpersonal payoffs $x \in \mathbb{R}^I$	Payoff-allocation mixing

Table 2: Literature audit summary, with illustrative objects and mixture-operation types by domain.

Notes. All counts refer to reviewed-paper design capabilities rather than observed participant frequencies. Convex and concave denote local weak comparisons. Within each merged pair, the paper lists coincide. Here $\Delta(O)$ denotes lotteries over consequences, S states, T dates, and AA denotes Anscombe–Aumann; $\alpha \in [0, 1]$. In particular, AA acts $f, g : S \to \Delta(O)$ are mixed according to $(f_\alpha g)(s) = \alpha f(s) + (1 - \alpha)g(s)$.

Other examples. Statewise monetary acts $u, v \in \mathbb{R}^S$ use $(u_\alpha v)_s = \alpha u_s + (1 - \alpha)v_s$; contingent dated receipts $c, d \in \mathbb{R}^{S\times T}$ use $(c_\alpha d)_{st} = \alpha c_{st} + (1 - \alpha)d_{st}$. Portfolio holdings $h, k \in \mathbb{R}^J$ use $h_\alpha k = \alpha h + (1 - \alpha)k$; funding-source allocations $b, b' \in \mathbb{R}^F_+$ are combined as $b_\alpha b' = \alpha b + (1 - \alpha)b'$; lotteries over interpersonal allocations $P, Q \in \Delta(\mathbb{R}^I)$ use $P_\alpha Q = \alpha P + (1 - \alpha)Q$.

Our classifications concern identification from the *directly* elicited comparisons without imposing a particular preference representation. We therefore do not count indirect identification based on an estimated model. For example, weighting functions displaying both curvatures have been estimated from valuations of certainty equivalents. However, this does not imply subjects will indeed display MMA, WNB or SNB attitudes, as the model may be misspecified.

Among the papers we audit, ten are classified as capable of identifying SNB under the conditions detailed in the appendix. Three study risk and randomization: Hey and Strazzera (1989), Camerer and Ho (1994) and Cubitt, Navarro-Martinez and Starmer (2015).[12] We highlight Hey and Strazzera (1989), who ask subjects to construct indifferent lotteries and plot the resulting indifference curves in the Marschak–Machina triangle. Our reading of these plots suggests that six of their nine subjects display SNB. The other two papers elicit comparisons at several mixture weights and also permit indirect identification, but their reported summaries do not establish the prevalence of SNB.[13]

Finally, Bernasconi (1994) studies a violation of Betweenness that he calls the *squiggle* hypothesis. This pattern of indifference curves can generate SNB, although it is more specific than our definition. His design does not establish the required strict preferences and is therefore not classified as SNB-capable. If we additionally interpret his choices as strict and stable, mixture continuity allows us to infer from his Table 1 a lower bound of 20 out of 100 subjects displaying SNB.

4 Experiment

Motivation. Our theory is mostly a descriptive analysis of SNB behavior. It is silent on how and why it might occur beyond the fact that it can occur due to aggregation of motives. But what might these motives be, and are there natural patterns of SNB behavior?

We think there is a clear candidate for the shape of SNB behavior. In particular, the benefit of mixing is often highest when objects are evenly mixed, while the cost of mixing can be strongest when departing from the endpoints. In general, we think that "half-holding" a mixture can be costly. Examples include fairness, which is maximized at an even split, versus efficiency, which may be stronger near skewed allocations. Splitting your time evenly between two projects might help with novelty, but an extreme split leaves little work done on one project while pulling focus from the other. Thus, these intuitions suggest mixture lovingness near balanced mixtures while mixture aversion near extreme mixtures.

[12]The other qualifying studies are MacCrimmon and Toda (1969); Rousseas and Hart (1951); Brocas et al. (2019); Pastor-Bernier et al. (2020) in consumption, Harbaugh, Mayr and Burghart (2007); Zhang et al. (2021) in charitable giving, and Fisman, Kariv and Markovits (2007) in interpersonal allocation.

[13]Both papers test Betweenness. Cubitt, Navarro-Martinez and Starmer (2015) elicit certainty equivalents at several mixture weights, from which we infer that at least 14 of 39 subjects display WNB. Their focus is preference imprecision, however, so these reversals may reflect noisy evaluations rather than stable preferences.

We operationalize this intuition by running an experiment in the domain of risky choice. In particular, we consider lotteries over holiday trips, where there is a delay in learning the destination. Surprise holiday trips, as well as surprise meals, movies, and other experiences, are becoming increasingly popular and common; thus, it is important empirically to understand them. However, we are also motivated by two additional conceptual reasons. First, choice under risk is a well-studied setting, and we wish to better understand the role of MMA preferences here. This domain has two natural forces that have been documented mostly separately: a preference for randomization and a preference for certainty. In particular, we expect these forces to be salient in the domain of surprise trips. This leads to our second conceptual rationale: trips are rich, multi-dimensional objects which may lead to rich tradeoffs concerning the experience of uncertainty. For example, rational subjects may display a preference for randomization because they know they will enjoy the experience of anticipation of surprise. This channel is naturally strongest when the randomization is even. On the other hand, not knowing the location may prevent research or planning, and even a small amount of mixing reduces this benefit of certainty. Naturally, these channels may not exist or would be much weaker were we to consider lotteries over monetary outcomes.

In the following, we describe our experimental design, introduce a model of trips that links this tradeoff between surprise and preparation to mixture attitudes, and showcase five results regarding the how and why of mixture attitudes.

4.1 Experimental design

4.1.1 Sample and incentives

The experiment, advertised as an "online study on preferences for travel," was conducted with 83 participants in March 2020.[14] Participants were recruited through the Behavioural Research Lab of the London School of Economics (87%) and online ads (**Table B.1** reports sample characteristics). Eligibility required being at least 18, living in London, and holding valid documents for travel to the European Union. Respondents had to answer comprehension questions correctly throughout the survey, which took 44 minutes on average.

All participants who completed the survey received a £25 voucher with the travel partner, whose identity was revealed at the end of the survey. They could also win a free, non-transferable trip of 4 days and 3 nights to a European destination, worth £420. With an announced cap of 100 responses and 5 trips to give away, the chance of winning was 1 in 20; non-winners received either a discount voucher for a specific trip or a monetary payment. The prize was determined by one randomly selected decision.[15] Ignoring vouchers, expected

[14] All responses were completed between March 5 and 19, 2020, amid growing concerns over the COVID-19 pandemic. The UK had not yet imposed restrictions, which made these concerns less salient: only one respondent raised them in the survey, and about 75% chose a travel date before the end of 2020. An exploratory analysis in Toussaert (2024) finds that the completion date has a limited effect on decisions.

[15] The random incentive system is standard in experimental economics and incentive compatible under ex-

earnings were £34.80.

4.1.2 Structure of the study

Respondents first indicated a date at which they would consider traveling. They then completed the four parts summarized in **Figure 3** and described in more detail below.[16]

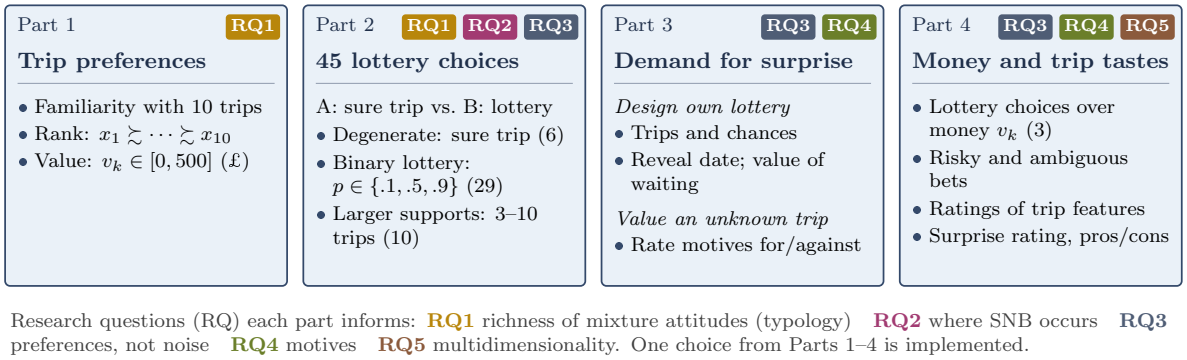


Figure 3: Structure of the survey. The four parts were completed in this order. Tags in each part give the research questions (RQ) its data are used for.

Part 1: Preferences over trips. Respondents considered trips to 10 European destinations: Düsseldorf, Gothenburg, Maastricht, Midi-Pyrénées, Porto, Prague, Santiago de Compostela, Sofia, Turku and Zakopane. The packages were designed so that they covered different countries and experiences, were easy to travel to and from, and had the same market value (£420).[17] Respondents saw a short description, picture, and accommodation for each trip (**Figure 4a**), and rated their familiarity with the destination.[18] They then ranked the trips, listed in random order, from x_1 (top) to x_{10} (bottom). A higher rank gave a trip a higher chance of being offered. Next to each destination in the ranked list, respondents also entered the minimum amount $v_k \in [0, 500]$ (£) they would accept to give up each trip, using the Becker, DeGroot and Marschak (1964) procedure (BDM).[19] Monotonicity of valuations in the ranking was not enforced; however, respondents were told "*logically you should have entered a higher price for*

pected utility (Azrieli, Chambers and Healy, 2018). A respondent who values uncertainty could, in principle, randomize across decision problems and treat the experiment as one big lottery. When asked after a sequence of decision problems how they had made their choices, all respondents reported treating each problem in isolation.

[16] For a complete description, see Toussaert (2024) Online Appendix.

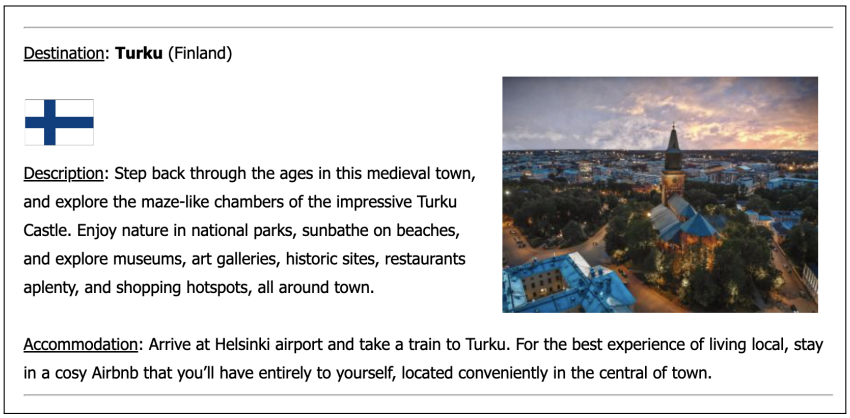
[17] Participants were told that "*the travel partner operates within a fixed budget such that if a destination is cheap to travel to, more money will be spent on the accommodation and vice versa.*"

[18] Because of an error in the survey, the first two sentences of the Prague description were incorrect. The analyses based on trip attributes (**Section 4.7**) use the information respondents actually saw.

[19] The ranking determined the probabilities of the trips which were $(.25, .20, .15, .12, .10, .08, .05, .03, .02, 0)$ from the top-ranked trip down, available through a link. If a respondent won trip x_k, their valuation v_k was compared with a random price $V \sim \mathcal{U}_{[0,500]}$. They received £$V$ if $V \geq v_k$ and kept the trip otherwise. The draw was revealed at the end of the survey, limiting the duration of anticipatory effects. Reporting the true ranking and valuations is incentive-compatible under FOSD and monotonicity in money, assuming the subject does isolate the two problems. Otherwise, if she does not isolate, EU would imply incentive compatibility but not otherwise. See Appendix B.1 in Toussaert (2024) for more details.

options you ranked higher" and asked to check their answers. The ranking and valuations jointly identify preferences over trips, with equal valuations allowing for indifferences that the ranking cannot reveal.

(a) Trip description



(b) Decision problem



Figure 4: Survey screens. (a) Description of one of the 10 trips, as shown in Part 1. (b) One of the 45 decision problems of Part 2: a sure trip (A) against a 50/50 surprise lottery (B); each of the 10 tickets is a 10% chance. The destinations in each problem depend on the respondent's ranking. Choosing B meant learning the destination no earlier than one week before travel.

Part 2: Binary choices with lotteries. The core of the experiment consists of 45 decision problems (DPs), tailored to each respondent's ranking. Most problems offered a choice between Option A, a trip to a given destination for sure (x_1, x_2, x_3, x_5 or x_{10}), and Option B, a "surprise lottery" over two or more destinations (**Figure 4b**). With a surprise lottery, participants would discover their destination only in the week of departure, in a sealed envelope received about 7 days before departure, with the suggestion to open it at the airport.

The 45 problems are listed in **Table 3**. Six compare two sure trips to check the ranking; 29 use a binary lottery $(x_j, p;\ x_k, 1-p)$ with $p \in \{.1, .5, .9\}$; and the remaining 10 use lotteries over 3, 5 or 10 destinations, reserved for out-of-sample tests.

We sort the problems by the direction of dominance. In $\mathcal{D}^+$ (24 problems), the sure trip is weakly preferred to every lottery outcome and strictly preferred to at least one; in $\mathcal{B}^-$ (11 problems), every lottery outcome is weakly preferred to the sure trip, with at least one strict preference. Choosing the lottery in $\mathcal{D}^+$ provides a mixture-loving witness; choosing the sure trip in $\mathcal{B}^-$ provides a mixture-averse witness. Since $\mathcal{B}^-$ contains only binary problems, we

Table 3: List of 45 decision problems

DP #	Core problems (35 choices)	Use
	Degenerate lotteries (6 choices)	
1, 4, 10, 13	A: $(x_1, 1)$ vs. B: $(x_k, 1)$ for $k \in \{2, 3, 5, 10\}$	Ranking
7	A: $(x_2, 1)$ vs. B: $(x_3, 1)$	Ranking
16	A: $(x_5, 1)$ vs. B: $(x_{10}, 1)$	Ranking
	Binary $\mathcal{D}^+$ problems (15 choices)	
2, 18, 27	A: $(x_1, 1)$ vs. B: $(x_1, p; x_2, 1-p)$ for $p \in \{0.5, 0.9, 0.1\}$	ML
5, 20, 29	A: $(x_1, 1)$ vs. B: $(x_1, p; x_3, 1-p)$ for $p \in \{0.5, 0.9, 0.1\}$	ML
8, 22, 31	A: $(x_1, 1)$ vs. B: $(x_2, p; x_3, 1-p)$ for $p \in \{0.5, 0.9, 0.1\}$	ML*
11, 24, 33	A: $(x_1, 1)$ vs. B: $(x_1, p; x_5, 1-p)$ for $p \in \{0.5, 0.9, 0.1\}$	ML
14, 25, 34	A: $(x_1, 1)$ vs. B: $(x_1, p; x_{10}, 1-p)$ for $p \in \{0.5, 0.9, 0.1\}$	ML
	$\mathcal{B}^-$ problems (11 choices)	
3, 19, 28	A: $(x_2, 1)$ vs. B: $(x_1, p; x_2, 1-p)$ for $p \in \{0.5, 0.9, 0.1\}$	MA
6, 21, 30	A: $(x_3, 1)$ vs. B: $(x_1, p; x_3, 1-p)$ for $p \in \{0.5, 0.9, 0.1\}$	MA
9, 23, 32	A: $(x_3, 1)$ vs. B: $(x_1, p; x_2, 1-p)$ for $p \in \{0.5, 0.9, 0.1\}$	MA*
12	A: $(x_5, 1)$ vs. B: $(x_1, p; x_5, 1-p)$ for $p = 0.5$	MA
15	A: $(x_{10}, 1)$ vs. B: $(x_1, p; x_{10}, 1-p)$ for $p = 0.5$	MA
	Sure trip between lottery outcomes (3 choices)	
17, 26, 35	A: $(x_5, 1)$ vs. B: $(x_1, p; x_{10}, 1-p)$ for $p \in \{0.5, 0.9, 0.1\}$	Level
	Out-of-sample problems (10 choices)	
	$\mathcal{D}^+$ over 3, 5 or 10 trips (9 choices)	
36	A: $(x_1, 1)$ vs. B: $(x_1, 0.4; x_2, 0.3; x_3, 0.3)$	ML
37	A: $(x_1, 1)$ vs. B: $(x_1, 0.4; x_3, 0.3; x_5, 0.3)$	ML
38	A: $(x_1, 1)$ vs. B: $(x_2, 0.4; x_3, 0.3; x_4, 0.3)$	ML*
39	A: $(x_5, 1)$ vs. B: $(x_6, 0.4; x_7, 0.3; x_8, 0.3)$	ML*
40	A: $(x_1, 1)$ vs. B: $(x_1, 0.2; x_2, 0.2; x_3, 0.2; x_4, 0.2; x_5, 0.2)$	ML
41	A: $(x_1, 1)$ vs. B: $(x_1, 0.2; x_3, 0.2; x_5, 0.2; x_7, 0.2; x_{10}, 0.2)$	ML
42	A: $(x_1, 1)$ vs. B: $(x_6, 0.2; x_7, 0.2; x_8, 0.2; x_9, 0.2; x_{10}, 0.2)$	ML*
43	A: $(x_5, 1)$ vs. B: $(x_6, 0.2; x_7, 0.2; x_8, 0.2; x_9, 0.2; x_{10}, 0.2)$	ML*
44	A: $(x_1, 1)$ vs. B: $(x_1, 0.1; x_2, 0.1; \ldots; x_9, 0.1; x_{10}, 0.1)$	ML
	Sure trip between lottery outcomes (1 choice)	
45	A: $(x_5, 1)$ vs. B: $(x_1, 0.1; x_2, 0.1; \ldots; x_9, 0.1; x_{10}, 0.1)$	–

Notes: DP # refers to the decision problem number as shown on the respondent's screen; x_k is the destination ranked # k by the respondent. The groups $\mathcal{D}^+$ (24 problems), $\mathcal{B}^-$ (11) and $\mathcal{B}^+$ (11) are defined in the text. The $\mathcal{B}^+$ problems are DPs 2, 5, 8, 11, 14, 18, 20, 22, 27, 29 and 31. Use: *Ranking*: checks that binary choices agree with the ranking. *ML* (mixture-loving witness): choosing B means that a lottery is preferred to the best trip it contains or to a trip ranked above it. *MA* (mixture-averse witness): choosing A means that a trip ranked at or below the worst trip in the lottery is preferred to the lottery. A single choice reveals only a weak preference; with the asterisk (*), the trip in A is ranked strictly above (ML*) or below (MA*) every outcome of the lottery, so the ranking makes the revealed mixture attitude strict. *Level*: places a lottery between x_1 and x_{10} above or below x_5, adding a comparison point on that chord, which can reveal SNB. Only the core problems are used to classify respondents; the out-of-sample problems test the classification.

compare like with like using $\mathcal{B}^+ \subset \mathcal{D}^+$, the 11 binary problems that mirror $\mathcal{B}^-$: same p, but with dominance reversed.[20]

[20]For example, let $x_1 \succ x_2$ and $L = (x_1, .5; x_2, .5)$. Choosing L over x_1 (DP 2, in $\mathcal{B}^+$) reveals mixture loving;

The DPs appeared on one page in a fixed order, from simple to more complex lotteries. Respondents were reminded of their ranking before starting, could consult the trip descriptions again, and were asked to review their 45 choices before moving on.

Parts 3 and 4: Demand for surprise, money and trip tastes. Binary choices reveal mixture loving and mixture aversion, but neither pin down subjects' preferred lotteries nor identify the underlying mechanisms. We therefore conduct three additional tasks.

Lottery construction and revelation. In Part 3, respondents chose between their favorite destination for sure and a surprise lottery of their own design, built by selecting 2 to 10 destinations and allocating 100 tickets among them. Builders then chose when to learn the outcome, from "today, after I complete the survey" to dates closer to departure. Respondents who chose a sure destination were offered the choice again, now with a revelation date of their choosing; those who switched built their lottery in the same way. Builders who delayed the resolution valued the delay in a multiple price list, choosing between revealing at the chosen date and revealing at the end of the survey + £X, with $X \in \{5, 10, 15, 20, 30, 40, 50\}$.

Wildcard valuation and motives. Respondents valued a "wildcard trip" to a new, unknown European destination, $v^* \in [0, 500]$, again with BDM. In addition, they rated their agreement with various motives for and against the wildcard. The wildcard trip could be in one of the 10 countries or another one, and had the same features as the 10 trips in terms of duration, market value, and date of revelation.

Money and trip tastes. Part 4 asked whether these mixture attitudes persist when prizes are monetary, measured risk and ambiguity attitudes over money, and collected ratings of trip attributes, travel history, and an overall assessment of the surprise-trip concept with reasons for liking or disliking it. The decision selected for payment was revealed at the end.

4.2 Surprise trips and mixture attitudes

We first describe trips and lotteries over trips formally, and then give a simple model that captures the channels we have in mind. The model separates the instrumental value of the destination from the psychological enjoyment and costs of not yet knowing where one will go.

4.2.1 Attributes, lotteries, and dominance

Vertical and horizontal dimensions. A trip x is a vector $a(x) = (a_1(x), \ldots, a_K(x))$ of attributes. Each coordinate describes its position on a bipolar dimension, such as mountains–beach or quiet–vibrant, with $a_\ell(x) \in \{-1, 0, 1\}$ denoting the left pole, neither pole, or the

choosing x_2 over L (DP 3, its mirror in $\mathcal{B}^-$) reveals mixture aversion.

right pole. **Figure B.6** gives the full list of measured dimensions and the distribution of respondents' ratings on each scale. These positions reflect the trip's characteristics and, where relevant, the subject's experience: the same destination can be new to one person and familiar to another. We suppress the subject index throughout.

How these attributes are evaluated also depends on the subject. Write $s_\ell \in \{-1, 0, 1\}$ for her leaning toward the left pole, neither pole, or the right pole. If $s_\ell \neq 0$, dimension ℓ is *vertical*: a larger $s_\ell a_\ell(x)$ is better. If $s_\ell = 0$, it is *horizontal*: neither pole is better, but they offer different experiences. We denote these sets by $\mathcal{V}$ and $\mathcal{H}$. Thus mountains–beach is vertical for someone who prefers the beach and horizontal for someone with no leaning.

Value of Sure Trips and Surprise Trips. We assume that the value of a *sure trip* is based only on the vertical attributes. A simple and illustrative example we employ is $v(x) = \sum_{\ell \in \mathcal{V}} w_\ell s_\ell a_\ell(x) + \epsilon_x$ where w_ℓ denotes the weight on the dimension and ϵ_x collects everything the dimensions do not capture. On the other hand, the value of a *surprise trip* can depend on horizontal dimensions. Surprise trips can create a sense of wonder by opening information gaps about pleasurable experiences—for example, whether one will go to the beach or the mountains (Golman and Loewenstein, 2018; Golman, Gurney and Loewenstein, 2021). The subject may enjoy imagining the possibilities and waiting to discover which occurs. But not knowing can also make preparation difficult and generate anxiety. Let P be a lottery over trips, we represent these forces by

$$V(P) = \sum_x P(x)v(x) + S\big(h(P)\big) - C\big(h(P)\big),$$

where $v(x)$ is the value of the sure trip, $h(P)$ is a continuous, nonnegative measure of the uncertainty left open, S its enjoyment, and C its cost. Both S and C are continuous and nondecreasing, with $S(0) = C(0) = 0$.

We take $h(P)$ to depend only on the horizontal attributes that differ across trips in the lottery. One specification would be

$$h(P) = \sum_{\ell \in \mathcal{H}} w_\ell \mathrm{Var}_P(a_\ell(x)),$$

where $\mathrm{Var}_P(a_\ell(x))$ is the variance of the attribute dimension ℓ given the lottery P. Hence, $h(P)$ is the weighted sum of variances over horizontal attributes. In general, we set $h(P) = 0$ when they all coincide for trips in the lottery. Thus, uncertainty on horizontal dimensions generates both the direct enjoyment and preparation costs. For the binary case, we assume symmetry $h(p) = h(1-p)$, and $h(0) = h(1) = 0$. A 10/90 and a 90/10 lottery then leave the same amount unresolved, but the latter places more weight on the better trip. Note that symmetry implies that WNB on a two-trip chord necessarily generates SNB on that chord. Since

SNB implies WNB in our hierarchy, the two coincide here.[21] Uncertainty about quality on vertical dimensions enters through the expected value of the destination, without a separate anticipation or preparation term.

We say a trip x attribute dominates trip y, written $x \trianglerighteq y$, if it is the same on every horizontal dimension and at least as good on every vertical dimension:

$$x \trianglerighteq y \quad \Longleftrightarrow \quad \begin{cases} a_\ell(x) = a_\ell(y), & \ell \in \mathcal{H}, \\ s_\ell a_\ell(x) \geq s_\ell a_\ell(y), & \ell \in \mathcal{V}. \end{cases}$$

If $x \trianglerighteq y$ then the sure trip x is better than y. Furthermore, if $x \trianglerighteq y$ then a lottery of the two must be valued above y but below x. Similarly, if shifting a weight from an attribute-dominated trip to the trip that dominates always improves the lottery, FOSD holds with respect to attribute dominance. Thus, strict mixture lovingness and aversion can arise *only if* trips differ on horizontal attributes. We code nine measured attributes from the trip descriptions and other information, and measure leanings from respondents' ratings. We find that the attribute dominance relationship only ranks 9% of pairs of trips, which leaves room for mixing attitudes.

4.2.2 The five preference patterns

Figure 5 illustrates the five patterns prevalent in our data and how natural assumptions on S, C, and h can rationalize them. For illustration, suppose trips have two attributes: hotel quality and destination type. The subject prefers better hotels but has no leaning between the beach and the mountains. Hotel quality is therefore vertical and destination type horizontal. Let x_1 be a beach trip with a better hotel, valued at 450, and x_2 a mountain trip with a worse hotel, valued at 400. The value of a lottery $L(p) = (x_1, p; x_2, 1-p)$ is then

$$V(L(p)) = p\, v(x_1) + (1-p)v(x_2) + \alpha h(p)^\gamma - \kappa h(p),$$

where uncertainty is measured by $h(p) = 4p(1-p)$, and we set $S(h) = \alpha h^\gamma$ and $C(h) = \kappa h$.[22] The subject can display Betweenness if C and S cancel, possibly both 0 (if she cares only about expected value), or if S and C are too small. This is captured by the gray linear curve. Additionally, if the subject cares only about preparation cost or only about surprise in a linear fashion, we expect mixture aversion and mixture lovingness, respectively. The orange and blue curves depict these effects, respectively. We note that a convex surprise factor by itself

[21] Let $x_j \succ x_k$ and $L(p) = (x_j, p; x_k, 1-p)$, symmetry gives $V(L(1-p)) = V(L(p)) + (1-2p)(v(x_j) - v(x_k))$. Suppose an upper violation occurs at p and a lower violation at q: $V(L(p)) > v(x_j)$ and $V(L(q)) < v(x_k)$. The reflected lower violation also lies below x_j, while the reflected upper violation lies above x_k. If q or $1-q$ exceeds p, there is a reversal against x_j at a larger probability of the better trip. Otherwise, $1-p < q$, and there is a reversal against x_k. Either case yields SNB. This argument uses symmetry alone and places no curvature restriction on S, C, or h.

[22] If there are additional horizontal dimensions, we take a weighted sum.

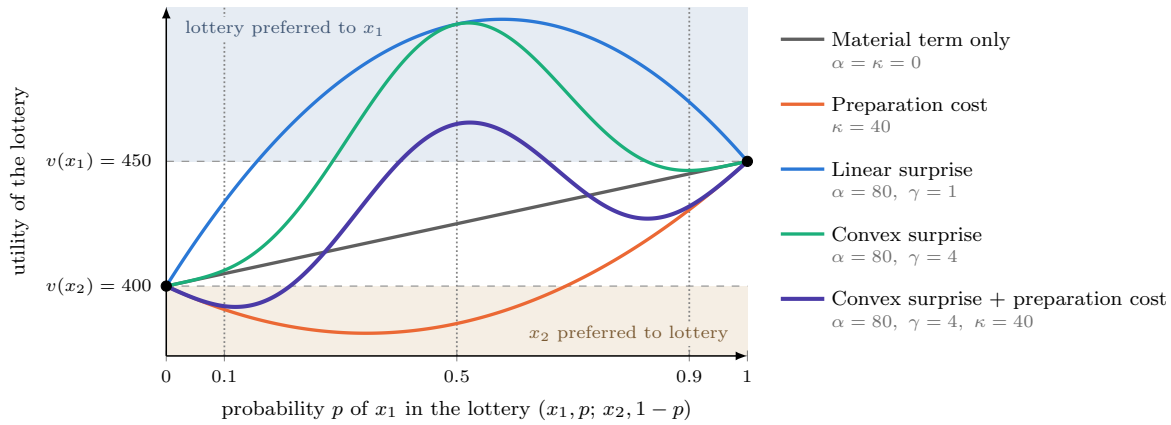


Figure 5: Utility of the lottery $(x_1, p; x_2, 1-p)$ under the five model cases. Shaded areas: lottery preferred to x_1 (top) and x_2 preferred to the lottery (bottom). Dotted lines: $p = .1$, .5 and .9, the probabilities used in the experiment.

is enough for SNB. As the green curve shows, we observe an SNB violation, but the curve always remains above the least preferred outcome. We call this SNB without a dip. The subject may alternatively display both a convex surprise and a preparation cost. Unlike the previous case, this also implies that the curve dips below the worst trip location. We call this SNB with a dip.

We now first use the main tasks (Part 2) to classify subjects based on their mixture attitudes. We then study the exact shape of the curves and show it is not driven by noise. Finally, we provide evidence that underlying motives for surprise and preparation drive the mixture attitudes, and the multidimensionality of attributes, vertical and horizontal, is related to choices in a way consistent with our model.

4.3 How rich are mixture attitudes? (RQ1)

Preferences over trips. We use the ranking and valuations together: if x_j is ranked above x_k, we record $x_j \succ x_k$ when $v_j > v_k$, indifference when $v_j = v_k$, and a conflict when $v_j < v_k$. Among adjacent ranked trips, 84% of comparisons are strict, 12% are indifferences and 4% are conflicts. **Table B.2** in the Appendix reports the consistency checks, including the direct choices between sure trips. We use a DP (see **Table 3**) to classify subjects only when the ranking, valuations, and, where available, direct choices establish the relevant strict preference between trips. Four respondents have no eligible chord and cannot be classified.

Identifying mixture attitudes. For an eligible pair of trips $x_j \succ x_k$, write $L(p) = (x_j, p; x_k, 1-p)$ for the lottery between them. Mixture loving is identified when $L(p) \succ x_j$, so the lottery is preferred to both trips; mixture aversion is identified when $x_k \succ L(p)$, so both trips are preferred to the lottery.

To identify SNB, we look for a reversal against a sure trip x_i lying between the two endpoints:

$$L(p) \succsim x_i \succsim L(p'), \qquad p < p', \quad x_j \succsim x_i \succsim x_k,$$

with at least one strict comparison. The typical case is $L(.5) \succ x_j \succ L(.9)$, as in the green and purple curves of **Figure 5**. This is SNB*, equivalent to SNB by **Proposition 3**. Reversals against trips outside the chord do not count.[23]

Among SNB respondents, those *with a dip* also prefer the worse sure trip to a lottery on at least one eligible chord. Those *without a dip* display no such choice on any eligible chord: their SNB is revealed by the reversal above, without an observed lottery below the worse trip.

Result 1. *Mixture loving and mixture aversion are common, and almost all respondents with mixed mixture attitudes display SNB.*

The choices depart from both global concavity and global convexity. On average, respondents choose the lottery in 22% of problems in $\mathcal{D}^+$ (25% in $\mathcal{B}^+$), revealing mixture loving, and choose the sure trip in 16% of problems in $\mathcal{B}^-$, revealing mixture aversion (**Table B.3** in the Appendix). There is substantial heterogeneity: 29% never choose the lottery in $\mathcal{D}^+$, while the top fifth do so in at least 44% of problems. In $\mathcal{B}^-$, 62% always choose the lottery, while the top fifth choose the sure trip at least 28% of the time.

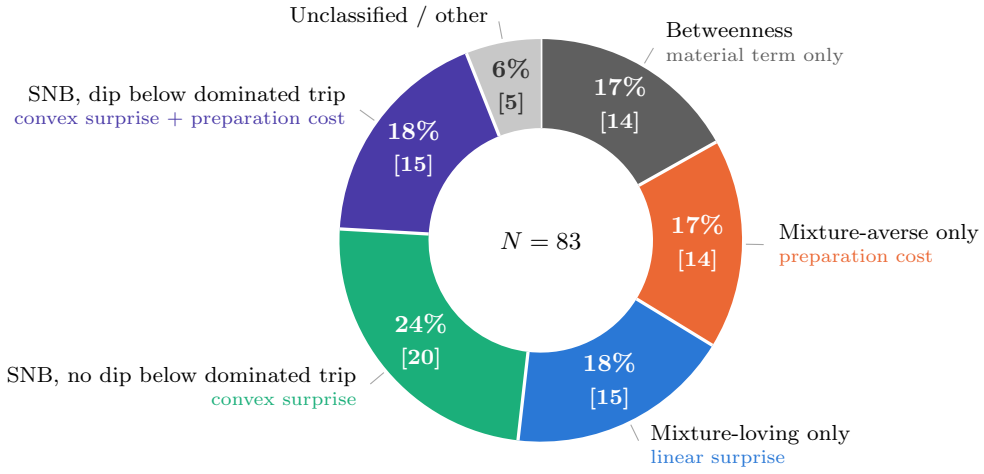


Figure 6: Respondents by mixture-attitude type ($N = 83$), based on the five binary-lottery chords. Inside the ring: share of respondents, with counts in brackets. Unclassified / other comprises 4 respondents with no eligible chord and 1 with MMA without SNB.

Figure 6 summarises the classifications across the five chords. Of the 79 classifiable respondents, 65 (82%) depart from betweenness: 15 display only mixture loving, 14 only mixture

[23] Binary choices reveal weak preferences. The classification treats choices at the fixed design points as strict, except where transitivity through strictly ranked trips already supplies the required strictness. Of the 35 SNB respondents, 9 satisfy the latter condition; for 8, overturning SNB would require two exact ties; for the remaining 18, a tie in the relevant endpoint comparison would suffice. The classification thus uses the absence of exact indifference at pre-specified comparisons.

aversion, and 36 mixed mixture attitudes. Of these 36, 35 display SNB; the remaining respondent displays mixture loving and aversion on different chords. The SNB group contains 20 respondents without a dip and 15 with one.[24]

All 10 respondents with both an upper and a lower witness on the same chord display SNB. This agrees with the equivalence of WNB and SNB on the symmetric chords of our model (**Section 4.2.2**), and with **Theorem 1**, which shows that MMA without SNB is meagre.

4.4 Which mixtures do people prefer? (RQ2)

We now turn to the shape of the mixing attitudes. The model in **Section 4.2** motivates two comparisons: how much uncertainty a lottery leaves open, and how much destination value is forgone by mixing.

Result 2. *Mixture-loving choices are most common at equal probabilities, whereas mixture-averse choices are most common when the better trip is unlikely.*

The most common SNB pattern arises when one prefers the 50/50 lottery to the better sure trip, but the sure trip to the 90/10 lottery (**Figure 7a**).[25]

The mixing attitudes also depend on the ranking of the two trips. At 50/50, the share choosing a lottery between the first- and second-ranked trips is 53%. Replacing the second trip with the third, fifth, or last lowers this share to 36%, 17%, and 6%. Mixture loving is thus most common between trips that are close in rank. This fits the trade-off in **Section 4.2**: enjoyment of uncertainty must compensate for the expected loss from sometimes receiving a less attractive destination.

Mixture aversion appears in a different region (**Figure 7b**). Respondents most often prefer the worse sure trip when the lottery offers only a 10% chance of the better trip, and less often when the chances are equal. A small chance of a better destination can therefore be less appealing than an even chance. This is consistent with the dip illustrated in **Section 4.2**: a little uncertainty already creates preparation costs, while providing little enjoyment of surprise.[26]

Paired tests and regressions support these patterns (**Table B.4** in the Appendix). The fall from 50/50 to 90/10 is significant between the two best trips (McNemar $p = .007$), but not on the other chords. Pooling across chords, the lottery is 6.7 percentage points less likely to be chosen over the sure trip at $p = .9$ than at $p = .5$. Controlling for value forgone and respondent fixed effects changes the estimate little, though it is significant only at 10%. On the aversion side, choosing the sure trip is 10.2 points more common at $p = .1$ than at $p = .5$, and this difference survives both controls.

[24] SNB without an observed dip still implies mixed mixture attitudes by **Proposition 3** and the hierarchy: the relevant comparisons involve indifferent lotteries that the design does not directly elicit.

[25] This reversal accounts for 39 of the 54 respondent–chord pairs with SNB (72%), and 25 of the 35 SNB respondents display it on at least one chord.

[26] The comparison $L(.5) \succ x_k \succ L(.1)$ does not itself establish SNB: the preferred mixture places more weight on the better trip.

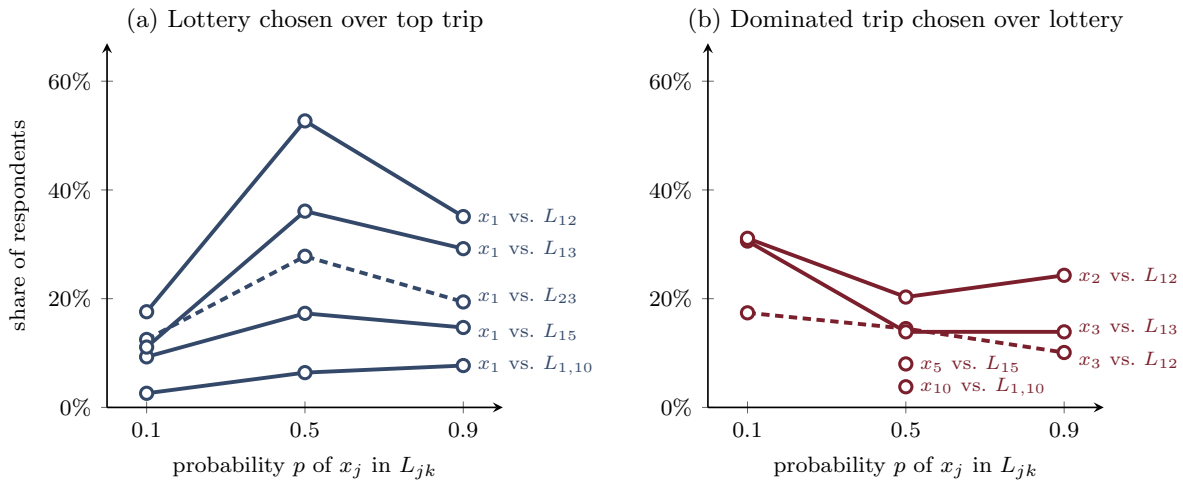


Figure 7: Choices along the chords. Share choosing the lottery over the sure trip (a) and the sure trip over the lottery (b), in problems meeting the trip-comparison criteria of **Section 4.3**. L_{jk} denotes the lottery $(x_j, p;\ x_k, 1-p)$, where x_j is ranked above x_k.

4.5 Is mixing behavior intentional or noisy? (RQ3)

Result 3. *The mixture-attitude types predict behavior in other tasks, while the calibrated noise benchmarks fail to reproduce the joint pattern of choices.*

We next examine whether the types predict behavior in three tasks that were not used to classify respondents: choices over larger lotteries, construction of a lottery, and ratings of the surprise-trip concept (**Figure 8**).

Out-of-sample choices. The share choosing the held-out lotteries rises from respondents who are only mixture averse to those who are only mixture loving, with the SNB types in between (panel a; Kruskal–Wallis, $p < .001$). The classification thus extends from binary lotteries to lotteries over three or more destinations.

Lottery construction. The same ordering appears when respondents decide whether to construct a lottery of their own (panel b; χ^2, $p < .001$). This task requires them to give up a sure destination and choose the trips and probabilities themselves. Those who construct lotteries also display more mixture loving and less mixture aversion in the binary choices (**Figure B.2f**). Their lotteries spread weight across several destinations while favoring better-ranked trips and usually leaving out the worst (**Figure B.2b–c**), consistent with valuing uncertainty while limiting the expected loss in destination value.

Concept ratings. Respondents who are only mixture loving give the surprise-trip concept the most favorable ratings, while those who are only mixture averse and those with no detected deviation from betweenness are less enthusiastic (panel c; Kruskal–Wallis, $p = .005$). The two SNB groups are intermediate and have more dispersed ratings. Their mean ratings are similar, consistent with enthusiasm for surprise that depends on the lottery offered rather than an unconditional preference for mixing.

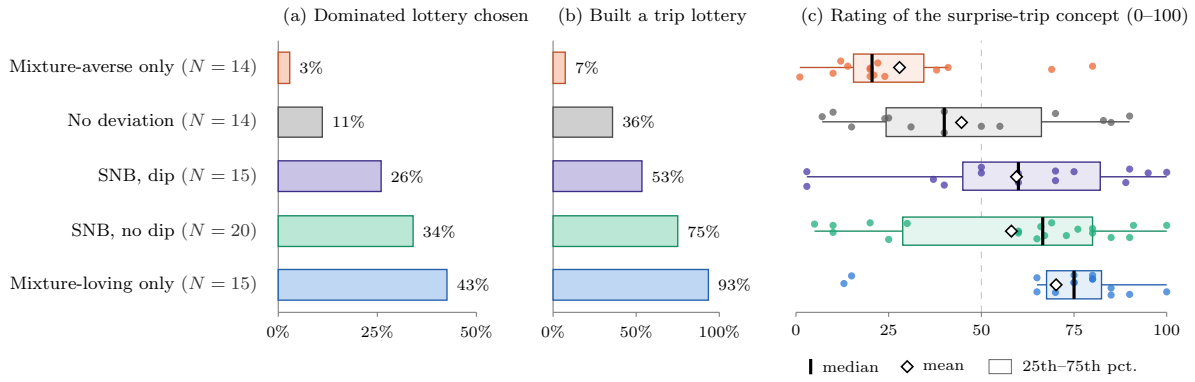


Figure 8: Validating the typology: choosing dominated multi-outcome lotteries not used in the classification (a), building a trip lottery in the design task (b), and rating the surprise-trip concept (c), by mixture-attitude type, in survey order. (a) DPs 36–44, lotteries over 3–10 trips that the sure trip dominates. (c) One dot per respondent. The 5 respondents not classified in **Figure 6** are omitted.

How much could noise explain? Inconsistencies between the ranking, the valuations and the direct choices are rare (**Section 4.3**), and the valuations separate trips by non-trivial amounts: on average, respondents value their top trip £38 more than their second and £218 more than their last (**Table B.5** in the Appendix). Violations are correspondingly costly: the value forgone in the choices that violate SD averages £41 (median £27), £46 on $\mathcal{D}^+$ and £29 on $\mathcal{B}^-$, about 10% of the market value of a trip (**Figure B.1** in the Appendix). Noise would have to be large to produce these choices.

We nonetheless confront the data with three models of random choice, each calibrated to the data and simulated 1,000 times on the 83 respondents (**Table 4**). Each simulated sample regenerates the 29 binary and 10 multi-outcome lottery choices from expected value at the respondent's valuations plus noise, and reapplies the classification. The *10% tremble* flips each choice with probability .1. Under *random utility*, each option receives an independent $N(0, \sigma^2)$ shock, with $\sigma = \$50$, close to its maximum-likelihood estimates from the sure-vs-sure choices (£55) and from the lottery choices (£50). The *person-specific tremble* flips each choice with probability equal to the respondent's own share of the 29 binary choices that go against expected value (mean .23).[27]

None of the benchmarks reproduces the data. First, all three reject the observed type distribution by wide margins: they predict too many SNB respondents with a dip (51–78% against 18%) and too few one-sided types. Second, they miss the comparative statics in p (**Figure 9a–b**). Under random utility, the lottery is chosen over x_j more often as p rises, since less expected value is at stake (18%, 25% and 41% at $p = .1$, .5 and .9), whereas in the data the share peaks at $p = .5$ (11%, 28% and 21%); the trembles predict no gradient at

[27] Two further benchmarks lead to the same conclusions: random utility with $\sigma = \$23$, which matches the overall violation rate (18.6%), and a tremble probability drawn for each respondent from a Beta(0.1, 0.9) distribution. They imply type distributions far from the data ($\chi^2 = 218$ and 374, against 90% ranges below 11) and positive correlations between the two kinds of violation (+0.29 and +0.82). The Beta tremble reproduces the out-of-sample gap (+31.5 points, range [+21.4, +42.9]), because it concentrates errors on a few respondents.

Table 4: Types and tests under noise benchmarks

	Data	10% tremble	Random utility (σ = £50)	Person-specific tremble
Share of respondents by type (%)				
SNB, with a dip	18	51 [43,59]	78 [72,84]	67 [60,72]
SNB, without a dip	24	19 [12,27]	3 [0,6]	10 [5,16]
Mixture-loving only	18	7 [2,12]	3 [0,6]	3 [0,7]
Mixture-averse only	17	5 [1,8]	1 [0,2]	2 [0,5]
No deviation	17	9 [4,13]	1 [0,2]	11 [7,14]
Other[a]	6	9 [6,13]	14 [8,18]	8 [5,11]
Tests				
Type distribution, χ^2		66.6 [1.0,9.9]	643.5 [1.0,9.4]	196.0 [0.6,8.4]
Correlation of violations, ρ	−0.33	+0.01 [−0.17,+0.21]	+0.23 [+0.05,+0.41]	+0.60 [+0.48,+0.71]
Out-of-sample gap[b] (pp)	+27.2	+0.1 [−6.7,+6.1]	+5.5 [−17.2,+20.5]	+20.0 [+14.4,+24.9]

Notes: Benchmarks: mean over 1,000 simulated samples of the 83 respondents; below, the 5th–95th percentile across samples, i.e. the range a sample of our size produces 90% of the time if the benchmark is true. Simulations regenerate the 29 binary and 10 multi-outcome lottery choices from expected value at the respondent's valuations plus noise, and reapply the classification. 10% tremble: each choice flipped with probability .1. Random utility: $N(0, \sigma^2)$ shocks on both options; σ =£50 is close to its maximum-likelihood estimates from the sure-vs-sure choices (£55) and the lottery choices (£50). Person-specific tremble: flip probability equal to the respondent's own share of the 29 binary choices that go against expected value (mean .23). [a] WNB without SNB, MMA without WNB, and unclassified. χ^2: Pearson statistic of the observed type counts against the benchmark's mean; below, its range across simulated samples. ρ: as in **Figure 9c**. [b] Share of dominated multi-outcome lotteries (DPs 36–44) chosen, mixture-loving types (SNB and mixture-loving only) minus all others.

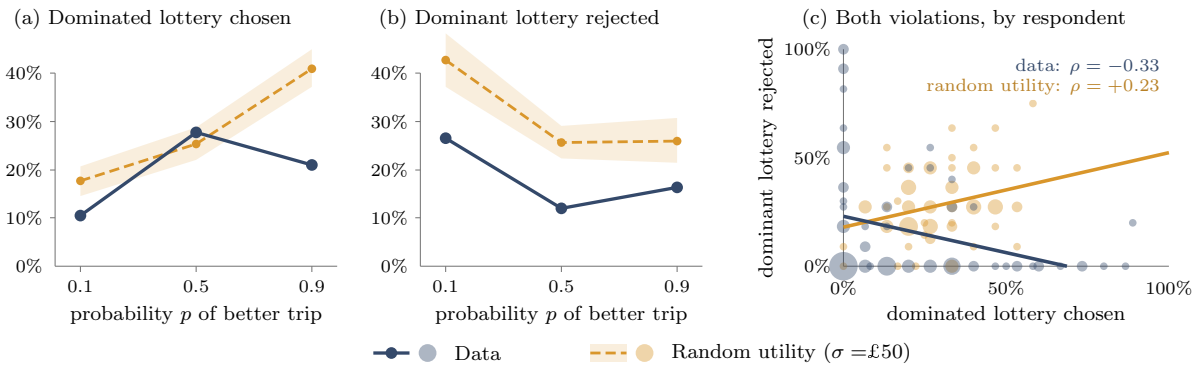


Figure 9: Violations of stochastic dominance against noise benchmarks. Clear-dominance choices. (a) Share of choices in which the lottery is chosen over the better trip in it. (b) Share in which the worse trip is chosen over the lottery. Random utility: expected value at the respondent's valuations plus an independent $N(0, 50^2)$ shock (£) on each option; dashed line: mean over 1,000 simulated samples; shaded: 5th–95th percentile across samples. (c) One dot per respondent, in the data (N = 79) and in the simulated sample under random utility whose ρ is closest to the mean; larger dots: several respondents; lines: least-squares fits. ρ: Spearman correlation.

all. Third, every benchmark implies a zero or positive correlation between the two kinds of violation across respondents, since noisier respondents err in both directions, whereas the data show a negative one (Spearman $\rho = -0.33$, outside the 90% range of every benchmark; **Figure 9c**). Fourth, only the person-specific tremble comes close to the out-of-sample gap between mixture-loving and other types (+20.0 points against +27.2 in the data), and even its 90% range excludes the data.

4.6 Where does the (dis)taste for randomization come from? (RQ4)

Result 4. *The motives for randomization are consistent with utility from uncertainty itself, through excitement and the ability to dream; the motives against it reflect the anxiety of feeling unprepared and the fear of a worse destination.*

We now ask what motives make uncertainty attractive or unattractive. We study this in two ways. First, by asking whether respondents want to prolong uncertainty and study their preference for delaying learning the lottery outcome. Second, we relate their reported motives to their choices and valuations.

Delaying the resolution. The lottery-building task also let respondents choose when to learn their destination. A priori, the optimal delay is not obvious: a longer delay extends the enjoyment of open questions but also the period of not being able to prepare. Of the 46 builders, 74% chose to delay the resolution, on average by 94 days, and 37% chose to learn it only in the weeks before departure (**Figure B.2d**). Of those who delayed, 74% were willing to pay to keep the surprise rather than learn the destination at the end of the survey, and 54% of all builders gave up at least £5 for it (**Figure B.2e**).

Motives and choices. Respondents' rankings of reasons for and against surprise trips give a similar picture: anticipation and daydreaming are prominent reasons for liking them, while disappointment and being unprepared are prominent reasons against them (**Figure B.3**). The wildcard is also valued more by respondents who choose more lotteries in $\mathcal{D}^+$ and by those who build a lottery, although its valuation does not differ significantly across the five types (**Figure B.4**). **Section B.4** reports these comparisons, the full lottery-construction results (**Figure B.2**), and the motive ratings and correlations (**Table B.6**)

We also ask whether the motives respondents give for liking or disliking the wildcard trip are associated with its valuation and with their mixture attitudes. **Figure 10** relates motives about surprise, anticipation, and anxiety to the wildcard valuation, mixture loving choices in $\mathcal{D}^+$, and mixture averse choices in $\mathcal{B}^-$. The pattern is consistent across the three outcomes: motives for uncertainty generally go with a higher wildcard valuation, more mixture loving and less mixture aversion, while motives against it point the other way.

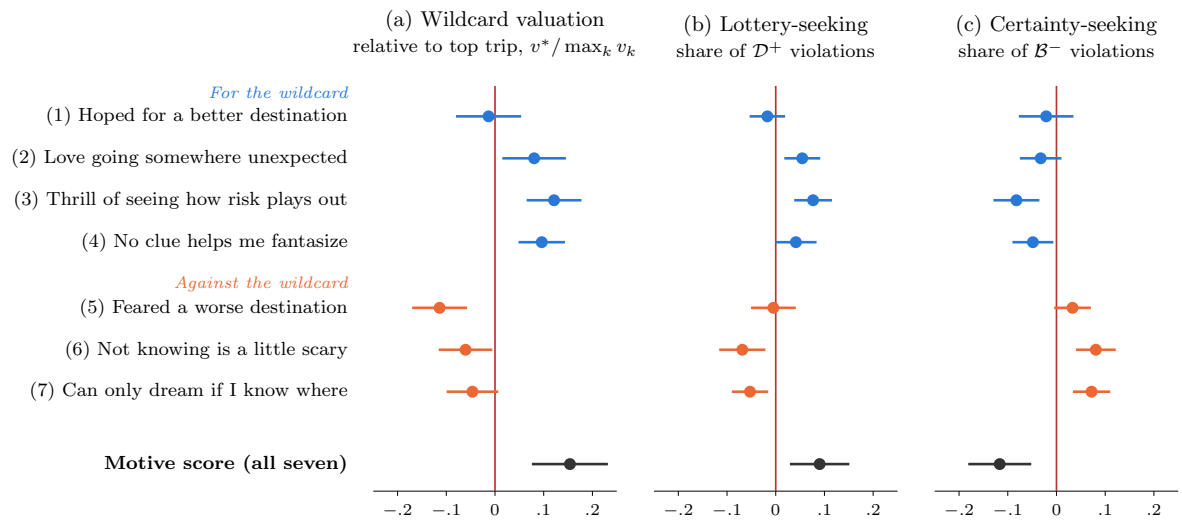


Figure 10: Same motives, three outcomes: wildcard valuation and SD violations. Coefficients from univariate OLS regressions of each outcome on each rating (1 = completely disagree, 5 = completely agree), or on the motive score (mean of the seven items, 5–7 reverse coded), with 95% confidence intervals (robust standard errors). (a) Wildcard valuation divided by the respondent's highest valuation of the ten trips ($N = 83$). (b) Share of $\mathcal{D}^+$ problems where the lottery is chosen over a sure trip that dominates it; (c) share of $\mathcal{B}^-$ problems where the sure trip is chosen over a lottery that dominates it ($N = 79$). Across the seven items, the coefficients in (a) correlate at 0.83 with those in (b) and at −0.86 with those in (c).

4.7 How does multidimensionality affect mixing? (RQ5)

Result 5. *Mixture loving almost disappears with money. For trips, horizontal differences are associated with more mixture loving, while vertical taste fit is associated with valuations.*

Money. Money is one dimensional and dominance is therefore complete. We repeated three decision problems with each trip x_k replaced by the respondent's valuation v_k, paid on the planned travel date and, for the lottery, revealed one week before. Mixture loving almost disappears (**Figure 11**): the share choosing the lottery over the best sure outcome falls from 53% to 4% in the binary comparison and from 38% to 1% in the five-outcome comparison. Mixture aversion falls less, from 20% to 11%, and this change is not statistically significant ($p = .21$). The contrast is consistent with the role of multidimensionality in our model: delayed resolution alone does not generate a taste for randomization when there is no multi-dimensionality in outcomes.

Heterogeneous tastes and trip multi-dimensionality. The trips, by contrast, differ on several dimensions, and respondents disagree about which features they value. Every destination is someone's favorite (**Figure B.5** in the Appendix). Preferences over the nine bipolar attributes are also heterogeneous: 90% of respondents have at least one horizontal dimension, on which they favor neither pole (**Figure B.6** in the Appendix). Using coded trip positions and respondents' leanings, neither trip dominates the other on the observed attributes in 91% of respondent-specific pairs.

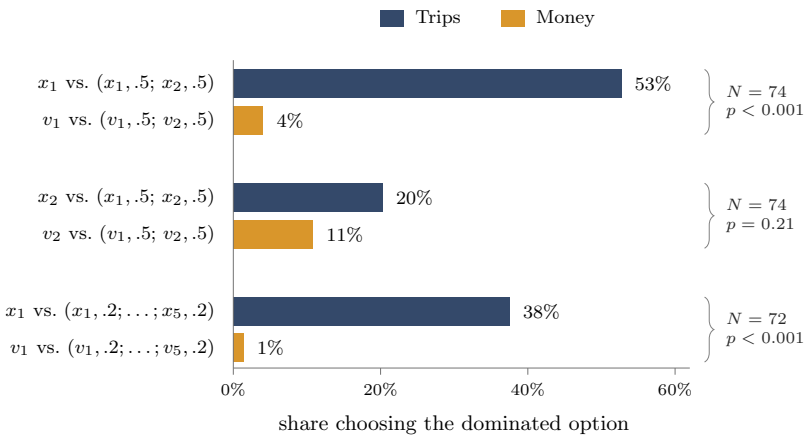


Figure 11: SD violations for trip vs. monetary lotteries, same respondents. Each brace groups a trip question and its money counterpart, answered by the same N respondents (clear-dominance trip problems only). Respondents choosing the dominated option for both trips and money / trips only / money only / neither: 2/37/1/34; 0/15/8/51; 1/26/0/45 (top to bottom). p: exact McNemar test, which compares the trips-only and money-only counts. Money prizes v_k equal the respondent's valuation of trip x_k.

Horizontal and vertical attributes. **Figure 12** relates attributes to mixing and valuations. We use the 12 problems in $\mathcal{D}^+$ in which the sure trip is one of the two lottery outcomes.[28] Mixture loving is more common when the trips differ on horizontal dimensions (panel a). We find no comparable pattern for horizontal dimensions on which the trips are equal or for vertical differences (panels b–c). Vertical attributes instead predict valuations: trips that better fit respondents' preferred poles receive higher valuations (panel d). As in our representation (**Section 4.2**), horizontal attributes correspond to psychological motives while vertical attributes correspond to material utility.

Regressions support this distinction (**Table B.7** in the Appendix). The positive association between horizontal differences and mixture loving remains statistically significant after controlling for the valuation gap and respondent fixed effects. Taste fit is significantly associated with trip valuations, and differences in taste fit with valuation gaps. Vertical differences have no statistically clear association with mixture-loving choices.

[28] In the 3 remaining binary problems in $\mathcal{D}^+$, the sure trip x_1 is compared with a lottery over x_2 and x_3, so the attributes of a third trip enter the comparison. Adding them, the effect of a horizontal difference between the two lottery trips falls to 2.5 points ($p = .085$); within these 3 problems alone it is -1.9 points ($p = .45$).

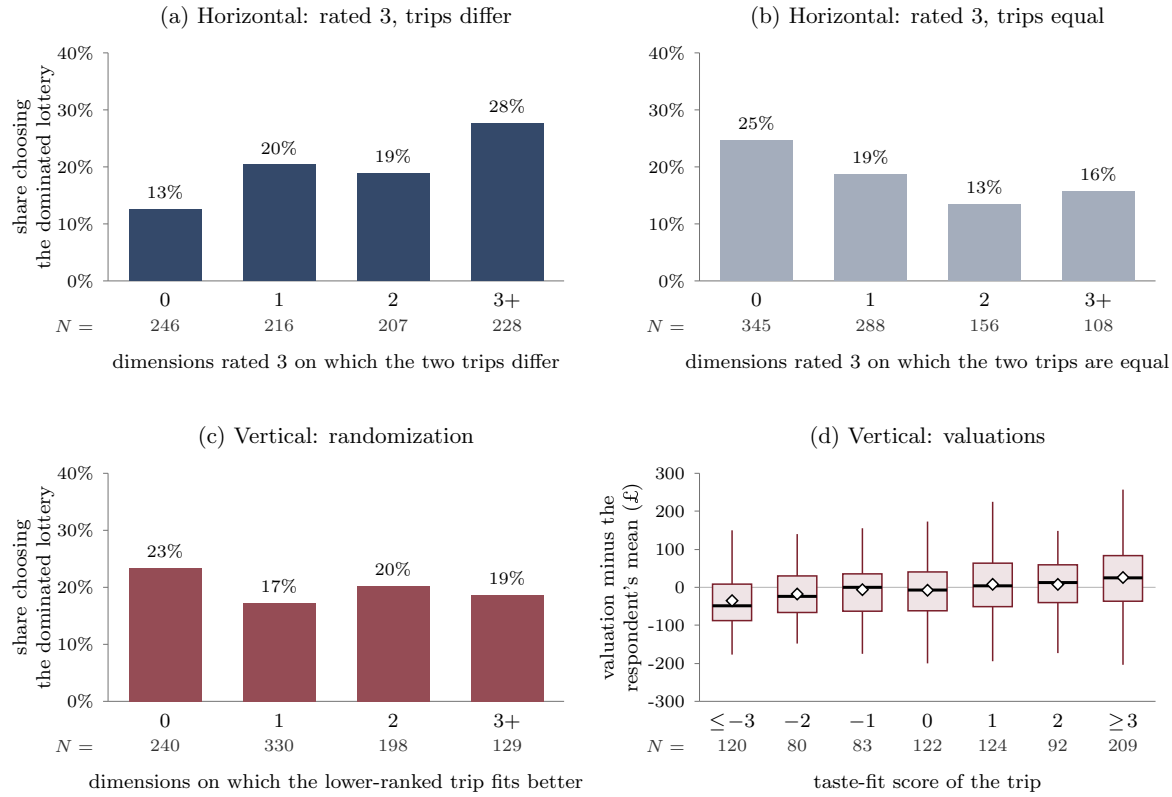


Figure 12: Attributes, randomization and valuations. Horizontal dimensions: rated 3 (no leaning); vertical: rated 1, 2, 4 or 5. (a)–(c): share of choices in which the lottery was chosen over the sure trip that dominates it, in the 12 $\mathcal{D}^+$ problems in which the sure trip is one of the two lottery trips (clear dominance only; 897 choices, 79 respondents); N = number of choices. (c) The lower-ranked trip fits better on a dimension if it is strictly closer than the sure trip to the preferred pole. (d) Valuation of each trip minus the respondent's mean valuation (830 observations), by taste-fit score: the number of vertical dimensions on which the trip sits at the preferred pole minus the number on which it sits at the other pole; N = number of valuations. Box: 25th–75th percentile; bar: median; diamond: mean; whiskers: most extreme value within 1.5 times the interquartile range. Counts and scores are top-coded at 3 (±3 in (d)).

5 Discussion

This paper studies preferences that are neither convex nor concave. We show that this class is both theoretically rich and empirically prevalent. Theoretically, we are intentionally descriptive and focus on the shape of behavior. We propose a hierarchy of mixed mixture attitudes, and show that its strongest form, SNB, in which a single indifference curve is itself neither convex nor concave, is not knife-edge. Our literature audit shows that existing experimental designs could rarely detect such attitudes, and our experiment then documents SNB in a realistic, high-stakes randomization setting. A simple rationale can explain this behavior: the tradeoff of two opposite motives which are strongest at different mixture weights. A preference for certainty, which is strongest near the endpoints, and a preference for randomization, which is strongest near the uniform randomization, can together generate SNB. Our work provides a common framework for studying mixed mixture attitudes and a targeted investigation of their prevalence and origins. We first discuss what our results imply for existing models and for experimental design. We then comment on the many directions we see for future work.

Models of Mixed Preferences. We see several promising avenues in studying models of mixed preferences further. First, our hierarchy separates models by how far they depart from one-sidedness, but it cannot tell apart two SNB preferences. For example, smooth ambiguity (Klibanoff, Marinacci and Mukerji, 2005) and α-maxmin can both display SNB, and neither model nests the other. Our hierarchy already separates them, as α-maxmin can only display mixed attitudes as SNB, while smooth ambiguity can display all three levels. Amongst SNB preferences, however, it is silent about how the two attitudes are arranged along a chord, for instance, where mixing helps and where it hurts. A finer classification would be necessary for an identification test of such models. Second, we believe that richer mixture attitudes arise out of richer motives. Some models, such as α-maxmin, intentionally build in a tradeoff between motives, such as optimism and pessimism. Others, such as smooth ambiguity, achieve mixed attitudes through the curvature of their functional form. There is no consensus about what a "proper" model of mixed attitudes should look like, as the question itself has not been posed. We think this leaves room for both axiomatic representations and experimental identification. Of course, the proper model depends on the domain, but we believe that comparing domains is valuable in and of itself.

On a related note, our illustrative model of surprise value and preparation cost assumes attributes are independently valued. This misses complementarity and substitute effects which are naturally another channel through which mixture preferences may occur. We think that given mixtures are probabilistic (instead of a trip that averages the attributes), this channel may not be consequential. However, in many environments, disentangling the "psychological" value of mixing from these utilitarian motives may be a challenge.

Domains besides Randomization. Our literature audit reveals many domains where mixed mixture attitudes have not been tested thoroughly yet. We think it is reasonable to expect evidence of such attitudes in most of them. We give a few examples and show how they fit under our motives tradeoff framework.

Example 1 (Social Preferences). A social planner chooses $x_a + x_b = m$ amounts to give to Ann and Bob. Their utilities are $u_A(x) = x^\rho$ and $u_B(x) = x^\rho$ with $\rho > 1$. The social planner's preference is:

$$V(x_a, x_b) = u_A(x_a) + u_B(x_b) - \lambda|u_A(x_a) - u_B(x_b)|$$

Then the social planner's preference satisfies all our basic axioms whenever $\lambda \in (0, 1)$ and displays SNB for all $m > 0$. She has a motive for efficiency due to the convexity of x^ρ which favors unequal allocation and hence is mixture averse. She has also a motive for fairness from $\lambda > 0$, which pushes in favor of mixture lovingness.

A similar behavior is noted by Zhang et al. (2026) in charity donations. They find that people may prefer giving nothing to giving a small amount, even though they are willing to give a larger amount. Their explanation is that giving too little can make someone feel selfish or appear ungenerous, whereas giving nothing may be less revealing.

Example 2 (Time Preferences). Consider a DM choosing her consumption stream. Her payoff for a consumption stream $c = (c_1, c_2, ..., c_T)$ is

$$U(c) = \sum_t^T \beta^t u(c_t) - \min\{(r - c_t)_+, h\}$$

where u is strictly increasing and concave, $\beta \in (0, 1)$, and $r > 0$ is a threshold that the DM must meet every period, i.e., rent, or she suffers a penalty of at most λh. Here it is easy to check that consumption smoothing via the concavity of u generates mixture lovingness, while meeting rent can generate mixture aversion. This interaction will lead to SNB.

Example 3 (Bayesian Persuasion). Consider the prosecutor and judge example of Kamenica and Gentzkow (2011). The defendant is guilty with probability 0.3. The judge convicts whenever her posterior probability of guilt is at least 0.5, and the prosecutor wishes to maximize the probability of conviction.

Consider two fully revealing information structures, x and y, which differ only by reversing the labels of the reports. The prosecutor is indifferent between them, with

$V(x) = V(y) = 0.3$.

Suppose the prosecutor mixes these information structures without revealing which was used. Under $z_\alpha = \alpha x + (1-\alpha)y$, the probability of a guilty report is α if the defendant is guilty and $1-\alpha$ if innocent. The prosecutor displays the following preference:

$$V(0.9x + 0.1y) = 0.34 > V(x) = V(y) = 0.3 > V(0.5x + 0.5y) = 0.$$

And since she is indifferent between x and y this satisfies our original SNB definition. A small amount of garbling increases the probability of conviction, while excessive garbling leaves the judge uninformed and leads to acquittal.[29]

Information structures are natural mixture spaces. Besides Bayesian persuasion, a preference for information may trade off against an aversion to bad news (Kreps and Porteus, 1978; Dillenberger, 2010; Masatlioglu, Orhun and Raymond, 2023).

Experimental Design. The large majority of works on mixture preferences allow subjects to choose the 50-50 midpoints. Choice on a convex budget, deliberate randomization, and many other designs have this feature. Hence, the literature does suggest a strong mixture-lovingness towards such midpoints. Our experimental results highlight that this is not a global mixture attitude in at least one context. We elicit via binary choice the mixture preference at multiple interior points. Hence our approach is, in some sense, a minimal augmentation of the standard approach. Therefore, given that mixed attitudes are a local feature, our measurement may not have been powerful enough and may have underestimated the number of subjects with such attitudes. A natural question is how best to elicit such attitudes; it may vary by context and environment. One avenue could be to elicit certainty equivalents within a range of mixture weights. Similarly, there are technical difficulties when accounting for subjects who either 1) do not isolate individual decision problems and 2) are not probabilistically sophisticated. These concerns arise in regular randomization and risk designs but may be aggravated here, as 1) multiple elicitations are needed and 2) if subjects have mixed attitudes, then we cannot even upper or lower bound the measurements.

Preference for Surprise and Certainty. In our experiment, we have strong reasons to believe that the preference for randomization is 1) strongest near an even split, and 2) driven partially by anticipatory utility and a taste for surprise (Loewenstein, 1987; Caplin and Leahy, 2001; Ely, Frankel and Kamenica, 2015). Existing work finds a preference for randomization mainly in environments that allow an even split (Agranov and Ortoleva, 2017; Dwenger,

[29] In the original example, the judge's action is binary, leading to a non-continuous value function and thus falling outside of our framework. One can however instead consider a judge whose decision is probabilistic and monotone in her posterior.

Kübler and Weizsäcker, 2018; Feldman and Rehbeck, 2022), while the certainty effect is well documented for mixtures close to the endpoints (Allais, 1953; Kahneman and Tversky, 1979; Dillenberger, 2010; Cerreia-Vioglio, Dillenberger and Ortoleva, 2015). In some cases, both attitudes are documented together (Chew and Waller, 1986; McGranaghan et al., 2026). We see two ways to continue this work. First, a more comprehensive study of mixture attitudes in the monetary risk domain could investigate the extent and shape of SNB behaviors. Second, contrasting the drivers of the preferences for surprise and for certainty across different, richer domains could lead to a better understanding of both. For example, do these attitudes get stronger or weaker for objects that carry more or less anticipatory utility?

Complexity and Rich Mixtures. Are richness and complexity of choice objects related? Work on complexity typically finds that complex choices produce noisy behavior, often compressed towards the midpoint (Enke and Graeber, 2023); well-known risk anomalies reappear in complex choices that involve no risk at all due to noise and randomization (Oprea, 2024; Banki et al., 2025); often deliberate randomization is also more prevalent (Agranov and Ortoleva, 2017). We hypothesize that when options are complex and unfamiliar[30] while stakes are low, subjects may not deliberate over the various motives at play. They may instead leave the tradeoffs unresolved, through indifference, incomplete preferences (Bewley, 2002; Dubra, Maccheroni and Ok, 2004), delegation, or randomization, and display one-sided attitudes. In our experiment, subjects are familiar with trips, and we consider simple lotteries between two trips, with the richness coming from psychological channels rather than from an artificial complexity of outcomes. Subjects appear to engage deliberately with these tradeoffs, and exhibit rich mixture attitudes.

[30]Lotteries over monetary outcomes may be simple, but become increasingly unfamiliar as the number of outcomes increases.

Appendix

A Theory Appendix

A.1 Proposition 1

*Proof of **Proposition 1**.* The inclusions follow from the definitions. To establish strictness, consider preferences on $[0,1]^2$.

1. MMA without WNB: Quadratic utility. Consider

$$u_M(x_1, x_2) = x_1^2 - (1 - x_2)^2.$$

Quadratic utility cannot display WNB, since the deviation from the linear interpolation of endpoint utilities has the same sign along any chord. It can, however, display MMA. Indeed,

$$u_M\left(\frac{3}{4}, \frac{3}{4}\right) = u_M\left(\frac{11}{12}, \frac{5}{12}\right) = \frac{1}{2} < \frac{25}{48} = u_M\left(\frac{5}{6}, \frac{7}{12}\right),$$
$$u_M\left(\frac{1}{4}, \frac{1}{4}\right) = u_M\left(\frac{7}{12}, \frac{1}{12}\right) = -\frac{1}{2} > -\frac{25}{48} = u_M\left(\frac{5}{12}, \frac{1}{6}\right).$$

The last point in each line is the midpoint of the first two.

2. WNB without SNB: Mixture symmetry. Consider

$$u_W(x_1, x_2) = x_1 + x_2 - 1 + |x_1 + x_2 - 1|(x_1 - x_2).$$

This utility is quadratic on each side of $x_1 + x_2 = 1$, and its sign agrees with that of $x_1 + x_2 - 1$. Consequently, indifferent endpoints and their mixtures lie within the same quadratic region. Since quadratic utility is symmetric around the midpoint of any indifferent chord, u_W satisfies mixture symmetry, which rules out SNB.

Nevertheless, WNB is possible. For $x = (1/4, 1)$ and $y = (3/4, 0)$,

$$u_W(x\tfrac{3}{4}y) = \frac{5}{64} > u_W(x) = \frac{1}{16} > u_W(y) = -\frac{1}{16} > u_W(x\tfrac{1}{4}y) = -\frac{5}{64}.$$

Thus u_W displays WNB but not SNB.

3. SNB. Finally, consider

$$u_S(x_1, x_2) = x_1 + x_2 + \frac{1}{4}(x_1 - x_2)\big(1 - (x_1 - x_2)^2\big).$$

For $x = (1, 0)$ and $y = (0, 1)$,

$$u_S(x\tfrac{3}{4}y) = 1 + \frac{3}{32} > u_S(x) = u_S(y) = 1 > u_S(x\tfrac{1}{4}y) = 1 - \frac{3}{32}.$$

Hence u_S displays SNB.

All three utilities are continuous and strictly increasing in each coordinate, and therefore satisfy the Basic Axioms. We conclude that

$$\mathcal{P}^S \subsetneq \mathcal{P}^W \subsetneq \mathcal{P}^M.$$

□

A.2 MMA without WNB

Proposition 6. *Suppose $\succsim$ satisfies the Basic Axioms on a compact convex set $\mathcal{X} \subseteq [0,1]^D$ containing the diagonal. For $R \subseteq \mathcal{X}$, let $I_R(x) = \{z \in R : z \sim x\}$. The following are equivalent:*

1. *$\succsim$ does not display WNB.*
2. *For every compact convex $R \subseteq \mathcal{X}$ satisfying $\mathbf{1} \in \operatorname{span}(R - R)$, every $x, y \in \operatorname{relint} R$, and every $p \in \mathbb{R}^D_{++}$, if the hyperplanes*

$$p \cdot z = p \cdot x \qquad \text{and} \qquad p \cdot z = p \cdot y$$

support $I_R(x)$ and $I_R(y)$, respectively, then

$$x \succ y \quad \Longrightarrow \quad p \cdot x \geq p \cdot y.$$

Proof. We show that WNB occurs exactly when the condition fails.

First suppose the condition fails at $x, y \in \operatorname{relint} R$:

$$x \succ y \qquad \text{but} \qquad p \cdot x < p \cdot y.$$

The points of R strictly above x's price hyperplane form a convex set, as do those strictly below. Both are nonempty: since $\mathbf{1} \in \operatorname{span}(R - R)$ and $p \in \mathbb{R}^D_{++}$, the points $x \pm \delta\mathbf{1}$ lie in R on either side for small $\delta > 0$. By continuity, a convex set containing no point indifferent to x is uniformly better or uniformly worse than x, and by monotonicity the side above can only be uniformly better, the side below only uniformly worse. Since the hyperplane supports $I_R(x)$, one side contains no indifferent point, and it is not the side above, which contains $y \prec x$. Thus every point of R strictly below x's price hyperplane is worse than x. Similarly, every point of R strictly above y's price hyperplane is better than y, since the side below contains $x \succ y$.

Now extend the segment slightly beyond both ends:

$$x' = x + \eta(x - y), \qquad y' = y + \eta(y - x).$$

For small $\eta > 0$, both points remain in R, with $p \cdot x' < p \cdot x$ and $p \cdot y' > p \cdot y$. Hence $x' \prec x$

and $y' \succ y$, while continuity gives $x' \succ y$ and $y' \prec x$. Thus x is better than both endpoints of $[x', y']$ and y is worse than both. This is WNB.

Conversely, suppose WNB holds on a chord $C = [v, w]$, which by continuity we may take in relint $\mathcal{X}$. Monotonicity implies that $w - v$ has both positive and negative coordinates, so choose $p \in \mathbb{R}^D_{++}$ with $p \cdot (w - v) = 0$. For small $\varepsilon > 0$, take

$$R = \{c + s\mathbf{1} : c \in C,\ |s| \leq 2\varepsilon\} \subseteq \mathcal{X},$$

and let x be a best point of $C - \varepsilon\mathbf{1}$ and y a worst point of $C + \varepsilon\mathbf{1}$. The strict comparisons witnessing WNB persist for small ε, so $x \succ y$ and neither point is an endpoint of its chord. Hence $x, y \in$ relint R, while

$$p \cdot y - p \cdot x = 2\varepsilon\, p \cdot \mathbf{1} > 0.$$

The chords $C \mp \varepsilon\mathbf{1}$ lie in the price hyperplanes through x and y. Every point of R strictly below the first is worse than the point directly above it on $C - \varepsilon\mathbf{1}$, hence worse than x; likewise, every point strictly above the second is better than y. Both hyperplanes therefore support the respective indifference sets, and the condition fails. □

A.3 Proposition 2

*Proof of **Proposition 2**.* Fix $x \succsim y$. If their chord does not witness WNB, the first case holds. Otherwise, $x \succ y$, since indifferent endpoints would already witness SNB.

First, there cannot be $0 < s < t < 1$ and $x \succsim z \succsim y$ such that

$$x\, s\, y \succ z \succ x\, t\, y.$$

Indeed, mixture continuity gives a point before s and another after t, both indifferent to z. The chord between these points contains both $x\, s\, y$ and $x\, t\, y$, and therefore witnesses SNB.

Now define

$$\alpha = \sup\{t \in [0,1] : x\, t\, y \prec y\}, \qquad \beta = \inf\{t \in [0,1] : x\, t\, y \succ x\}.$$

Both sets are nonempty by WNB. Mixture continuity gives

$$0 < \alpha, \beta < 1, \qquad x\alpha y \sim y, \qquad x\beta y \sim x.$$

The preceding observation rules out $\beta < \alpha$: a mixture strictly above x would precede one strictly below y. Equality is also impossible, since $x \succ y$. Thus $\alpha < \beta$.

For $t < \alpha$, there is a later mixture strictly below y. Applying the observation with $z = y$ gives $x\, t\, y \precsim y$. Similarly, for $t > \beta$, there is an earlier mixture strictly above x, so applying it

with $z = x$ gives $x\,t\,y \succsim x$. Between α and β, their definitions give

$$x \succsim x\,t\,y \succsim y.$$

Each outer comparison is strict somewhere because the chord witnesses WNB. This establishes the second case, which is mutually exclusive with the first. □

A.4 Proposition 3

*Proof of **Proposition 3**.* Suppose SNB holds. Exchange the indifferent endpoint labels, if necessary, so that the strictly better mixture occurs at a smaller weight than the strictly worse mixture. The same pair then witnesses SNB*.

Conversely, suppose SNB* holds, with $x \succsim y$ and $0 < \alpha < \beta < 1$.

If

$$x\alpha y \succ x \succ x\beta y,$$

mixture continuity gives $\gamma \in [0, \alpha)$ such that $x\gamma y \sim x$. Both $x\alpha y$ and $x\beta y$ lie strictly between these indifferent objects, one strictly better and the other strictly worse. Thus they witness SNB.

If instead

$$x\alpha y \succ y \succ x\beta y,$$

mixture continuity gives $\gamma \in (\beta, 1]$ such that $x\gamma y \sim y$. Both $x\alpha y$ and $x\beta y$ lie strictly between these indifferent objects, again giving a strictly better and a strictly worse mixture. Thus SNB holds in either case. □

A.5 Theorem 1

Proof. Step 1. Openness. Fix a utility U displaying SNB. By definition, there exist distinct $x, y \in \mathcal{X}$ with $U(x) = U(y)$ and $\alpha < \beta$ such that

$$U(x\alpha y) > U(x) = U(y) > U(x\beta y).$$

The equality between x and y need not survive a small perturbation of U. We therefore first obtain a witness involving only strict comparisons.

For $\varepsilon > 0$, let

$$x^\varepsilon = (1 - \varepsilon)x + \varepsilon\mathbf{1}.$$

This point belongs to $\mathcal{X}$ by convexity. Since $x \neq \mathbf{1}$, strict monotonicity gives

$$U(x^\varepsilon) > U(x) = U(y).$$

Moreover, as $\varepsilon \downarrow 0$, x^ε converges to x. By continuity, the original strict mixture comparisons are preserved for all sufficiently small $\varepsilon > 0$:

$$U(x^\varepsilon \alpha y) > U(x^\varepsilon), \qquad U(y) > U(x^\varepsilon \beta y).$$

Combining these inequalities gives

$$U(x^\varepsilon \alpha y) > U(x^\varepsilon) > U(y) > U(x^\varepsilon \beta y).$$

Fix such an ε and relabel x^ε as x. We have thus obtained an SNB* witness in which all three comparisons are strict.

We now show that this witness survives small perturbations of U. Let $\rho > 0$ be the smallest of the three utility gaps:

$$\rho = \min\{U(x\alpha y) - U(x),\ U(x) - U(y),\ U(y) - U(x\beta y)\} > 0.$$

Consider any calibrated utility $\widetilde{U} \in \mathcal{U}$ satisfying

$$\|\widetilde{U} - U\|_\infty < \rho/3.$$

For any two alternatives a, b, their utility difference can change by at most twice the uniform distance:

$$\widetilde{U}(a) - \widetilde{U}(b) \geq U(a) - U(b) - 2\|\widetilde{U} - U\|_\infty.$$

Each of the three strict comparisons above has a utility gap of at least ρ. Consequently, under $\widetilde{U}$, each gap remains strictly positive:

$$\widetilde{U}(a) - \widetilde{U}(b) > \rho - \frac{2\rho}{3} > 0.$$

It follows that

$$\widetilde{U}(x\alpha y) > \widetilde{U}(x) > \widetilde{U}(y) > \widetilde{U}(x\beta y).$$

Thus the same alternatives continue to witness SNB*, and hence SNB by the SNB–SNB* equivalence.

We have shown that every preference displaying SNB has an open neighborhood, in the calibrated uniform topology, consisting entirely of preferences displaying SNB. Therefore $\mathcal{P}^S$ is open in $\mathcal{U}^0$, and in particular open relative to $\mathcal{P}^M$. Since the C^k topology is finer than the uniform topology on $\mathcal{U}^k$, $\mathcal{P}^S \cap \mathcal{U}^k$ is open in $\mathcal{U}^k$, and hence open relative to $\mathcal{P}^M \cap \mathcal{U}^k$, for every $k \in \{1, 2, \ldots, \infty\}$.

Step 2. Without SNB, MMA requires a flat chord. Suppose that U displays MMA but not SNB. Assume, towards a contradiction, that there is no flat chord with distinct endpoints.

Without SNB, mixtures of indifferent endpoints cannot be both better and worse than those endpoints. Since no chord is flat, at least one comparison must be strict. This strict comparison survives when the indifferent endpoints move slightly, by continuity. Thus a mixture-loving chord remains mixture loving under small changes in its indifferent endpoints, and the same holds for a mixture-averse chord.

By MMA, there are two indifferent chords (x^0, y^0) and (x^1, y^1), the first mixture loving and the second mixture averse. Move between their endpoints by setting

$$x_t = (1-t)x^0 + tx^1, \qquad y_t = (1-t)y^0 + ty^1, \qquad t \in [0,1].$$

We can label the endpoints so that, throughout this movement, neither point dominates the other coordinate by coordinate.[31]

The moving endpoints need not remain indifferent. Whenever one is better than the other, move the better endpoint toward 0 until they become indifferent. Both endpoints are strictly preferred to 0, so continuity and strict monotonicity guarantee a unique adjustment, which varies continuously with t. Convexity keeps all these points in $\mathcal{X}$. If the endpoints are already indifferent, leave them unchanged.

The adjusted endpoints remain distinct. Indeed, if moving x_t toward 0 made it equal to y_t, we would have

$$y_t = sx_t \leq x_t \qquad \text{for some } s \in (0,1],$$

contrary to our choice of endpoint labels. The same argument applies when y_t is adjusted.

At $t = 0$ and $t = 1$, no adjustment is needed. We have therefore moved continuously from the original mixture-loving chord to the original mixture-averse chord, always keeping the endpoints distinct and indifferent. But, as argued above, the mixture attitude cannot change along such a movement. This is a contradiction. Hence MMA without SNB requires a flat chord with distinct endpoints.

Step 3. Density. Fix an MMA utility $U \in \mathcal{U}^k$, with $0 \leq k < \infty$, and choose $\lambda > 0$ small.

If $k = 0$, extend U continuously and monotonically from $\mathcal{X}$ to the cube (Nachbin, 1965); if $\mathcal{X}$ is the cube, no extension is needed. Bernstein approximation gives an arbitrarily close polynomial P with $\partial_j P \geq 0$.

If $k \geq 1$, approximate U and its derivatives through order k by a polynomial P in the affine hull of $\mathcal{X}$. Such approximation is possible because U has a C^k extension to a neighborhood of $\mathcal{X}$: using a smooth cutoff, extend it to the affine hull and apply Bernstein approximation on a containing cube. Monotonicity gives $DU(x)[v] \geq 0$ in every increasing direction, first in the relative interior and then on $\mathcal{X}$ by continuity. Choose the first derivatives sufficiently close

[31] Write $u = x^0 - y^0$ and $v = x^1 - y^1$. If this property fails, then $u + \lambda v \geq 0$ or $u + \lambda v \leq 0$ for some $\lambda > 0$. Consider the first case and reverse the second chord. For any $\mu > 0$, we cannot have $u - \mu v \geq 0$: a positive combination with $u + \lambda v \geq 0$ would give $u \geq 0$. Nor can we have $u - \mu v \leq 0$: subtracting this inequality from $u + \lambda v \geq 0$ would give $v \geq 0$. Both conclusions contradict strict monotonicity and the indifference of the original endpoints. The second case follows by reversing the inequalities.

that

$$DP(x)[v] \geq -\lambda/2 \quad \text{for all } x \in \mathcal{X}, \quad v \in \operatorname{aff}\mathcal{X} \cap \mathbb{R}^d_+, \quad \|v\|_2 = 1.$$

The same bound holds when $k = 0$.

Set

$$Q(x) = P(x) + \lambda \sum_j x_j + \mu \sum_j x_j^M, \qquad \mu > 0, \quad M > \max\{\deg P, 1\} \text{ even}.$$

For every increasing unit direction v in the affine hull,

$$DQ(x)[v] \geq -\lambda/2 + \lambda \sum_j v_j \geq \lambda/2 > 0,$$

where the last inequality follows from $v \geq 0$ and $\|v\|_2 = 1$. The preference represented by Q is complete and transitive, mixture continuous because Q is continuous, and strictly monotone by this derivative bound. It therefore satisfies the Basic Axioms. Moreover, for every $x \neq y$,

$$\frac{d^M}{dt^M} Q\big(y + t(x-y)\big) = \mu M! \sum_j (x_j - y_j)^M > 0,$$

so no nondegenerate chord is flat. With P and M fixed, the last term can be made arbitrarily small in C^k by choosing μ sufficiently small.

Calibration preserves these properties. Recalling that $b(t) = t\mathbf{1}$ is the diagonal map, set

$$\widetilde{U} = (Q \circ b)^{-1} \circ Q.$$

For each finite k, choose λ sufficiently small, P sufficiently close to U in C^k, and then μ sufficiently small. The form of Q ensures that it has no nondegenerate flat chord while remaining arbitrarily close to U in C^k. Since U is calibrated, $Q \circ b$ approaches the identity, so calibration preserves this approximation. Thus $\widetilde{U}$ remains arbitrarily close to U, the strict comparisons witnessing MMA survive, and Step 2 gives SNB. The same argument applies to C^∞, since the calibrated approximations are smooth and each basic neighborhood controls only finitely many derivative orders. Hence $\mathcal{P}^S \cap \mathcal{U}^k$ is dense in $\mathcal{P}^M \cap \mathcal{U}^k$ for every $k \in \{0, 1, 2, \ldots, \infty\}$.

Failure of density from C^2 onward. Let

$$V(x) = \sum_j (x_j - \gamma x_j^2), \qquad 0 < \gamma < \tfrac{1}{2}, \qquad h = V \circ b, \qquad U = h^{-1} \circ V.$$

Then U is smooth, calibrated, and strictly monotone. Since $D^2V = -2\gamma I$, for every calibrated W sufficiently close to U in C^2, the representation $h \circ W$ remains strictly concave. Such preferences are convex and cannot display MMA or SNB. This gives an open neighborhood

without SNB in every C^k topology with $k \geq 2$, including C^∞. □

A.6 Proposition 4

Proof. The implication SNB $\Rightarrow$ MMA is immediate. For the converse, suppose that SNB fails. We show that MMA also fails.

1. Translate the benchmark to the origin. Write $\mathcal{X}' = \mathcal{X} - n$, keeping the same notation for the translated preference. Translation preserves mixtures, and neutrality becomes

$$v \succsim w \quad \Longleftrightarrow \quad \gamma v \succsim \gamma w, \qquad \gamma \in (0,1).$$

Let $E = \operatorname{span}(\mathcal{X}' - \mathcal{X}')$ and $e = \mathbf{1}$. We extend the preference from the bounded domain $\mathcal{X}'$ to the whole linear space E. Since $0 \in \operatorname{ri}(\mathcal{X}')$, any finite collection of points in E can be contracted sufficiently toward 0 to lie in $\mathcal{X}'$. Compare points in E after such a common contraction. Neutrality makes the comparison independent of the contraction chosen. The extension still fails SNB, since any witness in E could be contracted back into $\mathcal{X}'$.

For each $v \in E$, let $U(v)$ be the unique number a such that $v \sim ae$. Continuity and strict monotonicity ensure that U is well-defined, continuous, and represents the extended preference. By construction, $U(ae) = a$, and neutrality gives

$$U(tv) = tU(v), \qquad t > 0.$$

All indifference surfaces and sections below refer to this extension on E.

2. Without SNB*, each section bends only one way.* Choose a linear complement H to $\mathbb{R}e$, so $E = H \oplus \mathbb{R}e$. Think of H as the horizontal directions and e as the vertical direction. For each level $r \in \mathbb{R}$ and position $h \in H$, let $g_r(h)$ be the unique height satisfying

$$U\big(h + g_r(h)e\big) = r.$$

Thus g_r describes the level-r indifference surface. Continuity and strict monotonicity imply that $g_r(h)$ is jointly continuous in r and h. Homogeneity gives

$$g_{tr}(th) = tg_r(h), \qquad t > 0.$$

Fix r, h, and $d \in H \setminus \{0\}$, and consider the section

$$\phi(s) = g_r(h + sd), \qquad s \in \mathbb{R}.$$

Take $s_1 < s_2$ and the two points

$$x_1 = h + s_1 d + \phi(s_1)e, \qquad x_2 = h + s_2 d + \phi(s_2)e.$$

By construction, $U(x_1) = U(x_2) = r$. For any $\lambda \in (0,1)$, put $s_\lambda = (1-\lambda)s_1 + \lambda s_2$. Their mixture is

$$(1-\lambda)x_1 + \lambda x_2 = h + s_\lambda d + \big[(1-\lambda)\phi(s_1) + \lambda\phi(s_2)\big]e,$$

whereas the point on the indifference curve at the same horizontal position satisfies

$$U\big(h + s_\lambda d + \phi(s_\lambda)e\big) = r.$$

Strict monotonicity in the vertical direction therefore gives

$$U\big((1-\lambda)x_1 + \lambda x_2\big) \gtreqless r \quad \Longleftrightarrow \quad (1-\lambda)\phi(s_1) + \lambda\phi(s_2) \gtreqless \phi(s_\lambda).$$

Thus we compare the average of the values of ϕ with its value at the average. If the difference is positive for one weight and negative for another, the same endpoints give SNB. Since SNB fails, every chord must lie weakly above the graph throughout its interval or weakly below it.

Suppose ϕ were neither convex nor concave. There would then be one chord strictly above the graph somewhere and another strictly below it somewhere. Continuously enlarge each of their intervals to the same interval containing both. Every intermediate interval still contains the original interval, on which ϕ is not affine, so its chord cannot coincide with the graph throughout. It must therefore lie on one side, with a strict inequality somewhere. By continuity, that strict inequality persists under small changes of the endpoints, so the side cannot change during the enlargement. The final chord would consequently have to lie both above and below the graph, yet could not coincide with it throughout. This is a contradiction. Hence ϕ is convex or concave, possibly affine.

3. Curvature depends only on direction. Fix $d \in H\backslash\{0\}$. For any r and h, write the corresponding section as $\phi(s) = g_r(h + sd)$. Since we are working on E, this section is defined for every real s. By homogeneity and continuity,

$$\begin{aligned} \frac{\phi(R)}{R} &= g_{r/R}\left(\frac{h}{R} + d\right) \longrightarrow g_0(d), \\ \frac{\phi(-R)}{-R} &= -g_{r/R}\left(\frac{h}{R} - d\right) \longrightarrow -g_0(-d) \end{aligned} \qquad (R \to \infty).$$

Thus the limiting slopes at the two ends do not depend on the level r or position h. Write

$$\Delta(d) = g_0(d) + g_0(-d)$$

for the difference between the right and left limiting slopes.

If ϕ is convex, its secant slopes increase from left to right. Hence, for every $u < v$,

$$-g_0(-d) \le \frac{\phi(v) - \phi(u)}{v - u} \le g_0(d).$$

If ϕ is concave, the inequalities are reversed. Thus $\Delta(d) > 0$ rules out concavity and affinity, while $\Delta(d) < 0$ rules out convexity and affinity. If $\Delta(d) = 0$, the limiting slopes coincide, so every secant slope must be the same and ϕ is affine. Using Step 2, we therefore obtain

$$\begin{aligned} \Delta(d) > 0 &\implies \phi \text{ is convex,} \\ \Delta(d) < 0 &\implies \phi \text{ is concave,} \\ \Delta(d) = 0 &\implies \phi \text{ is affine.} \end{aligned}$$

This holds for all levels and positions. Every nondegenerate indifferent chord belongs to one of these sections, since its endpoints cannot lie on the same vertical line. Convexity makes mixtures weakly better than the endpoints, concavity makes them weakly worse, and affinity makes them indifferent. In particular, a strict loving comparison requires $\Delta(d) > 0$, and a strict averse comparison requires $\Delta(d) < 0$.

4. Opposite signs are impossible. If MMA held, Step 3 would give directions $p, q \in H$ with

$$\Delta(p) > 0 > \Delta(q).$$

Since Δ is even and positively homogeneous, p and q cannot be collinear. By continuity, there is $\theta \in (0, 1)$ such that

$$z = (1-\theta)p + \theta q \neq 0, \qquad \Delta(z) = 0.$$

Step 3 says that every section $s \mapsto g_0(h + sz)$ is affine. Its slope is $g_0(z)$, by the limiting-slope calculation above. Hence, for every $h \in H$ and $s \in \mathbb{R}$,

$$g_0(h + sz) = g_0(h) + sg_0(z).$$

Adding this identity to the one with h, s replaced by $-h, -s$ gives

$$\Delta(h + sz) = \Delta(h).$$

Consequently,

$$\begin{aligned} (1-\theta)\Delta(p) &= \Delta((1-\theta)p) = \Delta((1-\theta)p - z) \\ &= \Delta(-\theta q) = \theta\Delta(q). \end{aligned}$$

This is impossible: the left side is positive and the right side is negative. Thus MMA cannot hold when SNB fails, proving the converse. □

A.7 Proposition 5

*Proof of **Proposition 5**.* Choose $p \in \mathrm{ri}(\mathcal{X})$ and two affine functions L, M with strictly positive coordinate coefficients summing to one, independent linear parts on the affine hull of $\mathcal{X}$, and $L(p) = M(p)$.

For sufficiently small $h > 0$, let the two preferences be represented by

$$\min\{L(x), M(x) + h\}, \qquad \max\{L(x), M(x) - h\}.$$

Both representations are continuous and strictly increasing. The first is concave and the second convex, each with a strict mixture attitude across its kink. Hence they belong to $\mathcal{L}$ and $\mathcal{A}$, respectively.

Take a short chord through p, with endpoints x_0, y_0 symmetric about p, on which L is constant and

$$M(x_0) = L(p) - 2h, \qquad M(y_0) = L(p) + 2h.$$

Such a chord exists by relative interiority. The loving bend lies on its first half and the averse bend on its second; the other motive is affine in each case. At x_0, only the loving motive ranks the object below p; at y_0, only the averse motive ranks it above p. Thus, for every $F \in \mathcal{F}$,

$$x_0 \prec_F p \prec_F y_0.$$

Fix F. By continuity, these comparisons persist at

$$x = x_0 + \varepsilon\mathbf{1}, \qquad y = y_0 - \varepsilon\mathbf{1}$$

for sufficiently small $\varepsilon > 0$. Both representations equal $L(p) + \varepsilon/2$ at $x\frac{3}{4}y$, and $L(p) - \varepsilon/2$ at $x\frac{1}{4}y$. These comparisons survive calibration, so

$$x \prec_F p \prec_F x\tfrac{3}{4}y, \qquad x\tfrac{1}{4}y \prec_F p \prec_F y.$$

Mixture continuity gives a point between x and $x\frac{3}{4}y$, and another between $x\frac{1}{4}y$ and y, both indifferent to p. Their chord contains a strictly better and a strictly worse mixture. Hence $\succsim_F$ displays SNB. □

B Experiment Appendix

B.1 Experimental design

Table B.1: Sample characteristics

	Mean	SD	Min	Max	N
Male respondent	0.59	0.49	0	1	83
Age	27.1	7.8	18	50	83
Number of trips abroad in last 12 months	4.0	3.0	0	20	83
Number of EU countries visited in list of 10	4.9	2.2	0	9	83
Time to chosen travel date (in days)	229.7	158.4	23	636	83
Anticipated number of travel partners					
None	0.14	0.35	0	1	83
1 partner	0.73	0.44	0	1	83
2 partners or more	0.12	0.33	0	1	83
Recruitment channel					
LSE Lab (email)	0.87	0.34	0	1	83
Social media or other ads	0.13	0.34	0	1	83
Survey completed after Italian lockdown	0.51	0.50	0	1	83
Total survey time (in minutes)	44.0	19.7	11	120	83

Notes: Total survey time censored at 120 minutes for one respondent. Recruitment channel re-coded for 9 respondents based on text responses.

B.2 How rich are mixture attitudes? (RQ1)

Table B.2: Consistency of preferences over trips

	$x_j \succ x_k$ $(v_j > v_k)$	$x_j \sim x_k$ $(v_j = v_k)$	$x_j \bowtie x_k$ $(v_j < v_k)$	Direct choice agrees
Adjacent trips ($k = j + 1$)	84.1	12.1	3.9	92.2
	(628/747)	(90/747)	(29/747)	(153/166)
All pairs	91.4	4.3	4.3	91.8
	(3415/3735)	(161/3735)	(159/3735)	(457/498)

Notes: Percentages of comparisons; counts in parentheses. In every pair x_j is ranked above x_k. The first three columns compare the ranking with the valuations; the last column reports, for the 6 direct choices between two sure trips (DPs 1, 4, 7, 10, 13 and 16; 2 of them between adjacent trips), the share in which the higher-ranked trip is both valued strictly higher and chosen, or the two trips are valued equally. 52% of respondents have a strict ordering, 88% have no conflict between ranking and valuations, and 77% have no inconsistency of any kind, so that all their decision problems are clear-dominance problems.

Table B.3: Prevalence of SD violations

	Mean share of violations (%)	Never violate (%)	At least one (%)	80th percentile (%)
$\mathcal{D}^+$: dominated lotteries (24 problems)	22.0	29.1	70.9	44
$\mathcal{B}^+$: binary, mirroring $\mathcal{B}^-$ (11)	25.0	39.2	60.8	50
$\mathcal{B}^-$: dominant binary lotteries (11)	16.1	62.0	38.0	28

Notes: Clear-dominance problems only; $N = 79$ respondents with at least one clear-dominance problem. Shares are computed per respondent and then averaged. Of the 79 respondents, 11% never violate SD, 51% violate it only on $\mathcal{D}^+$, 18% only on $\mathcal{B}^-$, and 20% on both. Spearman correlation between the shares of violations on $\mathcal{B}^+$ and $\mathcal{B}^-$: -0.34 ($p = .002$).

B.3 Which mixtures do people prefer? (RQ2)

Table B.4: Mixture-loving and mixture-averse choices by probability

	Lottery chosen over x_j (upper)			x_k chosen over lottery (lower)		
	(1)	(2)	(3)	(4)	(5)	(6)
$p = .9$	-6.7^{**}	-5.4^{*}	-5.5^{*}	-0.0	0.6	2.2
	(2.9)	(3.3)	(3.1)	(3.1)	(3.5)	(3.3)
$p = .1$	-17.3^{***}	-18.6^{***}	-18.5^{***}	10.2^{***}	9.6^{**}	8.1^{**}
	(3.1)	(3.4)	(3.6)	(3.6)	(3.9)	(4.1)
Value forgone (£100)		3.3	3.2		-3.3	-11.4
		(3.9)	(2.4)		(7.8)	(7.6)
Problem-triple FE	Yes	Yes	Yes	Yes	Yes	Yes
Respondent FE	No	No	Yes	No	No	Yes
Share at $p = .5$ (%)	27.8	27.8	27.8	12.0	12.0	12.0
Observations	1,113	1,113	1,113	798	798	798
Respondents	79	79	79	79	79	79

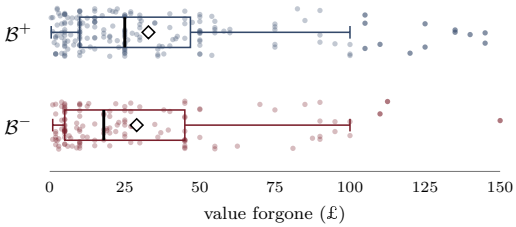
Notes: Linear probability models, in percentage points; binary choices meeting the trip-comparison criteria in **Section 4.3**. Columns (1)–(3): outcome is choosing the lottery $L_{jk}(p)$ over a sure trip ranked at or above x_j (mixture loving); columns (4)–(6): choosing a sure trip ranked at or below x_k over $L_{jk}(p)$ (mixture aversion). Reference category: $p = .5$. Problem-triple fixed effects: each triple of problems offers the same sure trip and a lottery over the same two trips, and differs only in p, so the coefficients compare the three probabilities within the same pair of options. In the upper half each chord appears with a single sure trip (x_1), so these are chord fixed effects; in the lower half the chord (x_1, x_2) appears with two sure trips, x_2 and x_3. Value forgone: $|v(x_A) - \mathrm{E}_{\mathbf{p}}[v(x_B)]|$ in the respondent's valuations. Standard errors clustered by respondent in parentheses; $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table B.5: Valuations by rank number

	Mean	SD	p25	p50	p75	Min	Max	N
v_1	373.2	88.6	320	400	450	80	500	83
v_2	335.5	93.9	300	350	400	10	480	83
v_3	310.6	90.3	260	300	380	50	490	83
v_4	278.1	100.2	200	290	350	8	454	83
v_5	260.8	96.4	190	270	340	7	480	83
v_6	242.1	98.9	170	240	320	6	498	83
v_7	216.8	100.7	150	200	300	4	485	83
v_8	193.6	104.7	120	180	250	3	500	83
v_9	174.1	104.5	100	150	240	0	499	83
v_{10}	155.2	105.0	80	135	220	0	475	83
v^*	275.4	134.4	180	295	400	10	500	83

Notes: Summary statistics about the distribution of valuations entered by respondents for the holiday trip they ranked in the kth position in their ranking, where $k \in \{1, 2, \ldots, 10\}$; v^* is the valuation for the wildcard trip. Valuations expressed in £. Market value of each trip equal to £420.

(a) Size of violations on $\mathcal{B}^+$ and $\mathcal{B}^-$



(b) Size of all violations

	Mean	SD	p25	p50	p75	Min	Max	N
$\mathcal{D}^+$	45.7	45.0	15	32	58	0.5	282	369
$\mathcal{B}^+$	32.9	33.1	10	25	47	0.5	145	194
$\mathcal{D}^+ \setminus \mathcal{B}^+$	59.9	51.7	27	44	73	3.3	282	175
$\mathcal{B}^-$	29.0	33.5	5	18	45	1	215	136
All	41.2	42.8	10	27	51	0.5	282	505

Figure B.1: Value forgone in SD violations. Value forgone in choices that violate stochastic dominance, $|v(x_A) - \mathrm{E}_{\mathbf{p}}[v(x_B)]|$ in £, computed from the respondent's reported valuations (clear-dominance problems only); market value of each trip: £420. $\mathcal{D}^+$: lottery chosen over a sure trip that dominates it; $\mathcal{B}^+$: the $\mathcal{D}^+$ problems that mirror $\mathcal{B}^-$; $\mathcal{D}^+ \setminus \mathcal{B}^+$: the remaining $\mathcal{D}^+$ problems; $\mathcal{B}^-$: sure trip chosen over a lottery that dominates it. N: number of violating choices. (a) Dots: individual choices, darker beyond the whiskers; box: 25th–75th percentile; bar: median; diamond: mean; whiskers: most extreme value within 1.5 times the interquartile range. One $\mathcal{B}^-$ violation (£215) lies beyond the axis and is not shown.

B.4 Where does the (dis)taste for randomization come from? (RQ4)

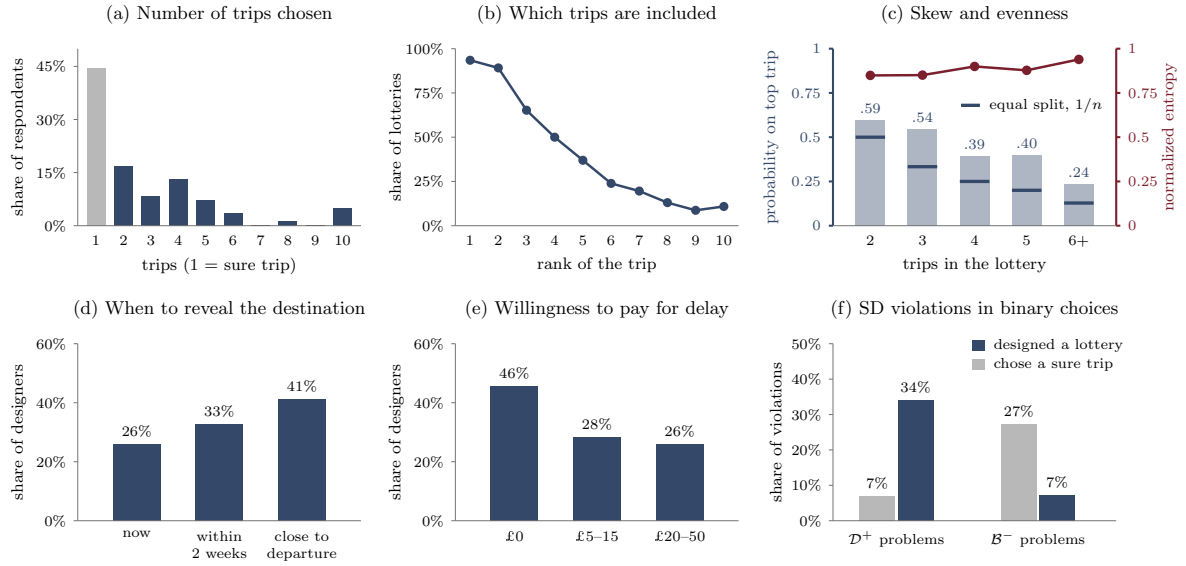


Figure B.2: The lottery-building exercise. Design task, $N = 83$; panels (b)–(e): the 46 respondents who designed a lottery, with trips ranked by each respondent's own ranking. (c) By number of trips in the lottery: bars, mean probability on the highest-ranked trip included (left axis), with dark lines at the equal split $1/n$ (for 6+, averaged over the lotteries with 6, 8 and 10 trips); line, mean entropy divided by its maximum given the number of trips (right axis; 1 = equal split). (d) Within two weeks of the decision, or at most four weeks before departure. (e) Elicited with a multiple price list; delay and willingness to pay are correlated (Spearman $\rho = 0.42$, $p = 0.013$, $N = 34$). (f) Share of SD violations in the binary choices: lottery chosen over a sure trip that dominates it ($\mathcal{D}^+$), sure trip chosen over a lottery that dominates it ($\mathcal{B}^-$); $N = 79$.

Reasons. Respondents who rated the surprise-trip concept at 50 or above ($N = 48$) ranked six possible reasons for liking it. The excitement until the destination is revealed (33%) or daydreaming about the possible destinations (19%) came first for 52% of them, and the unique holiday experience for another 27%. Respondents who rated it below 50 ($N = 35$) most often ranked first the disappointment if the destination is not the one they hoped for (34%) and the stress of going on holiday unprepared (26%), followed by the inability to plan activities (17%) (**Figure B.3** in the Appendix). These are the two forces of **Section 4.2**: the value of contemplating open questions, and the cost of uncertainty for preparation.

The wildcard trip. The wildcard trip is a trip to a completely unknown destination. Its valuation correlates with choosing dominated lotteries (Spearman $\rho = 0.22$, $p = .047$) and with building a lottery (£308 for builders against £234, $p = .017$), but not with the types (**Figure B.4** in the Appendix; Kruskal–Wallis, $p = .52$). **Table B.6** and **Figure 10** relate the motives respondents gave for and against the wildcard trip to three outcomes. A summary motive score is positively associated with the wildcard valuation and the share of mixture-loving choices in $\mathcal{D}^+$, and negatively associated with the share of mixture-averse choices in $\mathcal{B}^-$. Item by item, the motives for the wildcard predict the three outcomes in the expected direction, most

clearly the thrill of seeing how chance plays out, which is significant for all three, while the motives against it, such as not knowing being a little scary, predict them in the opposite direction. The two outcome-dependent motives behave differently. Fearing a worse destination is associated with a lower wildcard valuation (-0.11, 95% CI $[-0.17, -0.06]$), but hoping for a better one, although the most endorsed motive for the wildcard (mean 3.8 out of 5), has no statistically clear association with any of the three outcomes. This is relevant for the theory, since outcome-dependent uncertainty utility is what it would take to break the symmetry of **Section 4.2** and allow WNB without SNB; we find no statistically clear association on the hope side of that mechanism.

Table B.6: Wildcard motives: ratings and correlations

	Mean	Agree	(1)	(2)	(3)	(4)	(5)	(6)
For the wildcard								
(1) Hoped for a better destination	3.77	66%						
(2) Love going somewhere unexpected	3.27	42%	.03					
(3) Thrill of seeing how risk plays out	2.92	37%	−.08	.55				
(4) No clue helps me fantasize	2.75	29%	.03	.50	.62			
Against the wildcard								
(5) Feared a worse destination	3.71	60%	−.10	−.30	−.44	−.54		
(6) Not knowing is a little scary	3.47	55%	.05	−.43	−.62	−.45	.30	
(7) Can only dream if I know where	3.33	54%	.08	−.44	−.56	−.56	.51	.54

Spearman correlations of raw ratings (1 = completely disagree, 5 = completely agree); $N = 83$. Agree: share rating 4 or 5. Correlations among items 2–7 are significant at 1%, those with item 1 not at 10%. Motive score: mean of the seven items, 5–7 reverse coded (Cronbach's $\alpha = 0.81$).

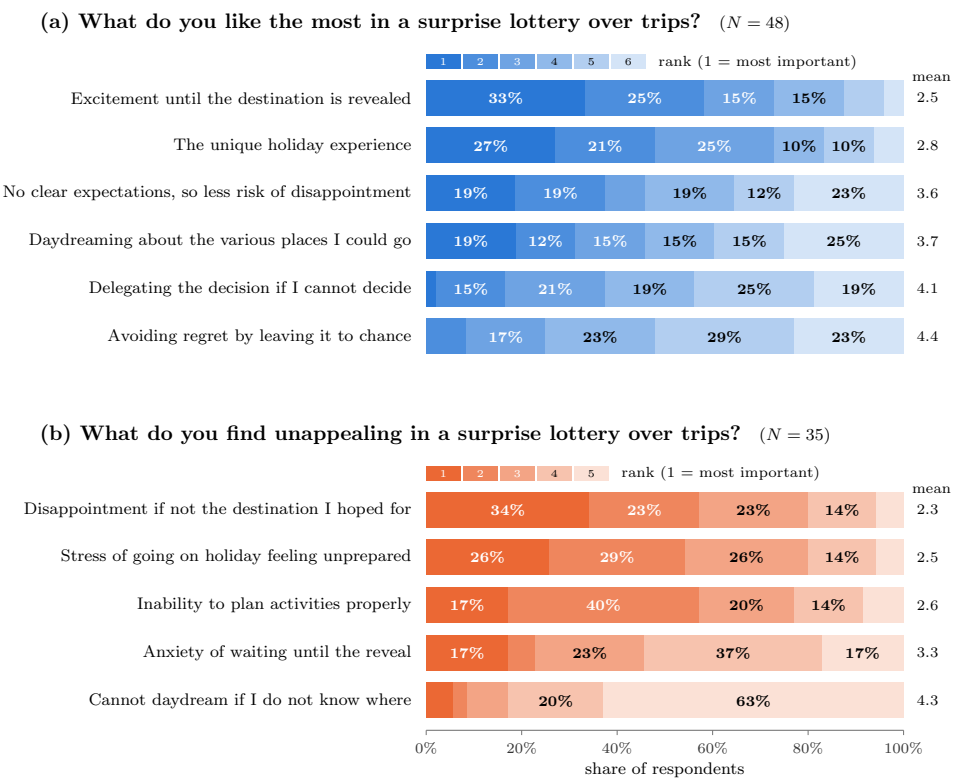


Figure B.3: What respondents like and dislike about surprise lotteries over trips (rankings). Panel (a): respondents who rated the concept of a surprise trip at 50 or above (out of 100); panel (b): those who rated it below 50. Each respondent ranked all aspects; bars show the distribution of ranks, with aspects ordered by mean rank.

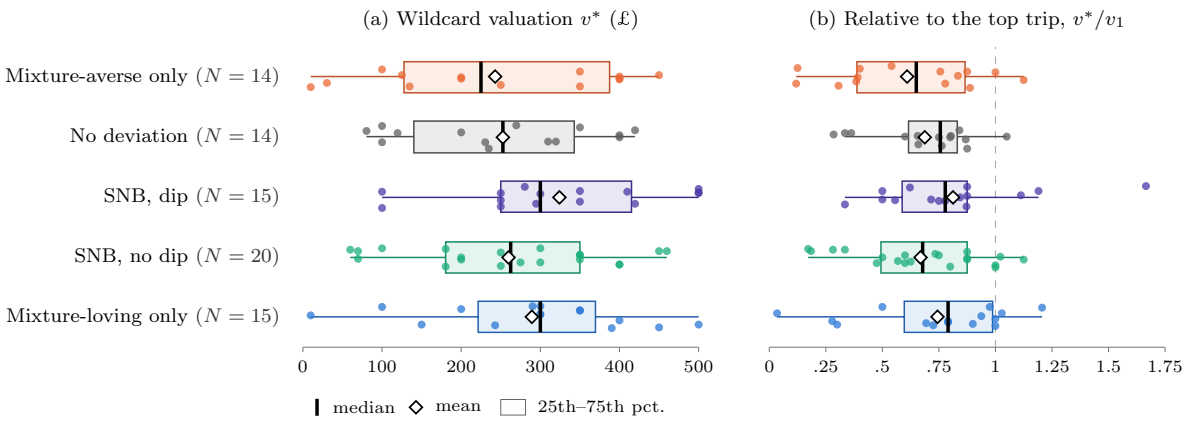


Figure B.4: Wildcard valuations by type, in pounds (a) and relative to the valuation of the top-ranked trip (b). v_1: valuation of the top-ranked trip. Each dot is one respondent; whiskers extend to the most extreme value within 1.5 times the interquartile range. Kruskal–Wallis test of equal distributions across the five types: $p = 0.52$ in (a), $p = 0.65$ in (b). Types ordered as in **Figure 8**. The 5 respondents not classified in **Figure 6** are omitted.

B.5 How does multidimensionality affect mixing? (RQ5)

No objective winner. The framework of **Section 4.2.1** attributes SD violations to the lack of dominance between trips. The trips were designed to offer different experiences at the same market value, and indeed no trip is an objective winner (**Figure B.5**). Every destination is someone's favorite, average ranks range only from 4.8 (Gothenburg) to 6.4 (Düsseldorf), and mean valuations from £227 to £274.

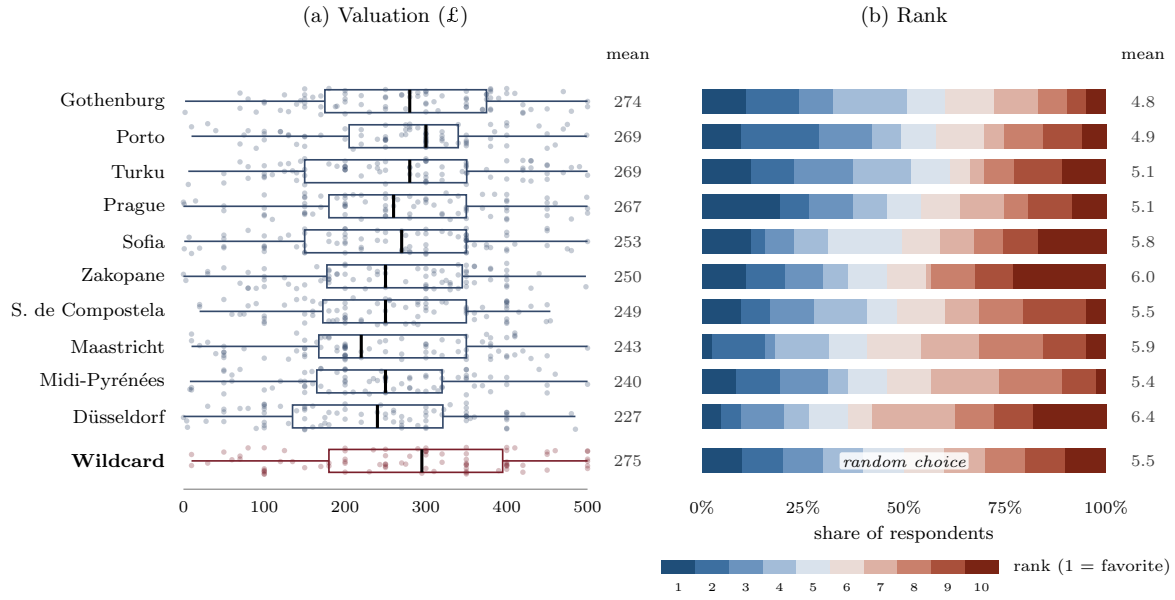


Figure B.5: No objective winner: valuations (a) and rank distribution (b) of each destination, ordered by mean valuation, and valuation of the wildcard. $N = 83$. Destinations ordered by mean valuation. (a) Dots: respondents' valuations; box: 25th–75th percentile; bar: median; whiskers: most extreme value within 1.5 times the interquartile range. (b) Each bar splits respondents by the rank they gave the destination; bottom row: ranks assigned at random, each with probability 1/10 (mean 5.5).

Attributes. Trips are vectors of attributes (**Section 4.2.1**). Respondents rated their preference on 9 bipolar dimensions (**Figure B.6**) from 1 to 5. We translate the score out of 5 to a bipolar leaning with 1- 2, 3, and 4-5 to -1,0 and 1, respectively. They lean, on average, toward new and warm destinations, but every dimension is horizontal for some respondents: the share with no leaning ranges from 18% (cozy apartment–luxury hotel) to 43% (rural–urban), and 90% of respondents have at least one horizontal dimension (2.8 on average). We coded each trip's position on the first 8 dimensions from its description (left pole, right pole or neither); its position on new–familiar comes from the respondent's own familiarity with the destination. With these positions and each respondent's leanings, the dominance order $\trianglerighteq$ of **Section 4.2.1** fails to rank 91% of the 3,735 respondent-specific pairs of trips: in 75% of pairs the trips differ on a horizontal dimension, and in another 16% each is better on some vertical one. Lack of dominance is thus the norm, as the framework requires for SD violations to be admissible.

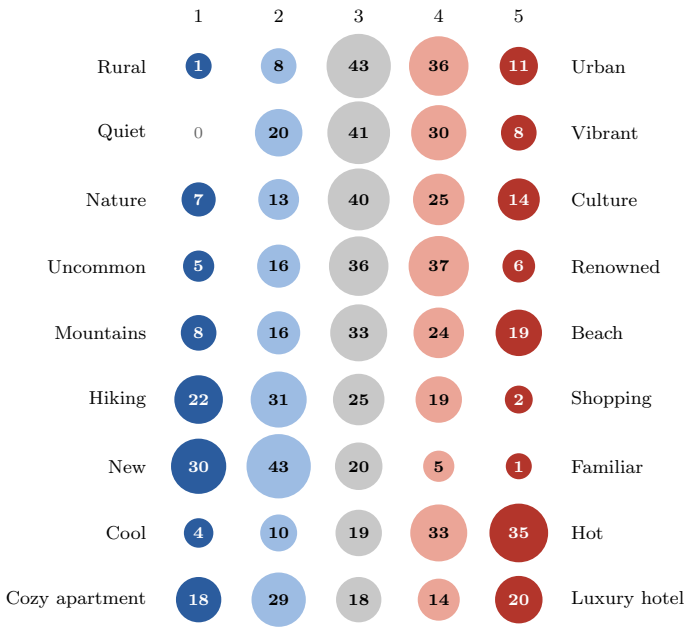


Figure B.6: Distribution of attribute ratings on each bipolar scale (1 = left pole, 5 = right pole), ordered by the share rating 3. Circle area increases linearly with the share of respondents giving each rating (numbers in %), plus a small constant so that every label fits. $N = 83$.

Figure 12 relates attributes to choices. We use the 12 problems in $\mathcal{D}^+$ in which the sure trip is one of the two trips in the lottery, so that the attributes of only two trips are involved. The more horizontal dimensions on which the two trips differ, the more often respondents choose the dominated lottery: 13% when they differ on none, 28% when they differ on three or more (panel a). The number of horizontal dimensions on which the two trips are *equal* does not raise it (panel b), and the number of vertical dimensions on which the lower-ranked trip fits better has no effect (panel c). Vertical dimensions instead shape valuations: a trip's valuation rises with its taste-fit score (panel d). This is the split of the representation (**Section 4.2**): horizontal differences enter the amount left open, D, and lower the price of uncertainty $\Delta/(Dp)$, while vertical ones enter the value of each sure trip.

Table B.7 considers a regression framework, with problem fixed effects and clustered standard errors. Each horizontal dimension on which the trips differ is associated with a higher probability of choosing the lottery by 3.8 percentage points (column 1), against a mean of 19.7%. Vertical dimensions have no statistically clear association with lottery choices, with or without the horizontal ones (columns 2–3), and the association with horizontal differences survives controlling for the value gap (column 3) and for respondent fixed effects (3.3 points, column 4). The association thus also holds within respondent. In 50/50 lotteries, with the most left open, the estimated association is 5.6 points ($p = .004$, column 5); the interaction with 50/50 is significant at the 10% level. Vertical taste fit is associated with valuations: each additional point increases trip valuation by £8.4 (column 6), and the difference between two trips' taste-fit scores is associated with a larger gap between their valuations (column 7).

Table B.7: Attributes, randomization and valuations: regressions

	Chose the dominated lottery (pp)					v (£)	$\lvert\Delta v\rvert$ (£)
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
No. of dimensions rated 3 on which							
the two trips differ	3.8**		3.8**	3.3**	2.1		−2.8
	(1.6)		(1.5)	(1.5)	(1.6)		(1.9)
× lottery is 50–50					3.5*		
					(1.8)		
No. of dimensions rated 1, 2, 4 or 5 on which							
H fits better		−0.3	0.4	0.5	0.5		
		(1.3)	(1.2)	(1.2)	(1.2)		
L fits better		−1.0	−0.4	0.2	0.2		
		(1.7)	(1.7)	(1.6)	(1.6)		
Taste-fit score							
of the trip						8.4***	
						(1.2)	
|difference between the two trips|							2.4***
							(0.8)
Value gap			2.6	2.8	2.8		
			(15.7)	(14.4)	(14.5)		
Problem FE	Yes	Yes	Yes	Yes	Yes	–	–
Respondent FE	No	No	No	Yes	Yes	Yes	Yes
Destination FE	–	–	–	–	–	Yes	–
Destination-pair FE	–	–	–	–	–	–	Yes
Mean of outcome	19.7	19.7	19.7	19.7	19.7	254.0	92.6
Observations	897	897	897	897	897	830	3,735
Respondents	79	79	79	79	79	83	83

Notes: Columns (1)–(5): linear probability models on the 12 $\mathcal{D}^+$ problems in which the sure trip is one of the two trips in the lottery (clear-dominance choices only); the outcome is choosing the lottery over the sure trip that dominates it, in percentage points. H is the sure, higher-ranked trip and L the other trip in the lottery. The columns add in turn the horizontal dimensions (1), the vertical dimensions (2), both together with the value gap (3), respondent fixed effects (4), and the interaction with 50–50 lotteries (5). Column (6): OLS of the respondent's valuation of each of the 10 trips. Column (7): OLS of the absolute difference between the respondent's valuations of two trips, over all 45 pairs of trips. Dimensions rated 3 are those on which the respondent has no leaning. On a dimension rated 1, 2, 4 or 5, a trip fits better than another if it is strictly closer to the pole the respondent prefers; the omitted categories in (1)–(5) are dimensions on which the two trips are equal. Taste-fit score of a trip: over the dimensions rated 1, 2, 4 or 5, the number on which the trip sits at the pole the respondent prefers minus the number on which it sits at the other pole (between −9 and 9). Lottery is 50–50: each trip has probability .5 (the other problems use .9/.1 and .1/.9). Value gap: $1 - \mathrm{E}_{\mathbf{p}}[v(x_B)]/v(x_A)$ in the respondent's own valuations; with problem fixed effects it is identified only from variation within a problem. In (5), the effect of a dimension on which the trips differ in a 50–50 lottery is 5.6pp ($p = 0.004$). Standard errors clustered by respondent in parentheses; $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Online Appendix: How Rich are Mixture Attitudes? Theory and Experimental Evidence

En Hua Hu Séverine Toussaert

Abstract

This online appendix details the classification of theoretical models and the literature audit in our manuscript.

JEL Codes: C91, D81, D11

Keywords: Mixture attitudes, betweenness, convex preferences, quasi-convexity, experiment

Literature Audit Appendix

OA.1 Paper selection

The review combines database searches on 26–27 August 2026 with citation searches and paper reviews through 15 October 2026. Section OA.4 gives the archived queries and hit counts.[32]

1. **Search the database.** Use Web of Science Core Collection, the Topic field, publication years 1900–2026, and no language or document-type restrictions. Seven queries cover randomization and lotteries; eleven cover the expanded domains.

2. **Deduplicate.** Combine results within each search set and deduplicate by accession identifier. Randomization produced 391 query occurrences and 367 unique records; the expanded set produced 2,654 occurrences and 2,560 records. The two sets may overlap.

3. **Screen titles and abstracts.** Retain human experiments that might compare endpoints with mixtures, including uncertain cases requiring full-text review. Randomization advanced 54 records and excluded 313; the expanded set advanced 559 and excluded 2,001.

4. **Expand through citations.** Follow backward and forward citations and search IDEAS/RePEc, SSRN, Google Scholar, publisher sites and authors' pages. These searches added 58 randomization candidates: 54 outside the database union and four reinstated after screening, bringing the total to 112. Supplementary searches without recorded hit counts are listed separately and are excluded from database totals.

[32] The search was assisted by ChatGPT 6 Astra, a future version of the draft will have complete human verification.

5. **Inspect designs and consolidate versions.** Examine choices, mixtures and within-participant comparisons; match working-paper and published versions by title, authors and identifiers. The initial design screen retained 59 of the 112 randomization candidates. A separate tally of the expanded search and citation additions records 638 candidate publications before PDF review.

6. **Establish the reviewed set.** Inspect the published article, an identified earlier manuscript or an experimental appendix. Exclude papers whose designs could not be inspected. The final tables contain 617 paper–domain entries from 589 publications, identified by linked citations.

OA.2 Mixture attitudes

For endpoints $x \neq y$ and $t \in (0,1)$, write $m_t(x,y) = (1-t)x + ty$. The comparisons below concern one participant's complete, transitive and stable preference. Label the endpoints so that $x \succeq y$.

Category	Required comparison
CV: convex	$m_t(x,y) \succeq y$.
CC: concave	$x \succeq m_t(x,y)$.
NCC: nonconcave	$m_t(x,y) \succ x \succeq y$.
NCV: nonconvex	$x \succeq y \succ m_t(x,y)$.
MMA	$m_a(x^+,y^+) \succ x^+ \succeq y^+$ and $x^- \succeq y^- \succ m_b(x^-,y^-)$. The two endpoint pairs may differ.
WNB	$m_a(x,y) \succ x \succeq y \succ m_b(x,y)$.
SNB	$x \sim y$ and $m_a(x,y) \succ x \succ m_b(x,y)$.

MMA denotes mixed mixture attitudes: mixture lovingness and mixture aversion under the same preference. WNB and SNB denote weak and strong non-betweenness, which require both signs on the same chord. CV and CC record local comparisons rather than global convexity or concavity.

A design is classified by the responses it permits, whether or not the paper reports participants displaying the attitude. Choice alone reveals weak preference; strict classifications require an explicit report or a further inference using the maintained continuity and monotonicity assumptions. MMA, WNB and SNB require the two strict signs to be jointly feasible for one preference. We used ChatGPT 6 Astra to classify the papers, future drafts will have a human RA verify.

Domain	N	Convex/ concave	Nonconcave or nonconvex	MMA	WNB/SNB	No local comparison
Randomization and lotteries	54	**26** (48%)	**15** (28%)	**4** (7%)	**3** (6%)	**6** (11%)
Consumption budgets	23	**3** (13%)	**8** (35%)	**0** (0%)	**4** (17%)	**8** (35%)
Time preferences	85	**66** (78%)	**5** (6%)	**3** (4%)	**0** (0%)	**11** (13%)
Portfolio and asset choice	98	**73** (74%)	**14** (14%)	**1** (1%)	**0** (0%)	**10** (10%)
Charitable giving	111	**91** (82%)	**6** (5%)	**0** (0%)	**2** (2%)	**12** (11%)
Effort allocation	57	**26** (46%)	**0** (0%)	**1** (2%)	**0** (0%)	**30** (53%)
Social/interpersonal allocation	189	**147** (78%)	**20** (11%)	**4** (2%)	**1** (1%)	**17** (9%)
Total	**617**	**432** (70%)	**68** (11%)	**13** (2%)	**10** (2%)	**94** (15%)

Table OA.1: Strongest identifiable mixture attitude for each paper–domain entry. Categories are mutually exclusive; percentages use all N reviewed entries in the domain. The MMA column excludes WNB/SNB, so the cumulative number of MMA-capable entries is 13 + 10 = 23.

OA.3 Papers by domain

Within each domain, each paper is listed only under the strongest mixture attitude its design can identify, using the priority WNB/SNB, MMA, nonconcave or nonconvex, and convex/concave. The categories are mutually exclusive. The merged nonconcave-or-nonconvex category contains papers that can identify at least one strict departure but cannot identify MMA; the convex/concave category contains the remaining papers with a local mixture comparison. Identifying the two departures separately does not suffice for MMA, which requires both signs to be jointly identifiable for one preference. Convex/concave and WNB/SNB remain combined because the original capability lists within each pair coincide. Papers without a local convex or concave comparison are listed separately in Table OA.3 and remain in the domain totals. Author–year citations link to the reviewed source or its publication record where available. Table OA.2 lists the papers in each category.

Table OA.2: Papers by domain and strongest identifiable mixture attitude.

Attitude (n)	**Papers**
Randomization and lotteries ($N = 54$)	
Convex/concave (26)	Adeney (2026); Becker et al. (1963); Bernasconi (1994); Burghart (2020); Calford (2026); Chew & Waller (1986); Conlisk (1987); Coombs & Huang (1976); Drichoutis et al. (2012); Dwenger et al. (2018); Eskinazi et al. (2025); Filiz-Ozbay et al. (2025); Gneezy et al. (2006); Harless (1993); Hoelzemann & Loewenfeld (2026); Krawczyk & Le Lec (2010); Kuzmics et al. (2024); Miao & Zhong (2018); Permana (2020b); Prelec (1990); Rehbeck & Stelnicki (2025); Schmidt (2025); Sopher & Gigliotti (1993); Sopher & Narramore (2000); Wang et al. (2013); Zhang & Zhong (2020)
Nonconcave or nonconvex (15)	Agranov & Ortoleva (2017); Agranov & Ortoleva (2025); Agranov et al. (2023); Alós-Ferrer et al. (2026); Arts et al. (2024); Arts et al. (2026); Butler & Loomes (2011); Cettolin & Riedl (2019); Dominiak & Schnedler (2011); Feldman & Rehbeck (2022); Feldman & Zhou (2024); Oechssler et al. (2019); Ong & Qiu (2023); Sandroni et al. (2013); Shi et al. (2024)
MMA (4)	Benzion et al. (2013); Camerer (1989); Kobayashi & Lucia (2026); McGranaghan et al. (2025)

Continued on the next page

Papers by domain and strongest identifiable mixture attitude (continued)

Attitude (n)	Papers
WNB/SNB (3)	Camerer & Ho (1994); Cubitt et al. (2015); Hey & Strazzera (1989)
Consumption budgets ($N = 23$)	
Convex/concave (3)	Bruyneel et al. (2017); Chen et al. (2023); İpek & Akyazı (2021)
Nonconcave or nonconvex (8)	Banerjee & Murphy (2011); Bruyneel et al. (2012); Burghart et al. (2013); Cherchye et al. (2013); Ellis & Freeman (2024); Harbaugh et al. (2001); Sippel (1997); Weinrabe et al. (2020)
WNB/SNB (4)	Brocas et al. (2019a); MacCrimmon & Toda (1969); Pastor-Bernier et al. (2020); Rousseas & Hart (1951)
Time preferences ($N = 85$)	
Convex/concave (66)	Alan & Ertac (2015); Alan & Ertac (2017); Alan & Ertac (2018); Andreoni & Sprenger (2012a); Andreoni & Sprenger (2012b); Andreoni et al. (2015); Andreoni et al. (2018); Angerer et al. (2015); Ashton (2015); Atalay et al. (2014); Augenblick et al. (2015); Aycinena & Rentschler (2018); Aycinena et al. (2019); Aycinena et al. (2022); Balakrishnan et al. (2020); Banerji et al. (2018); Barcellos & Carvalho (2014); Bartoš et al. (2021); Bechtel et al. (2024); Belot et al. (2025); Blumenstock et al. (2018); Boonmanunt et al. (2020); Bousquet (2016); Bover et al. (2018); Brocas et al. (2018); Carvalho et al. (2016a); Carvalho et al. (2016b); Cerrone & Lades (2017); Chakraborty et al. (2017); Chen & Zhao (2024); Chen et al. (2019); Cheung (2015); Cheung et al. (2022); Clot et al. (2017); Dantas et al. (2022); de Quidt et al. (2018); Dertwinkel-Kalt et al. (2022); Ersoy (2017); Esopo et al. (2018); Franco & Mahadevan (2017); Gerhardt & Suchy (2024); Giné et al. (2018); Grijalva et al. (2018); Hvide & Lee (2016); Janssens et al. (2017); Kilcullen et al. (2025); Kuhn et al. (2017); Kuroishi & Sawada (2024); Lindner & Rose (2017); Liu et al. (2014); Liu et al. (2025); Lührmann et al. (2018); Mayer et al. (2015); Mudzingiri et al. (2018); Penczynski & Santana (2024); Potters et al. (2016); Prissé (2023); Prissé & Jorrat (2025); Sawada et al. (2025); Sconti et al. (2024); Sengupta & Savani (2022); Slonim et al. (2013); Stango & Zinman (2026); Stango et al. (2017); Sutter et al. (2018); Yamamoto et al. (2025)
Nonconcave or nonconvex (5)	Chen et al. (2023); Güth et al. (2009); Imai & Camerer (2018); Lanier et al. (2024); Sun & Potters (2022)
MMA (3)	Amankwa-Poku & Kijima (2026); Bigoni et al. (2021); Kiessling et al. (2026)
Portfolio and asset choice ($N = 98$)	
Convex/concave (73)	Ackermann et al. (2013); Ackermann et al. (2016); Ackert et al. (2016); Ahn et al. (2014); Armona et al. (2019); Baltussen & Post (2011); Beaud & Willinger (2015); Bellemare et al. (2005); Bellemare et al. (2022); Benartzi & Thaler (2001); Beshears et al. (2017); Borsboom et al. (2024); Breunig et al. (2021); Brocas et al. (2019b); Brocas et al. (2026); Burghart et al. (2020); Buyukboyaci et al. (2019); Cardella et al. (2023); Carvalho & Silverman (2024); Castillo et al. (2017); Cettolin et al. (2020); Charness & Gneezy (2010); Choi et al. (2007a); Choi et al. (2007b); Choi et al. (2014); Dasgupta et al. (2019); de Castro et al. (2022); Dijk (2017); Dijk et al. (2014); Dlugosch et al. (2014); Eyster & Weizsäcker (2016); Fellner (2004); Fellner & Sutter (2009); Filiz et al. (2018); Filiz et al. (2022); Garagnani (2023); Georgalos (2019); Georgalos (2021); Gneezy & Potters (1997); Haigh & List (2005); Haisley et al. (2011); Halevy et al. (2018); Hammond & Traub (2012); Hayes & O'Brien (2021); Holden & Tilahun (2022); Hueber & Schwaiger (2022); Jia & McMahon (2020); Kallir & Sonsino (2006); Kirchler et al. (2018); Kroll & Levy (1992); Kroll et al. (1988); Langer & Weber (2008); Levy & Levy (2021); Lian et al. (2019); Lim et al. (2021); Lindner et al. (2021); Looney & Hardin (2009); Lucarelli et al. (2026); Magnani et al. (2022); Mol et al. (2022); Morrin et al. (2012); Nov & Su (2018); Papadovasilaki et al. (2018); Permana (2020a); Rapoport et al. (1988); Ridinger et al. (2026); Safford et al. (2018); Schwaiger et al. (2026); Sundali & Guerrero (2006); Sundali et al. (2012); Thaler et al. (1997); Tien & Miao (2013); Yamashita et al. (2008)
Nonconcave or nonconvex (14)	Banerjee & Das (2018); Doerrenberg et al. (2014); Ewing et al. (2004); Feldman & Rehbeck (2022); Fellner et al. (2006); Friedman et al. (2022); Glätzle-Rützler et al. (2013); Halevy & Mayraz (2020); Lambert et al. (2012); Laudenbach et al. (2023); Ponti & Tomás (2021); Schwaiger & Hueber (2021);

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Papers by domain and strongest identifiable mixture attitude (continued)

Attitude (n)	Papers
	Sutter (2007); Zrill (2021)
MMA (1)	Shavit & Rosenboim (2011)
Charitable giving ($N = 111$)	
Convex/concave (91)	Abraham et al. (2025); Adena & Huck (2017); Blanco et al. (2012); Blumenthal et al. (2012); Brañas-Garza et al. (2021); Brown et al. (2017); Butera & Horn (2020); Cappelen et al. (2017); Carpenter (2018); Caviola & Greene (2023); Chan & Feldman (2025); Chao (2017); Charness & Holder (2019); Chatterjee et al. (2020); Chlaß et al. (2023); Cho et al. (2023); Chugunova et al. (2025); Conte (2019); Corniciuc et al. (2026); Crumpler & Grossman (2008); Cryder & Loewenstein (2012); d'Albis et al. (2020); Davis (2006); Deb et al. (2014); Deck & Murphy (2019); Diederich et al. (2022); Drouvelis et al. (2026); Eckel & Grossman (2008); Eckel et al. (2005); Eckel et al. (2007); Eckel et al. (2020); Ekici & Shiri (2018); Esterzon et al. (2023); Feherova et al. (2022); Feicht et al. (2016); Feng et al. (2017); Fielding & Knowles (2015); Fielding et al. (2022); Filiz-Ozbay & Uler (2019); Fleiss et al. (2020); Gee & Schreck (2018); Grimalda et al. (2021); Gugenishvili & Nyström (2024); Haruvy et al. (2018); Hauge et al. (2016); Heger et al. (2021); Huber et al. (2011); Huck & Rasul (2011); Hysenbelli et al. (2013); Ibanez & Roussel (2021); Jin et al. (2021); Jung et al. (2017); Karlan & List (2007); Klinowski (2021); Lane (2021); Lilley & Slonim (2014); Luccasen & Grossman (2013); Luccasen & Thomas (2020); Montano-Campos & Perez-Truglia (2019); Moon & Vanepps (2023); Müller & Rau (2020); Newman & Shen (2012); Null (2011); Peng & Liu (2020); Rondeau & List (2008); Sabato (2025); Sadoff & Samek (2019); Sasaki et al. (2022); Sánchez (2022); Schmitz (2021); Schons et al. (2017); Seo et al. (2023); Septianto & Paramita (2021); Sharps & Schroeder (2019); Sheremeta & Uler (2021); Shreedhar & Mourato (2019); Simpson et al. (2018); Sinha (2021); Smith et al. (2016); Sudbury & Vossler (2022); Sussman et al. (2015); Tonin & Vlassopoulos (2013); Tonin & Vlassopoulos (2014); van Teunenbroek et al. (2021); Vieira et al. (2025); Weisz & Cikara (2025); Wu & Eftekhar (2024); Xu et al. (2023); Zhang et al. (2026); Zheng et al. (2024); Zylbersztejn et al. (2024)
Nonconcave or nonconvex (6)	Andreoni (2007); Andreoni & Miller (2002); Andreoni & Vesterlund (2001); Eckel & Grossman (2003); Eckel & Grossman (2004); Eckel & Grossman (2006)
WNB/SNB (2)	Harbaugh et al. (2007); Zhang et al. (2021)
Effort allocation ($N = 57$)	
Convex/concave (26)	Abebe et al. (2021); Andreoni et al. (2023); Augenblick & Rabin (2019); Augenblick et al. (2015); Bartoš et al. (2021); Breig et al. (2020); Cerrone et al. (2026); Chakraborty et al. (2017); Corbett (2016); Dal Forno & Merlone (2021); Dertwinkel-Kalt et al. (2022); Fedyk (2024); Freeman & Laughren (2024); He (2019); Hiemstra et al. (2019); Hur et al. (2021); Imas et al. (2022); Kantarci et al. (2025); Kool & Botvinick (2014); Kölle & Wenner (2023); Lévesque & Schade (2005); Lipman et al. (2026); Marrocoli et al. (2018); Reddinger (2024); Vladasel et al. (2024); Yamamoto et al. (2025)
MMA (1)	Bigoni et al. (2021)
Social/interpersonal allocation ($N = 189$)	
Convex/concave (147)	Abrahao et al. (2017); Ackert et al. (2011); Alevy et al. (2014); Angarita & Viciana (2022); Antinyan et al. (2024); Apicella et al. (2010); Auriol et al. (2020); Aycinena et al. (2026); Azar et al. (2015); Banker et al. (2017); Bao & Yu (2019); Barends et al. (2019); Becchetti et al. (2017); Begum et al. (2018); Bellemare et al. (2008); Ben-Ner et al. (2004); Ben-Ner et al. (2009); Benistant & Suchon (2021); Bezu & Holden (2015); Bilancini et al. (2020); Bilancini et al. (2023); Bjorvatn et al. (2020); Bolton et al. (1998); Bonan et al. (2023); Brañas-Garza et al. (2017); Brañas-Garza et al. (2026); Brock et al. (2013); Brook & Servátka (2016); Brosig-Koch et al. (2017); Bruhin et al. (2019); Cappelen et al. (2007); Cappelen et al. (2013); Carbó-Valverde et al. (2025); Carlsson et al. (2017); Cason & Mui (1997); Chapman et al. (2023); Charness (2000); Chiou et al. (2014); Chisadza et al. (2021); Choi et al. (2020); Chowdhury & Jeon (2014); Clot et al. (2013); Cochard et al. (2018); Dasgupta (2011); Dasgupta et al. (2020); De Arcangelis & Joxhe (2021); Demel et al. (2019); Dorin et al. (2021);

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Papers by domain and strongest identifiable mixture attitude (continued)

Attitude (n)	Papers
	Duch et al. (2015); Dutcher et al. (2024); Eckel et al. (2024); Ellingsen & Johannesson (2008); Ellis & Freeman (2024); Engelmann & Strobel (2004); Faillo et al. (2019); Feng et al. (2013); Forgas (2016); Fowler (2006); Franzen & Pointner (2014); Frohlich et al. (2004); Fromell et al. (2019); Fujii et al. (2015); Fukui & Toyoshima (2014); Gnagey et al. (2018); Godos-Díez et al. (2019); Goeree et al. (2010); Gomes & McCullough (2015); Gueth et al. (2014); Hayashi (2013); He et al. (2024); Heinrich & Weimann (2013); Heinz et al. (2012); Heinz et al. (2016); Hepp et al. (2022); Hoel (2015); Holm & Danielson (2005); Howe et al. (2023); Huang et al. (2023); Ioannou et al. (2015); Israel et al. (2009); Jakiela (2015); Jakiela et al. (2015); Julle-Danière et al. (2020); Kanagaretnam et al. (2009); Kawamura et al. (2025); Kim & Kim (2019); Kittel et al. (2020); Kleine et al. (2016); Klinowski (2018); Koenigs et al. (2010); Korenok et al. (2012); Koukoumelis et al. (2026); Kölle & Wenner (2023); Lang et al. (2019); Li et al. (2019); Li et al. (2022a); Liebe et al. (2019); Liebe et al. (2022); Lindová et al. (2010); List & Cherry (2008); Ludwig et al. (2025); Marini et al. (2025); Matarazzo et al. (2020); McNamara (2020); Monroe et al. (2017); Moore & Taylor (2007); Mosleh et al. (2020); My et al. (2018); Nettle et al. (2011); Noussair & Stoop (2015); Oberholzer-Gee & Eichenberger (2008); Peysakhovich et al. (2014); Potters & Xu (2020); Robbins et al. (2023); Rode & Le Menestrel (2011); Rodriguez-Lara (2015); Ross et al. (2024); Ru et al. (2022); Schulz et al. (2014); Schwaiger et al. (2022); Shariff & Norenzayan (2007); Sharma (2015); Sharps & Schroeder (2019); Sircar et al. (2018); Sun et al. (2020); Takagi & Terada (2021); Takahashi (2025); Thomae et al. (2013); Tinghög et al. (2017); Traub et al. (2023); Tutic & Liebe (2020); Uziel et al. (2020); van Dillen et al. (2021); Vladasel et al. (2024); Wang & Liu (2024); Wang et al. (2022); Wang et al. (2024); Whitt & Wilson (2007); Windrich et al. (2022); Wu et al. (2019); Xiao & Houser (2009); Yang et al. (2025); Yoo et al. (2026); Zhang et al. (2019); Zhang et al. (2026); Zhu et al. (2013); Ziegler (2021)
Nonconcave or nonconvex (20)	Andreoni (2007); Andreoni & Miller (2002); Andreoni & Vesterlund (2001); Efferson et al. (2007); Erkut (2022); Fisman et al. (2015a); Fisman et al. (2015b); Fisman et al. (2017); Gantner & Kerschbamer (2018); Hugh-Jones et al. (2011); Jakiela (2013); Kerschbamer (2015); Leider et al. (2010); Li et al. (2017); Molho et al. (2025); Murphy et al. (2011); Müller (2019); Porter & Adams (2016); Rigdon & Levine (2018); Visser & Roelofs (2011)
MMA (4)	Crawford & Harris (2018); Epper et al. (2024); Fehr et al. (2023); Fehr et al. (2026)
WNB/SNB (1)	Fisman et al. (2007)

OA.3.1 Papers without local mixture comparisons

These 94 paper–domain entries do not establish even local weak convexity or concavity in the domain assessed. In some designs, the relevant endpoints and an exact interior mixture are unavailable or insufficiently documented. In others, valuations come from different participants or contexts, observed outcomes reflect an evolving process, or the measures concern beliefs, norms or performance rather than preference over the required alternatives. Reviews, theoretical papers and methodological companions may also discuss relevant experiments without supplying a separate elicitation. We retain these papers as adjacent literature and sources for the search. The classification is specific to the domain: a paper listed here may provide qualifying comparisons in another domain.

Table OA.3: Reviewed papers without a local convex or concave comparison in the listed domain (94 entries).

Domain	N	Papers
Randomization and lotteries	6	Keren & Willemsen (2009); Mislavsky & Simonsohn (2018); Moon & Nelson (2020); Newman & Mochon (2012); Paolacci & André (2024); Simonsohn (2009)
Consumption budgets	8	Banerjee & Murphy (2006); Cherchye et al. (2024); Drichoutis & Nayga (2020); Février & Visser (2004); Freer & Castillo (2023); Gerasimou (2026); Huet-Vaughn et al. (2024); Tien & Miao (2013)
Time preferences	11	Batista et al. (2015); Bulte et al. (2016); Halevy et al. (2018); Hoel et al. (2022); Imai et al. (2021); Imas et al. (2022); Inukai et al. (2024); Miao & Zhong (2015); Reddinger (2024); Werthschulte (2023); Wuepper et al. (2023)
Portfolio and asset choice	10	Dahan et al. (2011); Di Cagno et al. (2018); Johnson & Staveley-O'Carroll (2020); Johnson & Tuckett (2022); Klos et al. (2005); Levy (1994); Mol et al. (2024); Pitkäjärvi & Vacca (2025); Polisson et al. (2020); Rötheli (1997)
Charitable giving	12	Daniels & Valdés (2021); Dubé et al. (2017); Hatsumi (2013); Kassirer et al. (2023); Kesternich et al. (2016); Köbrich Leon & Schobin (2023); Murphy et al. (2023); Pan & Houser (2011); Tremblay-Boire (2025); Vu et al. (2024); Wiepking & Heijnen (2011); Wiss et al. (2015)
Effort allocation	30	Ariel et al. (2009); Arnold et al. (2025); Blouzard et al. (2025); Bustamante et al. (2023); Cerrone & Lades (2017); Cherchye et al. (2024); Cochard et al. (2018); Dutcher et al. (2024); D'Exelle et al. (2024); Ersoy (2021); Farmer et al. (2018); Fisher et al. (2024); Hannan et al. (2013); Jakiela (2015); Kollins et al. (1997); Lange & Rieger (2025); Lewis et al. (2023); Meyniel et al. (2014); Murad et al. (2019); Orland & Rostam-Afschar (2021); Payne et al. (2007); Phillips & Schmidt (2016); Rattini (2023); Sun & Potters (2022); Suzuki et al. (2021); Takahashi et al. (2016); Werthschulte (2023); Wilkinson et al. (2012); Wu et al. (2009); Zhang et al. (2010)
Social/interpersonal allocation	17	Aguiar et al. (2008); Avinun et al. (2011); Charness & Rabin (2002); Filippin & Guala (2017); Gambetta & Przepiorka (2014); Haesevoets et al. (2013); Haruvy et al. (2017); Kovacs et al. (2023); Li et al. (2022b); Long et al. (2025); Lönnqvist & Walkowitz (2019); Niszczota & Antoniou (2026); Osório (2017); Reed et al. (2019); Rramani et al. (2020); Russell & Cain (2022); Shoji et al. (2020)

OA.4 Archived searches

OA.4.1 Database queries

The strings below reproduce the archived Web of Science searches. `TS` denotes Topic and `PY` the publication-year range. Counts precede deduplication; E7–E8 span several domains. R1–R7 and E1–E11 identify the archived searches.

Randomization and lotteries

R1: Explicit preference for randomization; 21 hits.

```
TS=(("preference* for randomi?ation" OR "prefer* randomi?ation" OR "paying for
↪  randomi?ation")
AND (experiment* OR empirical OR laborator* OR field))
AND PY=1900-2026
```

R2: Mixture preference under risk; 4 hits.

```
TS=(("preference* for mixture*" OR "preference* for mixing" OR "mixture choice" OR
↪  "mixture loving" OR "mixture aversion")
AND (lotter* OR gamble* OR risk OR uncertain*)
AND (experiment* OR empirical OR laborator*))
AND PY=1900-2026
```

R3: Deliberate stochastic choice; 77 hits.

```
TS=(("deliberate randomi?ation" OR "deliberate stochastic choice" OR "consciously
↪  stochastic" OR "randomi?ation advice" OR "choice frequenc*")
AND (choice OR preference* OR lotter* OR risk OR ambiguity)
AND (experiment* OR empirical OR laborator* OR field))
AND PY=1900-2026
```

R4: Betweenness and probability-triangle experiments; 153 hits.

```
TS=((betweenness OR "Marschak-Machina" OR "probability triangle" OR "unit triangle" OR
↪  "fanning out" OR "fanning in")
AND (lotter* OR gamble* OR risk OR uncertain* OR "convex budget*")
AND (experiment* OR empirical OR laborator*))
AND PY=1900-2026
```

R5: Random-lottery incentive systems; 8 hits.

```
TS=(("random lottery incentive system" OR "random-lottery incentive system" OR "random
↪  incentive system" OR "one task randomly selected")
AND (lotter* OR risk OR choice OR preference*)
AND (experiment* OR valid* OR empirical OR distort*))
AND PY=1900-2026
```

R6: Procedural randomization and delegation; 104 hits.

```
TS=((randomi?ation OR "coin flip*" OR "coin toss*" OR "random device*")
AND (ambiguity OR indecis* OR confidence OR regret OR responsibility OR autonomy)
AND (choice OR preference*)
AND (experiment* OR empirical OR laborator* OR field))
AND PY=1900-2026
```

R7: Independence and connected lottery anomalies; 24 hits.

```
TS=(("independence axiom" OR "certainty independence" OR "reversal of order" OR
↪  "uncertainty effect" OR "common ratio" OR "common consequence")
AND (mixture* OR randomi?ation OR compound*)
AND (lotter* OR gamble* OR risk OR ambiguity)
AND (experiment* OR empirical OR laborator*))
AND PY=1900-2026
```

Consumption budgets

E3: Consumption budgets; 380 hits.

```
TS=(("convex budget*" OR "linear budget*" OR "budget line*" OR "revealed preference*" OR
↪  "consumer choice experiment*")
AND (consumption OR consumer* OR goods OR bundle* OR demand)
AND (experiment* OR laborator* OR incentiv*))
AND PY=1900-2026
```

Time preferences and effort allocation

E1: Time preferences / convex time budgets; 30 hits.

```
TS=(("convex time budget*" OR "intertemporal allocation*" OR "time-dated allocation*" OR
↪  "dated allocation*" OR "allocate* across time" OR ("convex budget*"
AND (sooner OR later)))
AND ("time preference*" OR intertemporal OR discount* OR sooner OR later)
AND (experiment* OR laborator* OR incentiv*))
AND PY=1900-2026
```

E2: Effort allocation; 328 hits.

```
TS=(("effort allocation*" OR "allocate* effort" OR "work allocation*" OR "task
↪  allocation*" OR "real effort" OR "effort budget*")
AND (time OR date* OR task* OR intertemporal)
AND (choice OR preference* OR discount* OR commitment OR "present bias")
AND (experiment* OR laborator* OR field))
AND PY=1900-2026
```

E11: Time/effort sensitivity; 515 hits.

```
TS=(("time allocation*" OR "allocate* time" OR "effort allocation*" OR "allocate*
↪  effort" OR "multi-task setting*" OR "multitask* effort")
AND (task* OR work OR labor OR leisure)
AND (experiment* OR laborator* OR incentiv*))
AND PY=1900-2026
```

Portfolio and asset choice

E4: Portfolio allocation; 559 hits.

```
TS=(("portfolio allocation*" OR "asset allocation*" OR "investment allocation*" OR
↪  "portfolio choice*" OR "Arrow securit*" OR "state-contingent claim*" OR "state
↪  claim*")
AND (experiment* OR laborator* OR incentiv*)
AND (risk OR ambiguity OR diversification OR preference*))
AND PY=1900-2026
```

E5: Convex asset budgets; 8 hits.

```
TS=(("convex budget*" OR "linear budget*" OR "allocation task*")
AND (portfolio* OR investment* OR "risky asset*" OR "safe asset*" OR "Arrow securit*" OR
↪  "state claim*")
AND (experiment* OR laborator*))
AND PY=1900-2026
```

E10: Investment-task sensitivity; 51 hits.

```
TS=(("investment game*" OR "risky investment task*" OR "portfolio experiment*" OR "asset
↪  allocation experiment*")
AND (allocat* OR invest* OR fraction OR amount OR share*)
AND ("safe asset*" OR "risky asset*" OR risk OR ambiguity)
AND (experiment* OR laborator*))
AND PY=1900-2026
```

Giving and social allocation

E6: Giving convex budgets; 183 hits.

```
TS=(("convex budget*" OR "linear budget*" OR "budget allocation*" OR "allocation game*"
↪  OR "generalized dictator game*" OR "modified dictator game*")
AND ("social preference*" OR altruism OR giving OR charitable OR charity OR donation* OR
↪  generosity OR recipient*)
AND (experiment* OR laborator* OR field))
AND PY=1900-2026
```

E9: Giving allocation sensitivity; 553 hits.

```
TS=(("dictator game*" OR "charitable giving" OR "donation decision*" OR "donation
↪  allocation*")
AND (continuous OR amount* OR allocat* OR split* OR budget* OR price*)
AND (experiment* OR laborator* OR field))
AND PY=1900-2026
```

Shared searches across allocation domains

E7: Diversification / splitting; 3 hits.

```
TS=(("preference* for diversification" OR "demand for diversification" OR "preference*
↪  for splitting" OR "split allocation*" OR "interior allocation*" OR "corner
↪  allocation*")
AND (consumption OR portfolio* OR asset* OR time OR effort OR charity OR donation* OR
↪  social)
AND (experiment* OR empirical OR laborator*))
AND PY=1900-2026
```

E8: Revealed-preference allocations; 44 hits.

```
TS=(("revealed preference*"
AND (experiment* OR laborator* OR incentiv*)
AND (allocat* OR budget*)
AND (time OR intertemporal OR consumption OR portfolio OR asset* OR social OR altruism
↪  OR giving OR effort)))
AND PY=1900-2026
```

OA.4.2 Supplementary update searches

The archive records these five supplementary routes on 27 August 2026. They are search descriptions, not Web of Science strings: U1–U4 target 2024–2026 publications, while U5 follows citations across domains. No hit counts were recorded.

U1: Time/effort update.

```
"convex time budget" OR "intertemporal allocation" OR "effort allocation"; publication
↪  years 2024--2026; targeted scholarly web and citation-chain update run 2026-08-27;
↪  hits not enumerated
```

U2: Consumption-budget update.

```
"convex budget" experiment consumption OR goods OR competition; publication years
↪  2024--2026; targeted scholarly web update run 2026-08-27; hits not enumerated
```

U3: Asset/portfolio update.

```
"portfolio allocation" experiment OR "asset allocation" convex budget; publication years
↪  2024--2026; targeted scholarly web update run 2026-08-27; hits not enumerated
```

U4: Giving/social update.

```
"social preferences" budget line OR interpersonal allocation OR charitable allocation;
↪  publication years 2024--2026; targeted scholarly web update run 2026-08-27; hits not
↪  enumerated
```

U5: Cross-domain citation update.

```
Forward/backward citation checks from recent CTB, real-effort, social-budget, and
↪  portfolio papers; update run 2026-08-27; hits not enumerated
```

OA.4.3 Supplementary searches: charitable giving

The giving search began with 128 candidates and added ten papers through citations and the queries below. Two archived records referred to the same publication (Eckel et al. (2007)) and were consolidated. Citation searches used Clochard–Dey–Sasaki–Imai's July 2026 review and the primary studies' references. These supplementary queries have no recorded hit counts or separate search dates.

```
"Clochard" "charitable" meta analysis
"donation" "bundling" "preferences" experiment charity diversification
"charitable giving" "mixture" experiment
"Self-Signaling and Prosocial Behavior" pdf
"Robbing Peter to Pay Paul" charitable giving pdf
"The shape of warm glow" pdf
"Charitable giving and intermediation" pdf
"Karlan" "List" "2007" "Does Price Matter" pdf
"Huck" "Rasul" "2011" "matched fundraising" pdf
"Eckel" "Grossman" "2008" "Subsidizing charitable contributions" pdf
"Harbaugh" "Mayr" "Burghart" "2007" pdf
```

Sharps & Schroeder (2019) and Weisz & Cikara (2025) were found in the references of Caviola & Greene (2023).

OA.4.4 Supplementary searches: effort allocation

The initial 52 effort candidates were supplemented by 15 papers found through these queries, citation searches and checks for effort tasks in the time-preference literature. Separate query dates and hit counts are not recorded.

```
experiment "effort allocation" "convex" tasks
"real effort" "convex time budget" experiment
"effort allocation" "mixture" experiment
"effort allocation" "preferences" allocation task experiment
experiment "effort allocation" "convex time" -site:researchgate.net
experiment "effort allocation" "indifference" -site:researchgate.net
"Augenblick" "Rabin" "2019" pdf effort
"Fedyk" "2024" "effort" pdf
"Effort Foraging Task" pdf
"Women compared with men work harder for small rewards" pdf
```

OA.4.5 Supplementary searches: social allocation

The database and shared allocation searches supplied 237 social-allocation candidates. We supplemented them through references, checks against the other domains and these queries. Separate query dates and hit counts are not recorded.

```
"social preferences" "convex" "allocation" experiment pdf
"distributional preferences" "nonconvex" experiment pdf
"interpersonal allocation" experiment "budget" pdf
"social preferences" "convex budgets" experiment
"Measuring Social Value Orientation" Murphy Ackermann Handgraaf pdf
"geometry of distributional preferences" pdf Kerschbamer
"Inequality aversion, efficiency, and maximin" experiment pdf
"Many Faces of Human Sociality" pdf Bruhin Fehr Schunk
```

```
"social preferences of future physicians" pdf
"distributional preferences of an elite" pdf
"pluralism of fairness ideals" pdf
"Econographics" pdf 2023
"Just Luck" Cappelen pdf 2013
"Measuring Inequity Aversion in a Heterogeneous Population" pdf
"Social interaction effects" "distributional" pdf
"Engelmann" "Strobel" "2004" "pdf" inequality aversion
"Kerschbamer" "geometry" pdf 2015
"Sharing the cake during a crisis" pdf
"Econographics" "pdf" Chapman Ortoleva
"Engelmann" "Strobel" "2004" filetype:pdf
"geometry of distributional preferences" filetype:pdf
"Econographics" filetype:pdf
"Measuring Inequity Aversion" Bellemare Kroger 2008 pdf
"Experimental Evidence of Physician Social Preferences" pdf
"Sharing the cake during a crisis" filetype:pdf
```

These searches added 24 papers: seven already reviewed in giving or effort and 17 further social-allocation papers. Exact-title searches located the articles and their working-paper versions.

Theory Appendix

This appendix establishes the model classifications in the manuscript. Mixtures are probability mixtures of lotteries under risk and statewise mixtures of Anscombe–Aumann acts under uncertainty.

OA.5 Models under risk

OA.5.1 Quadratic Utility and Mixture symmetry

We showed in the main text that quadratic utility can only display MMA but not WNB and SNB. Under the Basic Axioms, mixture symmetry is equivalent to

$$p \sim q \quad \Longrightarrow \quad (1-t)p + tq \sim tp + (1-t)q \qquad (t \in [0,1])$$

(Chew, Epstein and Segal, 1991, Theorem 1). It rules out SNB.

To see this, suppose an indifferent chord displays SNB and write its utility as $f(t)$. Its maximum is above $f(0) = f(1)$ and its minimum below. Symmetry gives $f(t) = f(1-t)$, so both extrema first occur in $(0, 1/2]$. Let a be the later of these first occurrences. Between the two, continuity gives $r \in (0, a)$ with $f(r) = f(0)$. Symmetry gives $f(1-r) = f(0)$ too. Apply mixture symmetry to the indifferent subchord $[0, 1-r]$ and then to the original chord:

$$f(a-r) = f(1-a) = f(a).$$

Since $0 < a - r < a$, the extremum first reached at a was reached earlier, a contradiction.

Quadratic utility satisfies mixture symmetry, so the preceding example gives MMA without WNB. WNB without SNB is possible as well. On a three-prize triangle, take

$$V(p_0, p_1) = \begin{cases} p_1^2 - p_0^2, & p_1 \le p_0, \\ (p_1 - p_0)(6 - 7p_0 - 3p_1), & p_1 \ge p_0. \end{cases}$$

The pieces agree at zero on $p_1 = p_0$. Each strictly decreases in p_0 and strictly increases in p_1 on its region, so the utility is continuous and strictly respects stochastic dominance. Its sign is the sign of $p_1 - p_0$. Thus indifferent endpoints lie in the same half-triangle, where utility is quadratic, or both lie on the zero line. In either case, complementary mixtures are indifferent, establishing mixture symmetry.

For $x = (2/5, 3/5)$ and $y = (1/5, 0)$,

$$V(x) = \frac{7}{25}, \qquad V(y) = -\frac{1}{25},$$
$$V\left(\frac{31x + y}{32}\right) = \frac{9}{32} > V(x), \qquad V\left(\frac{x + 7y}{8}\right) = -\frac{9}{200} < V(y).$$

Hence WNB holds, while mixture symmetry rules out SNB.

OA.5.2 Rank-dependent expected utility

For ordered prize utilities $u_1 < \cdots < u_n$, rank-dependent expected utility (Quiggin, 1982) is

$$V(p) = u_1 + \sum_{i=2}^{n} (u_i - u_{i-1}) w\big(\Pr_p(z \geq z_i)\big),$$

where w is continuous and strictly increasing, with $w(0) = 0$ and $w(1) = 1$. Cumulative probabilities are affine in lottery mixtures. Thus a convex or concave w makes mixture attitudes globally one-sided, ruling out MMA.

With three prizes and utilities $(0, c, 1)$, let p_0, p_1 be the worst- and best-outcome probabilities. Utility becomes

$$cw(1 - p_0) + (1 - c)w(p_1), \qquad p_0, p_1 \geq 0, \quad p_0 + p_1 \leq 1.$$

We use these Machina coordinates throughout the RDEU proofs.

Three prizes: MMA without WNB. Take $c = 1/2$ and $w(p) = (1 - \cos(\pi p))/2$. Utility is

$$V(p_0, p_1) = \frac{1}{2} + \frac{\cos(\pi p_0) - \cos(\pi p_1)}{4}.$$

An indifference graph $p_1 = g(p_0)$ at level v satisfies $\operatorname{sgn} g'' = \operatorname{sgn}(v - 1/2)$. Lower curves are concave, upper curves convex, and the middle curve $p_0 = p_1$ is straight. Thus there are strictly mixture-averse and strictly mixture-loving indifferent pairs, giving MMA. Every indifference curve bends in only one direction, so SNB is absent. The three-prize restriction proved in Section OA.7 then rules out WNB as well.

Four or more prizes.

Proposition OA.1. *With at least four strictly ordered prize utilities,*

$$\text{MMA} \iff \text{WNB} \iff \text{SNB} \iff w \textit{ is neither convex nor concave.}$$

With three prizes, WNB still implies SNB, although MMA need not.

Section OA.7 proves the result. The two middle prizes give different utility weights on two faces sharing the same binary edge. Avoiding SNB requires neutral chords on both faces that a weighting function with mixed curvature cannot support. The four-prize face remains available in every larger prize space.

Prelec weighting. For the two-parameter family (Prelec, 1998),

$$w(p) = \exp\{-\beta(-\log p)^\alpha\}, \quad p > 0, \qquad w(0) = 0, \qquad \alpha, \beta > 0,$$

MMA is equivalent to SNB already with three prizes. If $\alpha = 1$, then $w(p) = p^\beta$, so MMA is absent. If $\alpha \neq 1$, SNB holds for every triple of distinct prize utilities.

Proof. For $\alpha \neq 1$, w'' has opposite signs near zero and one. Choosing indifferent lotteries with

both cumulative probabilities in each of these regions gives opposite strict mixture attitudes. If SNB were absent, the geometric argument in Section OA.7 would give a neutral line.

Along this line, $w'(p_1)/w'(1-p_0)$ is a positive constant. Analyticity extends the neutrality identity throughout the line's intersection with the open unit square. Follow it towards $p_0 = 0$ or $p_1 = 0$. For $\alpha < 1$, w' tends to infinity at both probability endpoints; for $\alpha > 1$, it tends to zero at both. Thus the line cannot reach just one of these boundaries. If it reaches both, write $p_0 = kp_1$, with $k > 0$. Then

$$\log \frac{w'(p_1)}{w'(1-kp_1)} = \alpha(-\log p_1) - \beta(-\log p_1)^{\alpha} + O\big(\log(-\log p_1)\big).$$

This diverges as $p_1 \downarrow 0$, contradicting the constant ratio. The resulting three-prize SNB witness embeds in any larger prize space. □

OA.5.3 Cumulative prospect theory

We allow continuous, strictly increasing gain and loss weighting functions with endpoints zero and one, without imposing an inverse-S shape (Tversky and Kahneman, 1992). For the first two examples, take a fixed reference point and prize values $(-1, 0, 1)$. With loss probability p_0 and gain probability p_1,

$$V(p_0, p_1) = w^+(p_1) - w^-(p_0), \qquad p_0, p_1 \geq 0, \quad p_0 + p_1 \leq 1.$$

These restrictions ensure continuity and strict stochastic dominance.

MMA without WNB. Take $w^+(p) = w^-(p) = p^2$. This gives the quadratic utility $p_1^2 - p_0^2$ above, which displays MMA but cannot display WNB.

WNB without SNB. Take the same weighting function for gains and losses:

$$w(p) = \frac{\arctan(8p-2) + \arctan 2}{\arctan 6 + \arctan 2}.$$

Suppressing the positive normalizing factor, utility is $V(p_0, p_1) = \arctan(8p_1-2) - \arctan(8p_0-2)$. For $x = (3/8, 1/2)$ and $y = (1/8, 0)$,

$$V(x) = \arctan(1/3) = -V(y) > 0, \qquad V\big(\tfrac{5}{6}x + \tfrac{1}{6}y\big) = \arctan(6/17) > V(x).$$

The complementary mixture has the negative of this last value, so it is worse than y. Thus WNB holds.

For absence of SNB, write $u = 8p_1 - 2$ and $v = 2 - 8p_0$. Utility is $\arctan u + \arctan v$, and along an indifference curve,

$$\frac{d^2v}{du^2} = \frac{2(u+v)(1+v^2)}{(1+u^2)^2}.$$

The sign of $u + v$ is the sign of utility. Every positive-level curve is therefore convex, every negative-level curve concave, and the zero-level curve straight. No indifferent chord can

have mixtures on both sides of its curve, so SNB is absent. We use the same fact for smooth ambiguity below.

SNB. On three prizes on the gain side of the reference point, CPT reduces to RDEU. The three-prize Prelec example therefore gives SNB. Hence unrestricted CPT permits all three exclusive patterns.

OA.5.4 Bell-type linear disappointment

Bell's mechanism compares each outcome with the lottery's own expectation. We use its linear extension to finite lotteries (Bell, 1985; Jia, Dyer and Butler, 2001):

$$V(p) = \mu(p) - \kappa \sum_i p_i(\mu(p) - u_i)_+, \qquad \mu(p) = \sum_i p_i u_i, \qquad 0 \le \kappa < 1,$$

where $(r)_+ = \max\{r, 0\}$. The second term is expected disappointment. The following restriction concerns this linear specification.

With at least three distinct prizes, every $\kappa > 0$ gives SNB. Normalize three prize utilities to $(0, a, 1)$. In the affine coordinates (μ, p_0), an indifference curve of value c is

$$p_0 = \begin{cases} (1 - c/\mu)/\kappa, & \mu \le a, \\ 1 - \mu/a + \dfrac{(1-a)(\mu - c)}{a\kappa(1-\mu)}, & \mu \ge a. \end{cases}$$

Choose an interior lottery with mean a. Its value satisfies $0 < c < a$, and its indifference curve crosses $\mu = a$ in the interior. Below this mean, only the lowest prize disappoints; above it, the middle prize disappoints too. The first graph is strictly concave and the second strictly convex. The curve is therefore neither convex nor concave, and a crossing secant gives SNB. At $\kappa = 0$, preferences are expected utility. Thus neither of the two milder exclusive patterns is possible.

For a numerical example, take utilities $(0, 1/2, 1)$, $\kappa = 3/4$, and lotteries $p = (2/9, 2/3, 1/9)$ and $q = (4/9, 0, 5/9)$. Then

$$V(p) = V(q) = \frac{10}{27}, \qquad V\left(\frac{p+q}{2}\right) = \frac{3}{8} > \frac{10}{27} > V\left(\frac{p+4q}{5}\right) = \frac{37}{100}.$$

Continuity is immediate. To check strict dominance, couple a lottery with a dominating one whose mean is higher by $\delta > 0$. Expected shortfall can increase by at most δ, because the positive-part function is increasing and 1-Lipschitz. Utility therefore rises by at least $(1-\kappa)\delta > 0$.

OA.6 Models under uncertainty

OA.6.1 α-maxmin, Choquet, and invariant biseparable preferences

These models satisfy certainty independence (Schmeidler, 1989; Ghirardato, Maccheroni and Marinacci, 2004). Normalize the worst and best prize utilities to zero and one, and identify an Anscombe–Aumann act with its statewise expected utilities $x \in [0,1]^d$. Every point in this cube is attainable, and mixing acts mixes their statewise utilities. The constant act $n = (1/2, \ldots, 1/2)$ is therefore an interior neutral benchmark. The paper's neutral-benchmark result gives

$$\text{MMA} \iff \text{WNB} \iff \text{SNB}.$$

Neither of the two milder exclusive patterns is possible.

For an SNB example common to all three classes, take three states and

$$I_*(x) = \frac{1}{2}\bar{x} + \frac{1}{4}\big(\min_i x_i + \max_i x_i\big), \qquad \bar{x} = \frac{x_1 + x_2 + x_3}{3}. \tag{OA.6.1}$$

This utility is continuous and strictly increasing. It is α-maxmin with $\alpha = 1/2$ and prior set

$$P = \operatorname{co}\left\{\frac{1}{2}e_i + \frac{1}{2}(1/3, 1/3, 1/3) : i = 1, 2, 3\right\}.$$

It is also Choquet expected utility: assign capacity $5/12$ to each singleton and $7/12$ to each two-state event, with capacity zero and one on the empty and full events. Hence it is invariant biseparable as well. For $x = (0, 1/2, 1)$ and $y = (1/2, 1, 0)$,

$$I_*(x) = I_*(y) = \frac{1}{2}, \qquad I_*\left(\frac{2}{3}x + \frac{1}{3}y\right) = \frac{11}{24}, \qquad I_*\left(\frac{1}{3}x + \frac{2}{3}y\right) = \frac{13}{24}.$$

The endpoints are indifferent, with one mixture strictly better and another strictly worse.

OA.6.2 Smooth ambiguity

For these examples, normalize statewise expected utilities to $[-1, 1]$. Give equal second-order probability to the d degenerate priors, each concentrated on one state. Smooth ambiguity (Klibanoff, Marinacci and Mukerji, 2005) then evaluates an act by

$$V(x) = \frac{1}{d}\sum_{i=1}^{d} \phi(x_i). \tag{OA.6.2}$$

Each choice of ϕ below is smooth and strictly increasing, so the resulting preference satisfies the Basic Axioms.

MMA without WNB. Take two states and $\phi(t) = \arctan t$. The indifferent acts $(1, 0)$ and $(0, 1)$ strictly prefer their midpoint, since ϕ is strictly concave on $[0, 1]$. Their negatives strictly dislike their midpoint, since ϕ is strictly convex on $[-1, 0]$. Thus MMA holds.

To rule out WNB, consider any chord. Comparable endpoints give a monotone chord.

Otherwise, interchange states if necessary and write the line as $x_2 = a - bx_1$, with $0 < b \leq 1$. Its utility derivative has the sign of

$$1 - b + (a - bx_1)^2 - bx_1^2. \tag{OA.6.3}$$

For $b = 1$, this is linear or identically zero. For $b < 1$, it is strictly concave. Two roots in $(-1, 1)$ would make its values at -1 and 1 both negative, but their sum is $2[a^2 + (1-b)^2] \geq 0$. Thus a nonconstant chord has at most one interior stationary point. WNB requires both an interior maximum and an interior minimum, so it is impossible.

WNB without SNB. Keep two states and take $\phi(t) = \arctan(2t)$. Suppressing the positive factor $1/2$ in (OA.6.2), let $x = (1, -1/2)$ and $y = -x$. Then

$$V(x) = \arctan(1/3) > 0, \qquad V\left(\frac{5}{6}x + \frac{1}{6}y\right) = \arctan(6/17) > V(x).$$

Oddness gives $V(y) = -V(x)$ and a complementary mixture strictly worse than y, establishing WNB. As shown in Section OA.5.3, each indifference curve of an arctangent sum is convex, concave, or straight. Every indifferent chord stays on one side of its curve, so SNB is absent.

SNB. Take three states and $\phi(t) = t + t^3$. For $x = (1, 0, -1)$ and $y = (0, -1, 1)$, we have $V(x) = V(y) = 0$ and

$$V\big(\lambda x + (1-\lambda)y\big) = -\lambda(1-\lambda)(1-2\lambda).$$

This is negative at $\lambda = 1/4$ and positive at $\lambda = 3/4$, giving SNB.

These examples use the unrestricted model. A globally concave or convex ϕ makes utility respectively concave or convex in statewise expected utilities, ruling out MMA.

OA.6.3 Variational preferences without uncertainty aversion

Dropping uncertainty aversion from the variational axioms gives (Chandrasekher et al., 2022, Theorem 3)

$$I(x) = \max_{c \in C} \min_{p \in \Delta_d} \{p \cdot x + c(p)\}, \qquad \max_{c \in C} \min_{p} c(p) = 0, \tag{OA.6.4}$$

where each cost is convex. On the bounded domain $[0, 1]^d$, this class permits all three patterns.

MMA without WNB. On $[0, 1]^3$, take

$$I_Q(x) = \frac{x_1 + x_2 + x_3}{3} + \frac{(x_1 - x_3)^2 - (x_2 - x_3)^2}{12}. \tag{OA.6.5}$$

Its coordinate derivatives are at least $1/6$. Moreover,

$$I_Q(1, 1/2, 0) = I_Q(0, 1/2, 1) = 9/16,$$
$$I_Q(1/2, 1, 0) = I_Q(1/2, 0, 1) = 7/16.$$

Both pairs have midpoint $(1/2, 1/2, 1/2)$, with value $1/2$. One pair therefore dislikes its midpoint and the other likes it. Since utility is quadratic, WNB is impossible by Section OA.5.1.

WNB without SNB. On $[0,1]^2$, take

$$I_W(x) = \frac{x_1 + x_2}{2} + \frac{3}{4} \max\left\{-\frac{1}{3}, \min\left\{\frac{x_1 - x_2}{2}, \frac{1}{3}\right\}\right\}. \tag{OA.6.6}$$

This utility is continuous and strictly increasing. For $x = (4/5, 0)$ and $y = (1/5, 1)$,

$$I_W(y) = \frac{7}{20}, \qquad I_W(x) = \frac{13}{20}, \qquad I_W\left(\frac{x + 11y}{12}\right) = \frac{1}{3}, \qquad I_W\left(\frac{11x + y}{12}\right) = \frac{2}{3}.$$

Thus WNB holds. The two kink lines $x_1 - x_2 = \pm 2/3$ have disjoint utility ranges, $[1/12, 5/12]$ and $[7/12, 11/12]$. Each indifference curve meets at most one kink and is therefore convex, concave, or straight, ruling out SNB.

SNB. On $[0,1]^2$, take

$$I_S(x) = \frac{x_1 + x_2}{2} + \frac{1}{4}(x_1 - x_2)^4 - \frac{3}{8}(x_1 - x_2)^2. \tag{OA.6.7}$$

Its coordinate derivatives are at least $1/4$. For $x = (1, 0)$ and $y = (0, 1)$,

$$I_S(x) = I_S(y) = \frac{3}{8}, \qquad I_S\left(\frac{x + y}{2}\right) = \frac{1}{2}, \qquad I_S\left(\frac{7x + y}{8}\right) = \frac{377}{1024} < \frac{3}{8}.$$

These comparisons give SNB.

Why the examples belong to the class. All three utilities have nonnegative coordinate slopes that sum to one and equal r at every constant act $r\mathbf{1}$. Integrating along the segment from y to x gives $I(x) - I(y) \geq \min_i(x_i - y_i)$. The affine costs $c_y(p) = I(y) - p \cdot y$, indexed by $y \in [0,1]^d$, therefore satisfy

$$\max_y \min_{p \in \Delta_d} \{p \cdot x + c_y(p)\} = \max_y \{I(y) + \min_i (x_i - y_i)\} = I(x).$$

Equality follows by choosing $y = x$. Also, $\min_p c_y(p) = I(y) - \max_i y_i \leq 0$, with equality at constant y, so the costs are normalized.

The bounded domain matters. On $\mathbb{R}^d$, constant additivity makes WNB imply SNB: the acts $x - I(x)\mathbf{1}$ and $y - I(y)\mathbf{1}$ are indifferent at zero, and their mixture utility is

$$I(tx + (1-t)y) - tI(x) - (1-t)I(y).$$

A WNB witness makes this expression take both strict signs. On the cube, these translations may be infeasible.

OA.7 RDEU: the neutrality argument

We prove Proposition OA.1. The geometric steps are justified below using only continuity of w.

Proof. If w is convex or concave, RDEU is respectively convex or concave in cumulative prob-

abilities, ruling out MMA. For the converse, suppose w is neither convex nor concave but SNB is absent. It is enough to work with four prizes.

Normalize four outcome utilities to $0 < a < b < 1$. In each Machina triangle, let p_0 and p_1 be the worst- and best-outcome probabilities. On the face with middle utility c, utility is $cw(1-p_0) + (1-c)w(p_1)$, where $p_0, p_1 \geq 0$ and $p_0 + p_1 \leq 1$. As usual, $w(0) = 0$ and $w(1) = 1$.

The geometric argument below gives a full neutral curve of slope $a/(1-a)$ on the a-face between two curved levels. Write its endpoints as $P = (\bar{p}_0, \bar{p}_1)$ and $P_A = (1-p_A, p_A)$. Bars distinguish the fixed endpoint P from a moving point (p_0, p_1).

First, w cannot be affine on $[\bar{p}_1, 1-\bar{p}_0]$. Since the curve is full, either $\bar{p}_1 = 0$ or $\bar{p}_0 = 0$. In the first case, every lower-level lottery has $p_0 > \bar{p}_0$, so affinity on this interval would make every lower indifference curve straight. In the second case, every higher-level lottery has $p_1 > \bar{p}_1$, so every higher curve would be straight. Either conclusion contradicts one of the two curved levels.

Through every point of the original curve, the b-face has a neutral segment of slope $b/(1-b)$ leading to the binary edge. Let $P_B = (1-p_B, p_B)$ be the endpoint of the segment from P. Here p_A, p_B are the best-outcome probabilities at the binary endpoints:

$$p_A = a(1-\bar{p}_0) + (1-a)\bar{p}_1 < p_B = b(1-\bar{p}_0) + (1-b)\bar{p}_1.$$

These segments fill PP_AP_B in Figure OA.1. Throughout this triangle, neutrality gives

$$bw(1-p_0) + (1-b)w(p_1) = w\big(b(1-p_0) + (1-b)p_1\big). \tag{OA.7.1}$$

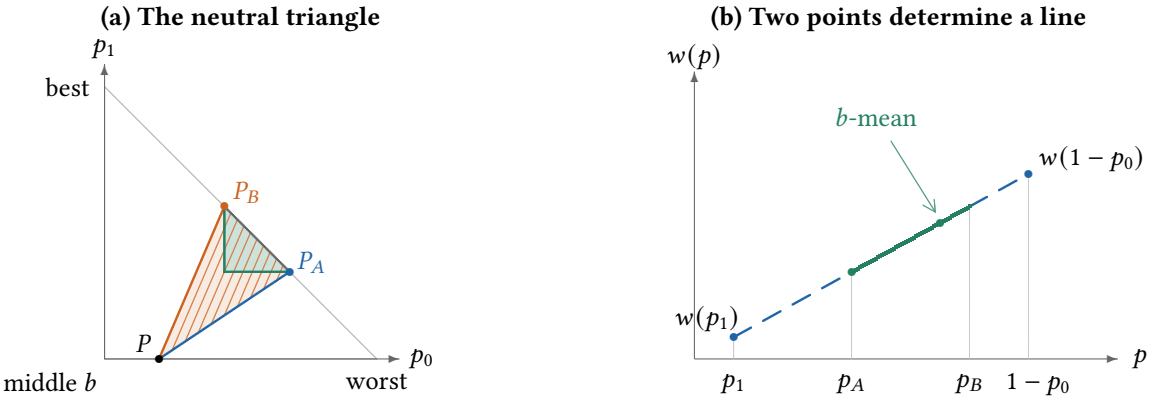


Figure OA.1: The orange segments fill a triangle containing all ordered pairs in $[p_A, p_B]$, where w must be affine. Each graph chord in panel (b) passes through two points of this straight part and extends the same line. Panel (a) shows $\bar{p}_1 = 0$; its blue segment is neutral on the a-face. The drawings are schematic.

The smaller green triangle contains every pair $p_A \leq p_1 \leq 1 - p_0 \leq p_B$. Thus (OA.7.1) is Jensen's equality on $[p_A, p_B]$, making w affine there. Repeatedly splitting the interval in proportions b and $1-b$ puts a dense set of graph points on its chord; continuity puts the whole graph on that line.

Now take any nonbinary point (p_0, p_1) on the original neutral curve. Its two neutrality

conditions place two points of the graph chord joining $(p_1, w(p_1))$ and $(1-p_0, w(1-p_0))$ on the graph of w, at probabilities p_A and $b(1-p_0)+(1-b)p_1$. These are distinct points in the known straight part. Two points determine a line, so both endpoints of the chord lie on that same line. As (p_0, p_1) moves along the original curve, the probabilities p_1 and $1-p_0$ cover $[\bar{p}_1, 1-\bar{p}_0]$. Hence w is affine on the interval where we just showed it cannot be affine, a contradiction. □

The geometric steps

Regularity. Without SNB, each indifference graph is convex or concave: otherwise a secant has graph points strictly on both sides, giving an indifferent chord with mixtures both better and worse than its endpoints. We first show that w is locally bi-Lipschitz on $(0,1)$. On the a-face, an indifference curve has the form

$$p_1 = h_v(p_0) = w^{-1}\left(\frac{v - aw(1-p_0)}{1-a}\right).$$

Near any interior point, choose a small interval of levels $[v_0, v_1]$ whose curves share a p_0-interval. On a smaller interval, convexity or concavity bounds inner secants between outer secants. Strict increase and compactness give uniform positive bounds on their slopes, above and away from zero. Integrating over v preserves these bounds and gives $H(w(1-p_0))$, where

$$H(z) = \int_{v_0}^{v_1} w^{-1}\left(\frac{v-az}{1-a}\right)\,dv,$$
$$H'(z) = -a\left[w^{-1}\left(\frac{v_1-az}{1-a}\right) - w^{-1}\left(\frac{v_0-az}{1-a}\right)\right] < 0.$$

A change of variables gives the derivative. It is continuous and bounded away from zero locally, so the secant bounds transfer to w. Thus w is locally absolutely continuous, with $0 < w' < \infty$ almost everywhere. Call these probabilities *regular*.

The neutral curve. On the face with middle utility $c \in \{a, b\}$, a Machina indifference graph $p_1 = g(p_0)$ has, when $1-p_0$ and p_1 are regular,

$$g'(p_0) = \frac{c}{1-c}\,\frac{w'(1-p_0)}{w'(p_1)}.$$

At a regular binary-edge point its slope is $c/(1-c)$. Since that point is the right endpoint, convex curves have slopes at most this value and concave curves have slopes at least this value. A strict bend persists at nearby levels, so approximation by regular binary-edge points extends these bounds to every curved level.

If all curves on the a-face were convex, the same approximation would give the upper bound on every level. Hence $w'(1-p_0) \le w'(p_1)$ for regular $0 < p_1 < 1-p_0 < 1$, making w concave. If all curves were concave, w would be convex. Thus, when w is neither, both curved shapes occur.

Every curved level lies on one side of the line of slope $a/(1-a)$ through its binary endpoint. A straight level also lies on one side, even if its slope differs. The levels with a strict positive gap and those with a strict negative gap form disjoint open sets. Between two curved levels of opposite shapes, some full curve must therefore coincide with its line. This gives the full neutral curve used in the proof.

Passing to the second face. At every point of this neutral curve where $1-p_0$ and p_1 are regular, differentiating neutrality gives $w'(1-p_0) = w'(p_1)$. The b-curve through the same probabilities con-

sequently has slope $b/(1-b)$. If it is curved, monotonicity of its slopes and the relevant bound force it to stay straight from this point to the binary edge. If it is already straight, the conclusion is immediate. Regular pairs have full measure along the original line, since both p_1 and $1-p_0$ vary affinely with nonzero slopes. Continuity therefore gives a b-neutral segment through every point, including the boundary endpoint. Its binary endpoint runs from P_B to P_A, so the segments fill precisely the triangle used in the proof.

The three-prize restriction. Suppose SNB is absent on a three-prize face and let $c/(1-c)$ be its binary-edge benchmark. If WNB held, choose an indifference level strictly between its peak and both endpoint values, and another strictly between its dip and both endpoint values. Choose these levels with regular binary-edge probabilities; such levels are dense. Continuity gives two indifferent subchords of the original line, one with a strictly better mixture and one with a strictly worse mixture. The first belongs to a convex curve and has slope strictly below $c/(1-c)$; the second belongs to a concave curve and has slope strictly above it. They cannot lie on the same line. Hence WNB implies SNB. Strict comparisons allow a boundary witness to be moved into the interior first if necessary.

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